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Grube

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(54) **SECURELY PROCESSING A CONTINGENT ACTION TOKEN**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 194 days.

This patent is subject to a terminal disclaimer.

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Related U.S. Application Data

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(51) **Int. Cl.**

H04L 29/06 (2006.01)

G06Q 20/38 (2012.01)

G06Q 20/40 (2012.01)

G06Q 40/04 (2012.01)

G06Q 40/06 (2012.01)

(Continued)

(52) **U.S. Cl.**

CPC **H04L 9/50** (2022.05); **G06Q 20/389** (2013.01); **G06Q 20/401** (2013.01); **G06Q 40/04** (2013.01); **G06Q 40/06** (2013.01);

G06Q 40/08 (2013.01); **H04L 9/0825** (2013.01); **H04L 9/3213** (2013.01); **H04L 9/3234** (2013.01); **H04L 9/3247** (2013.01); **G06Q 2220/00** (2013.01); **H04L 2209/56** (2013.01)

(58) **Field of Classification Search**

CPC H04L 9/50; H04L 9/0825; H04L 9/3213; H04L 9/3234; H04L 9/3247; H04L 2209/56; G06Q 20/389; G06Q 20/401; G06Q 40/04; G06Q 40/06; G06Q 40/08; G06Q 2220/00

See application file for complete search history.

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Primary Examiner — Mahfuzur Rahman

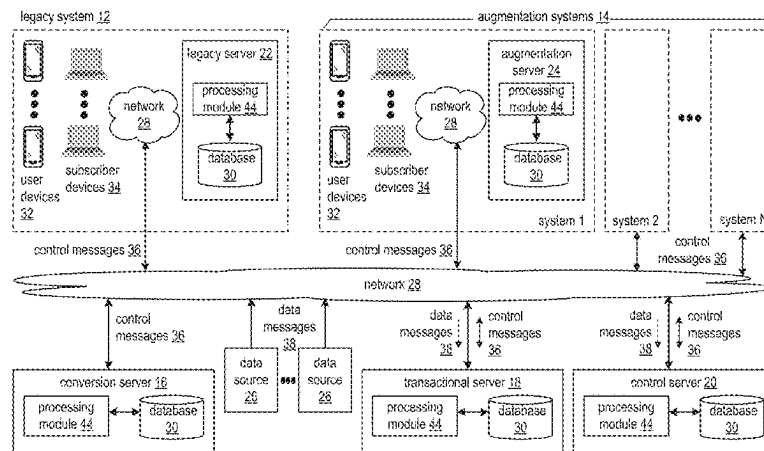
(74) *Attorney, Agent, or Firm* — Gary W. Grube

(57)

ABSTRACT

A method executed by a computing device includes identifying a set of contingency-action tokens (CATs) of an object distributed ledger that meets minimum CAT requirements. The method further includes determining to update the set of CATs based on an evaluation of each CAT. The method further includes determining reassignment information for each CAT and facilitating taking control of each CAT of a blockchain of the object distributed ledger using a securely passing process. The method further includes updating each CAT to include the reassignment information to produce an updated CAT. The method further includes causing generation of a new block affiliated with each corresponding updated CAT via the blockchain of the object distributed ledger using the securely passing process, where the new block includes the corresponding updated CAT.

12 Claims, 53 Drawing Sheets



- (51) **Int. Cl.**
G06Q 40/08 (2012.01)
H04L 9/00 (2022.01)
H04L 9/08 (2006.01)
H04L 9/32 (2006.01)

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				705/71
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				705/71
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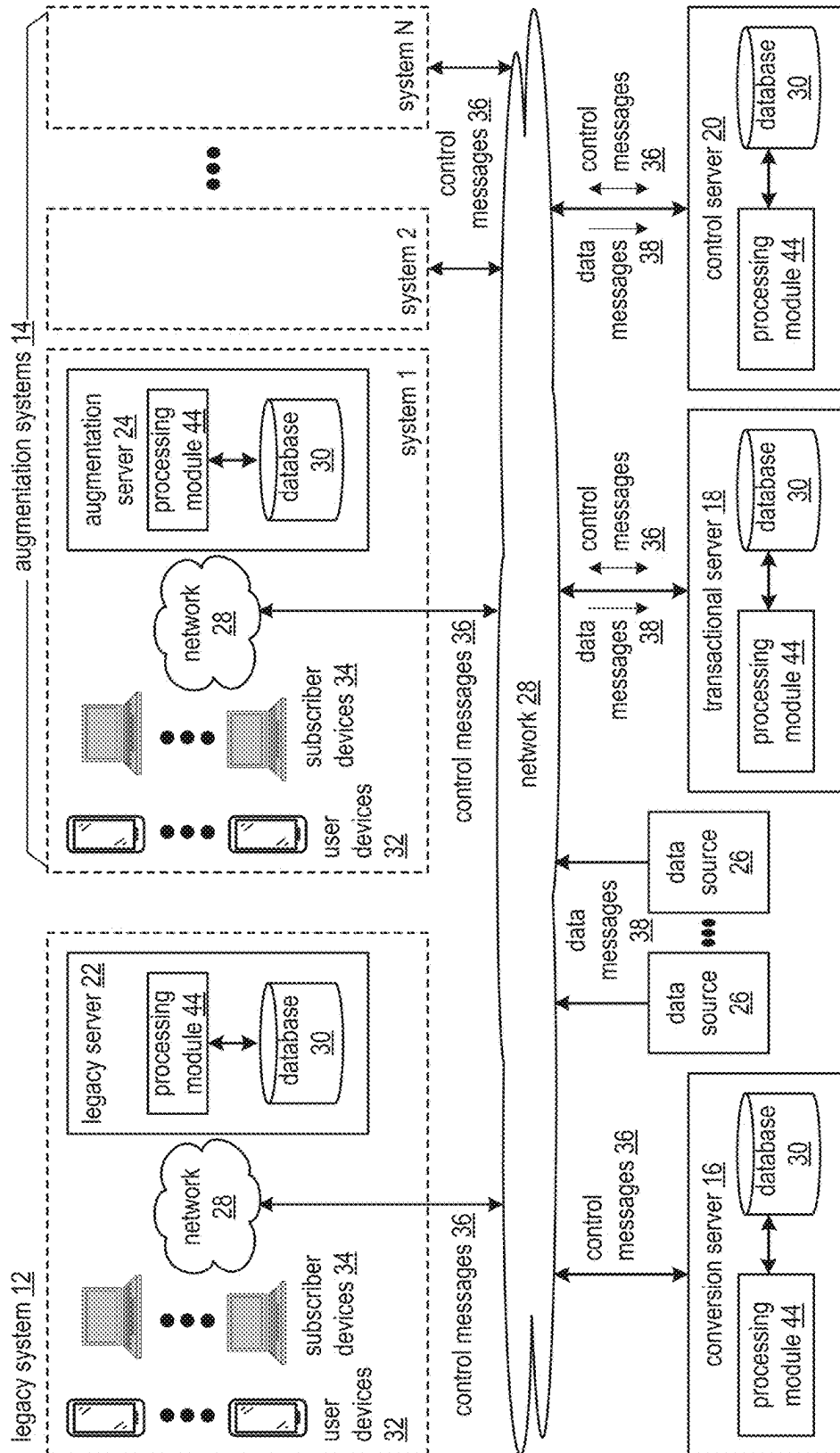


FIG. 1
communication system 10

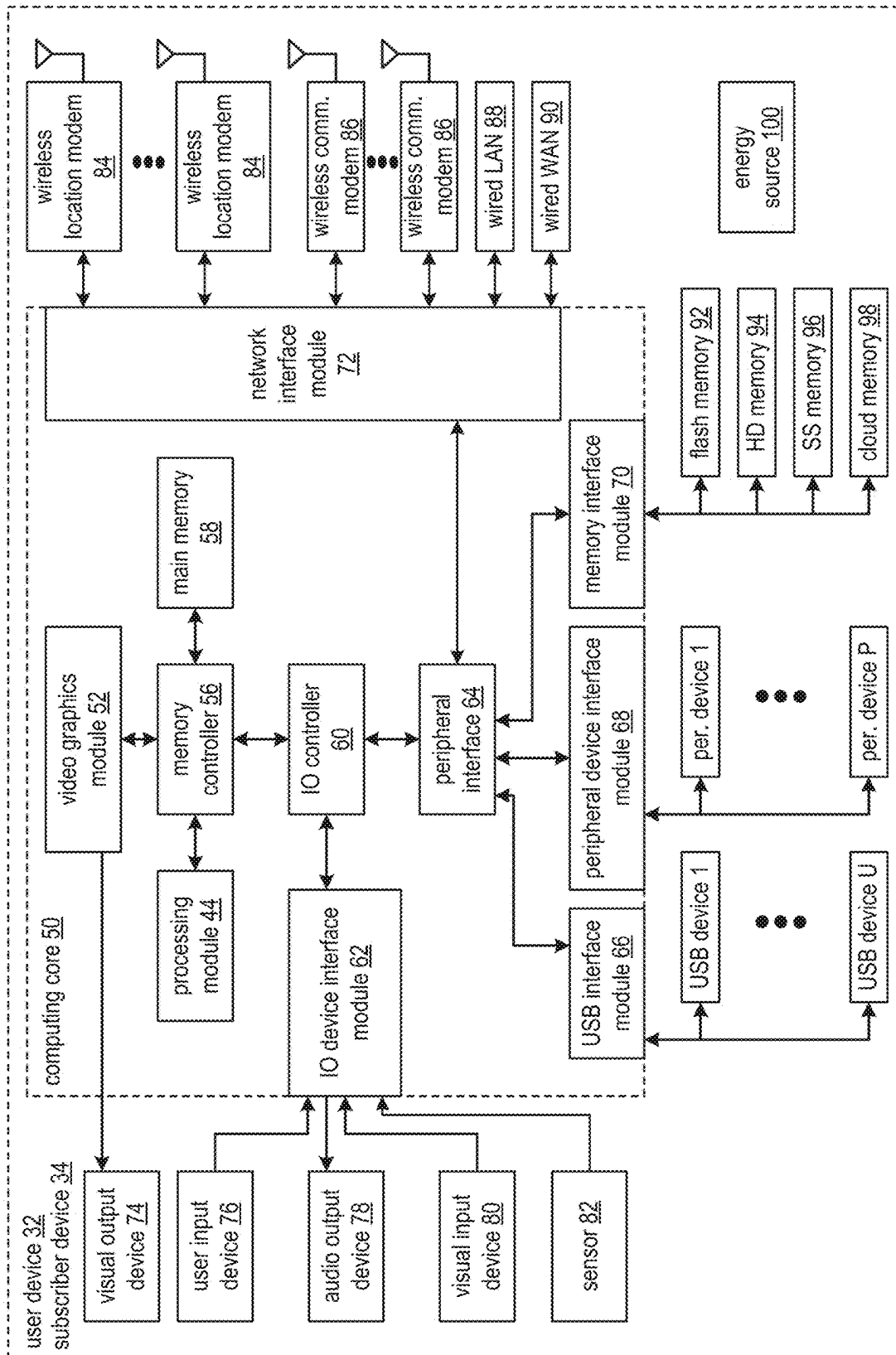


FIG. 2

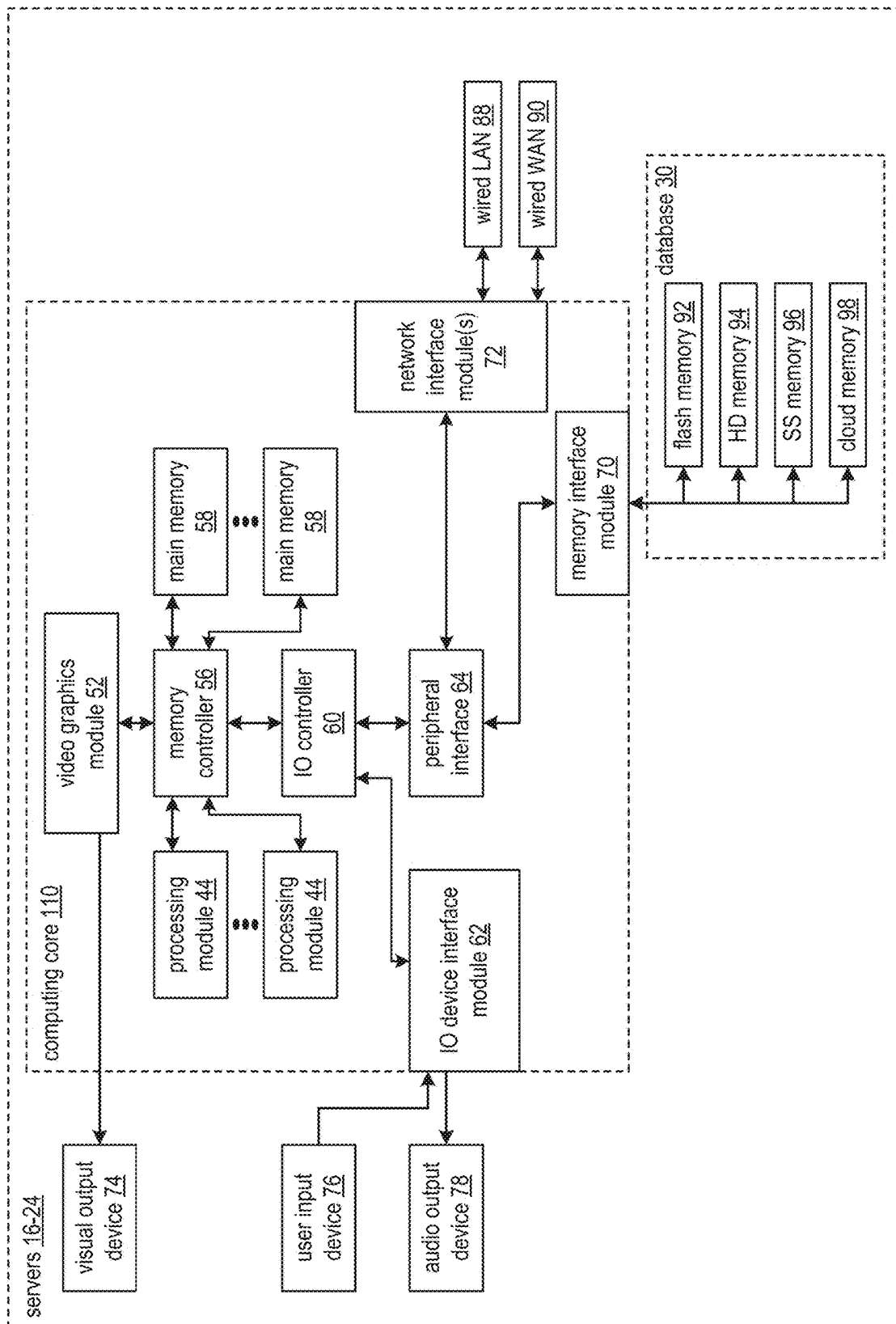


FIG. 3

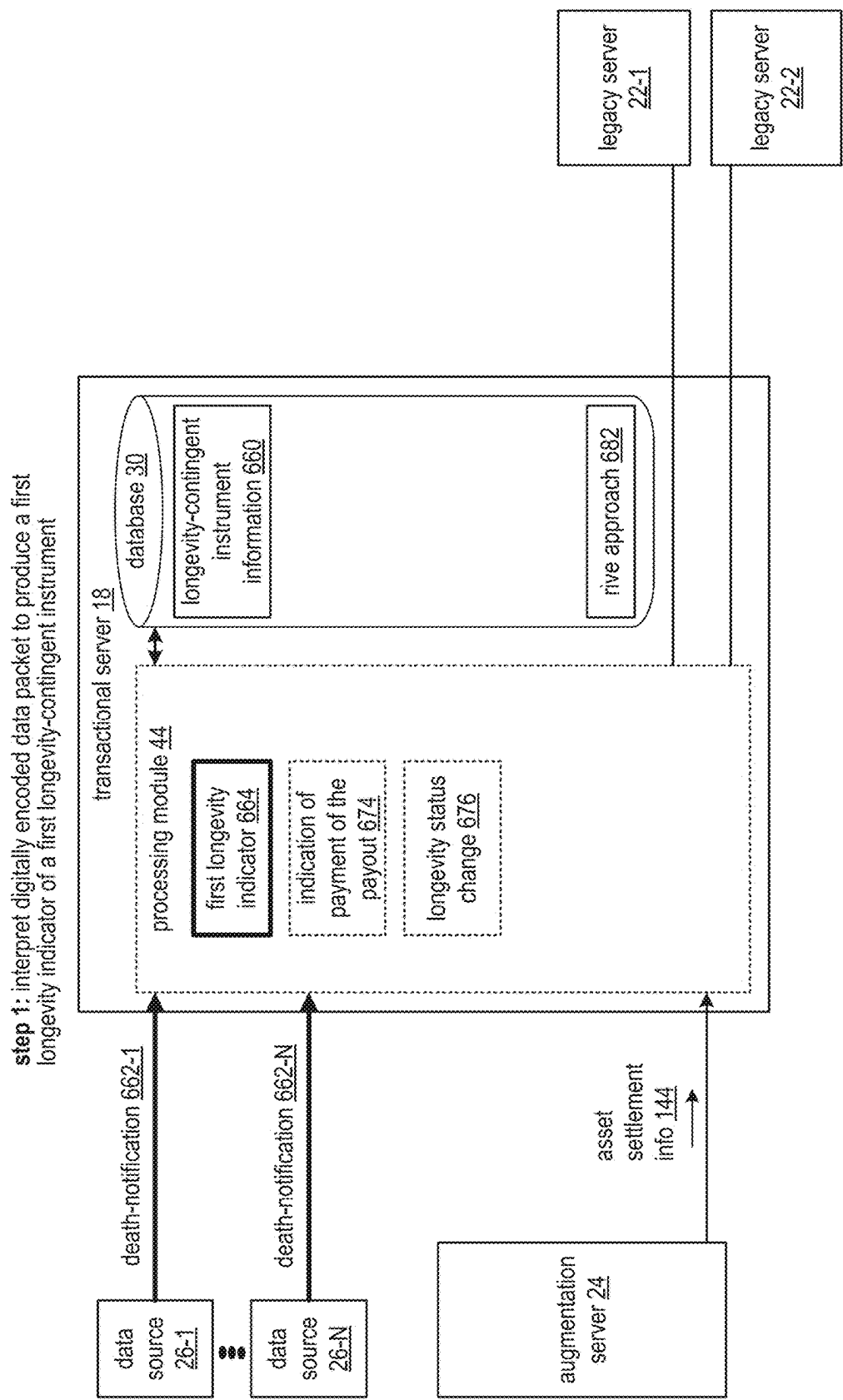


FIG. 4A

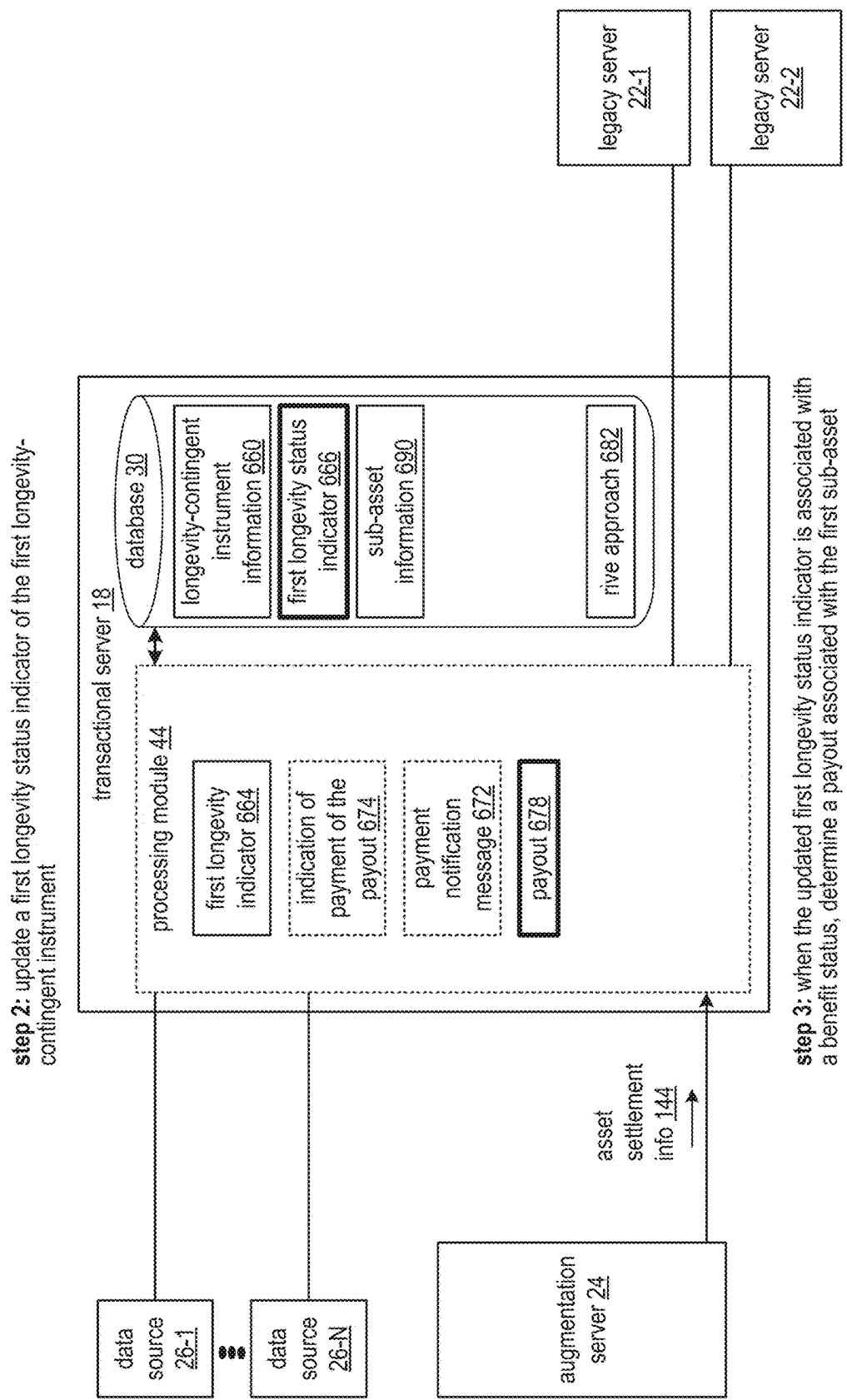


FIG. 4B

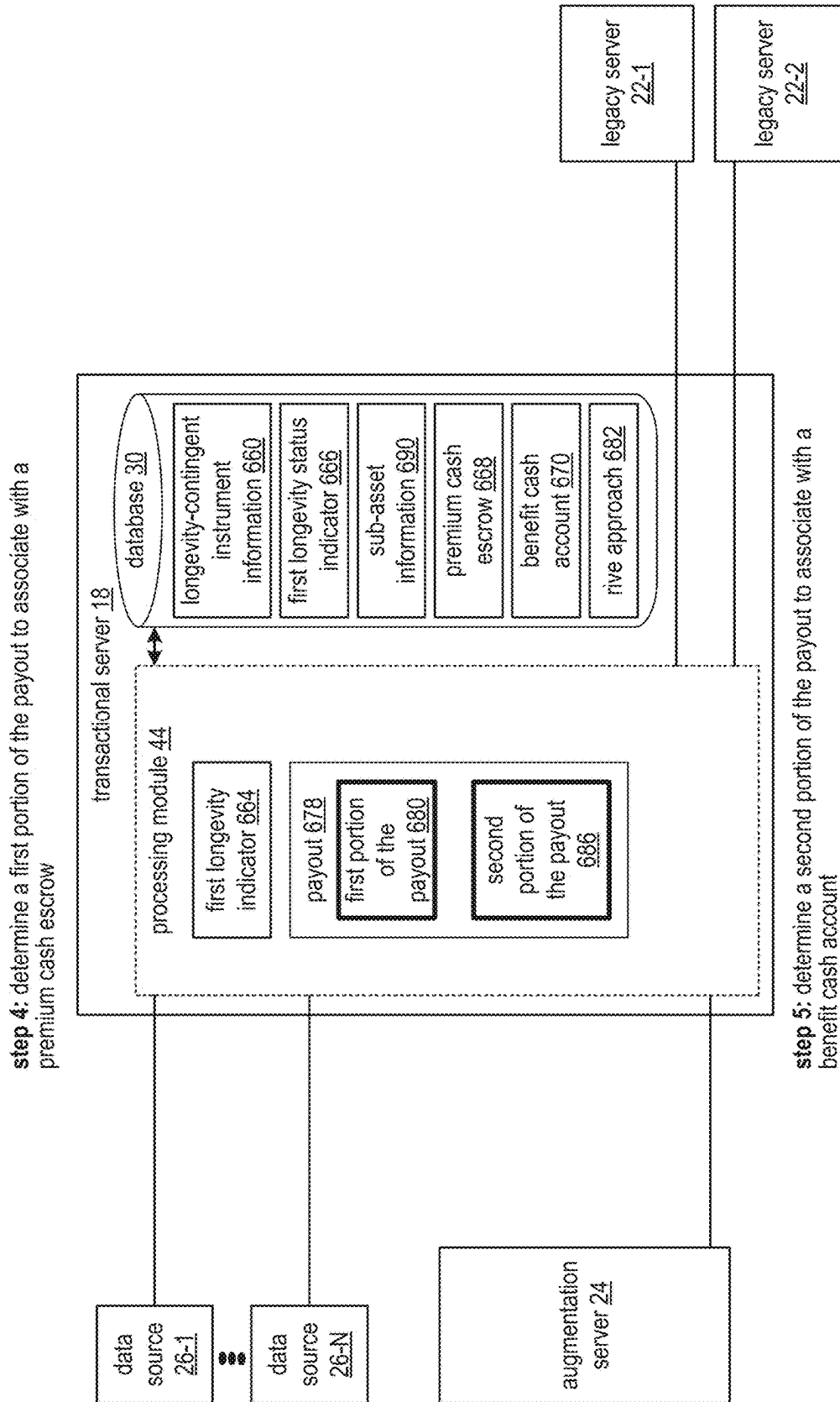


FIG. 4C

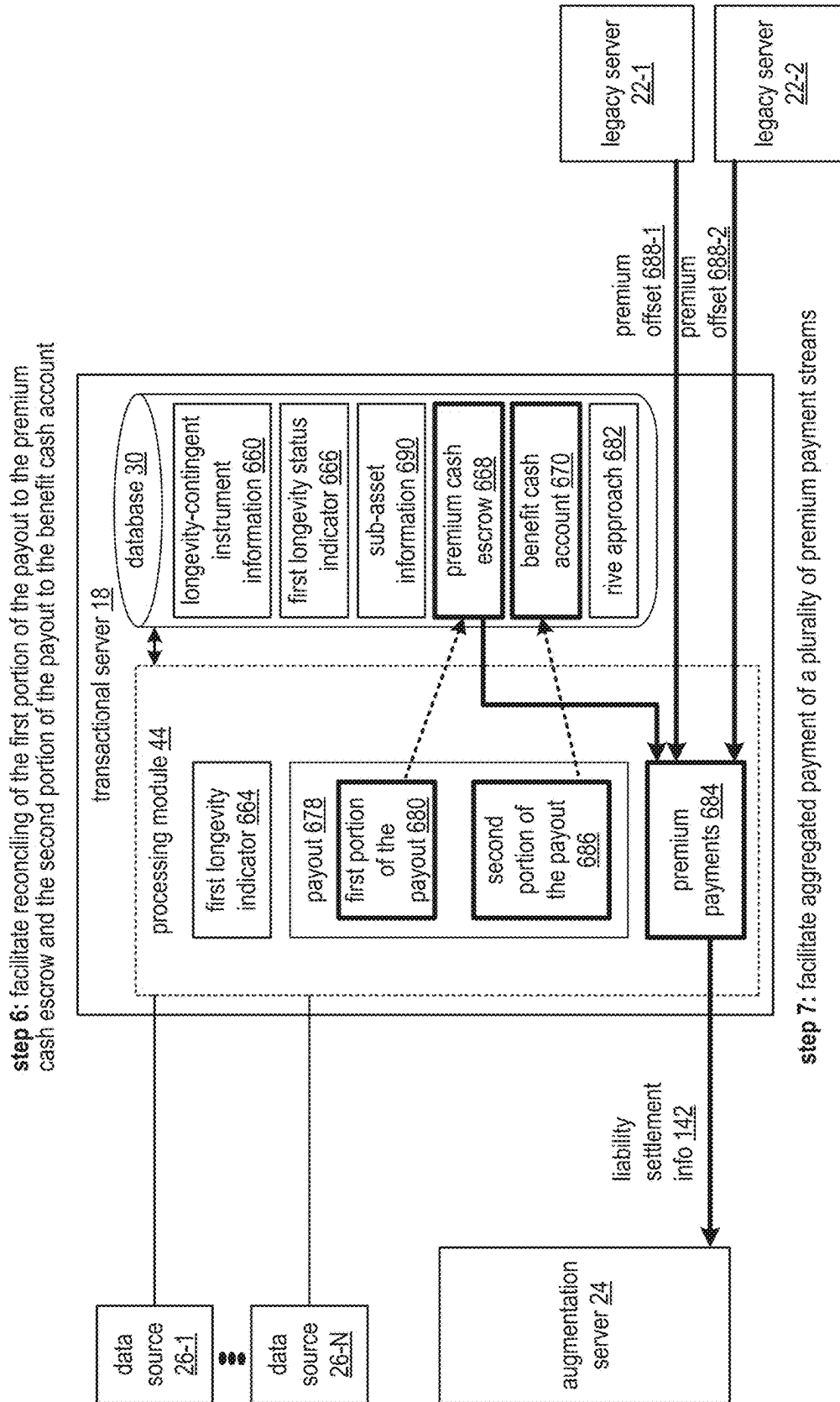


FIG. 4D

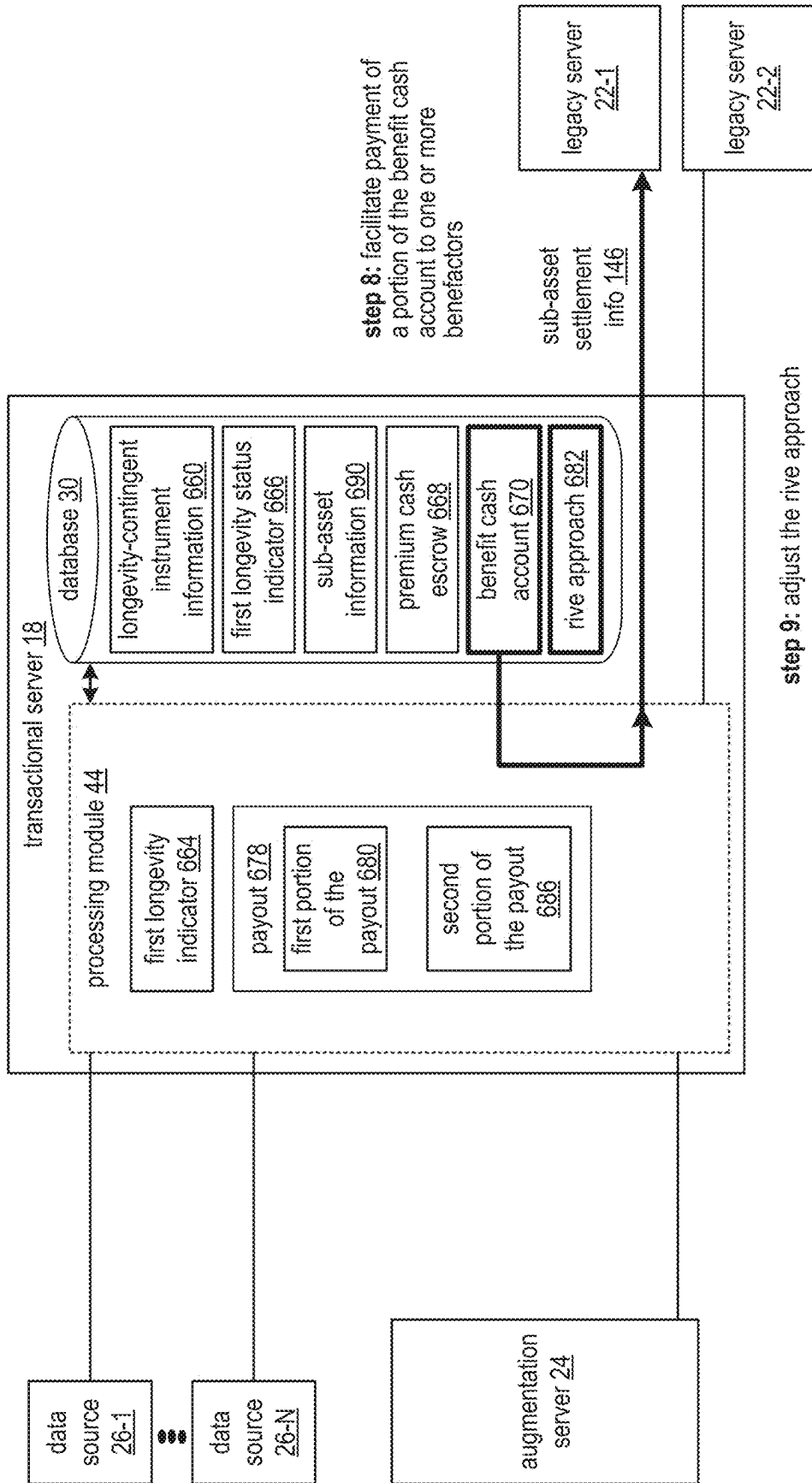


FIG. 4E

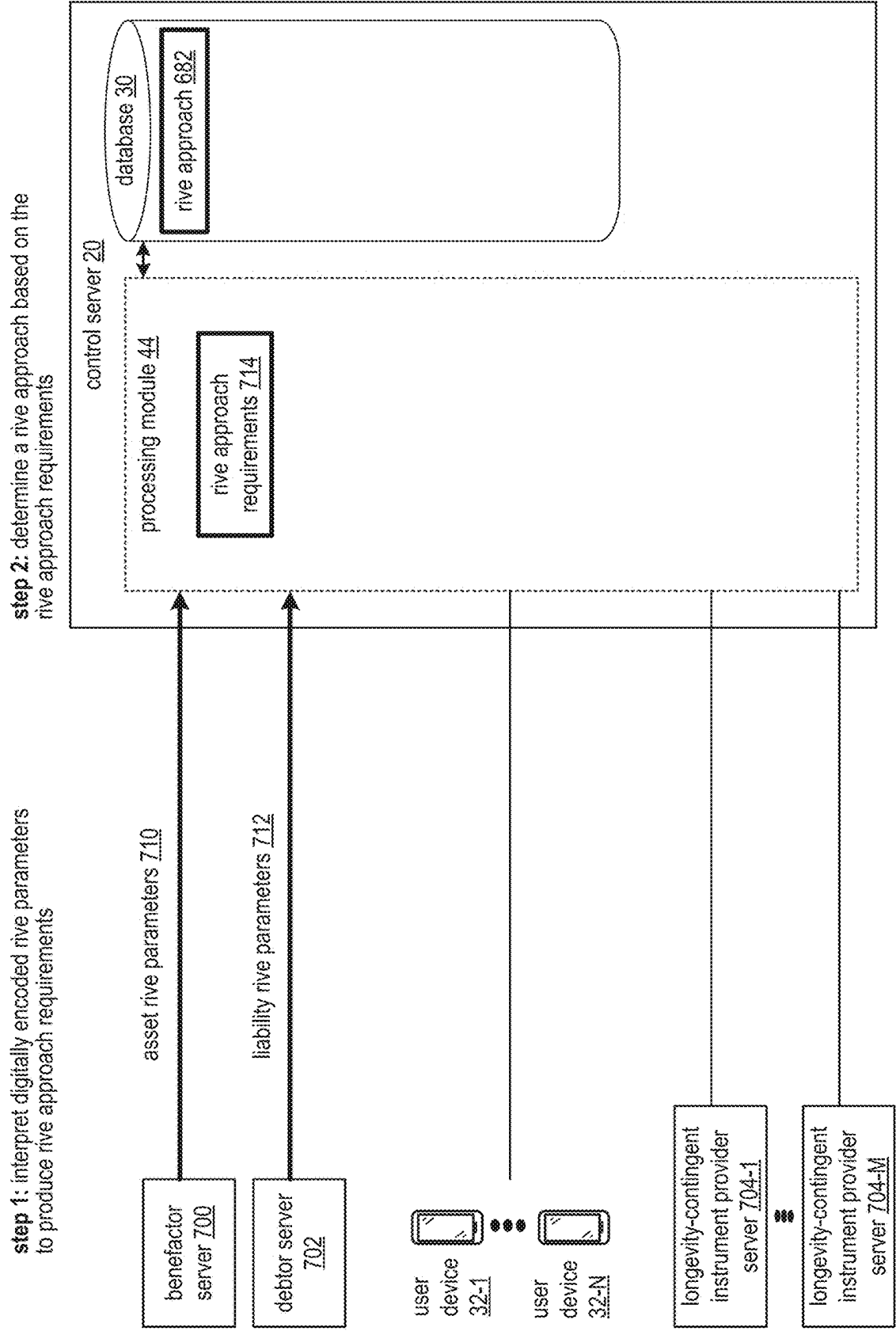


FIG. 5A

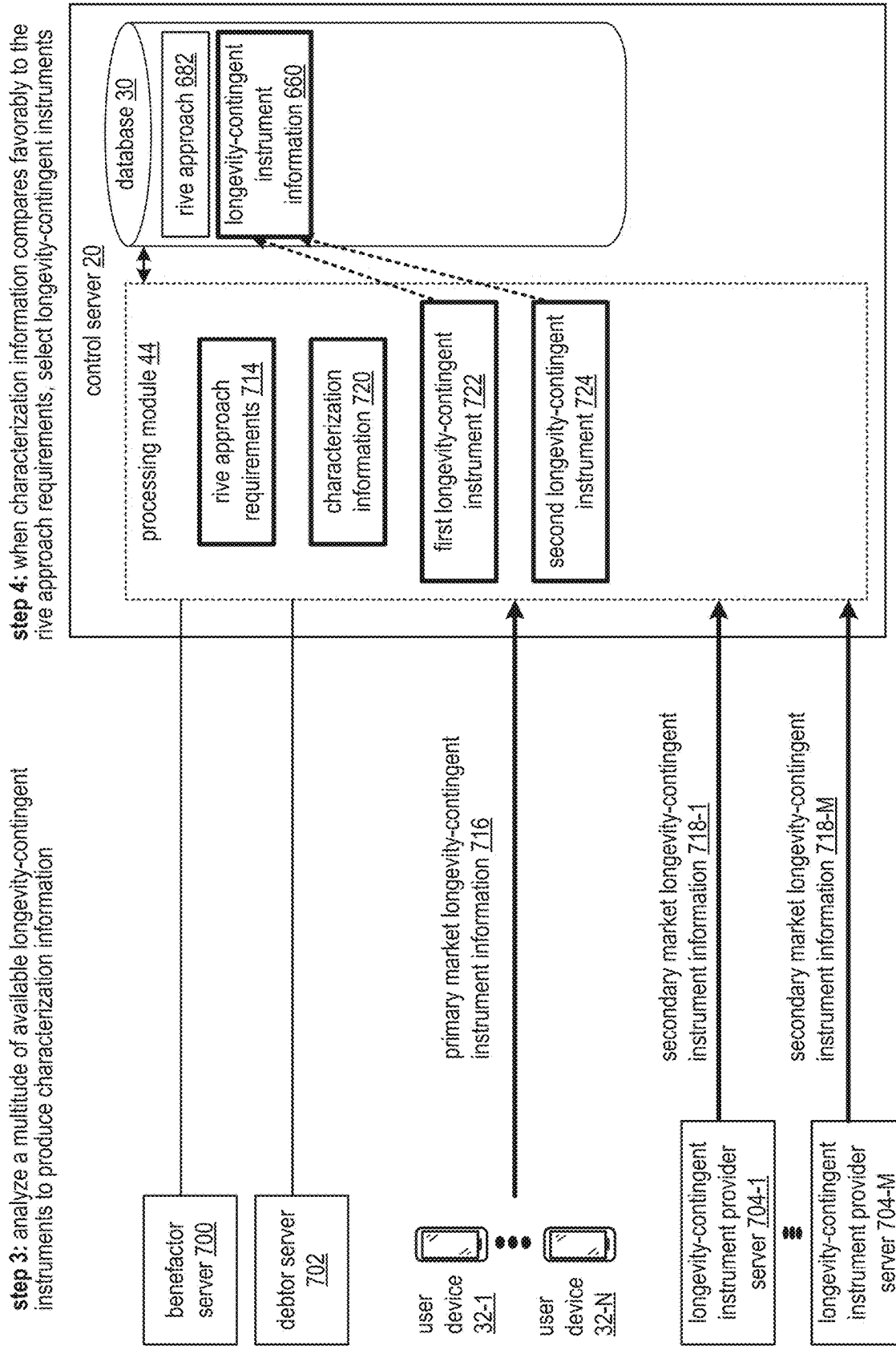


FIG. 5B

step 5: rive a first longevity-contingent instrument in accordance with the rive approach to produce a first sub-asset and a first sub-liability

step 6: rive a second longevity-contingent instrument in accordance with the rive approach to produce a second sub-asset and a second sub-liability

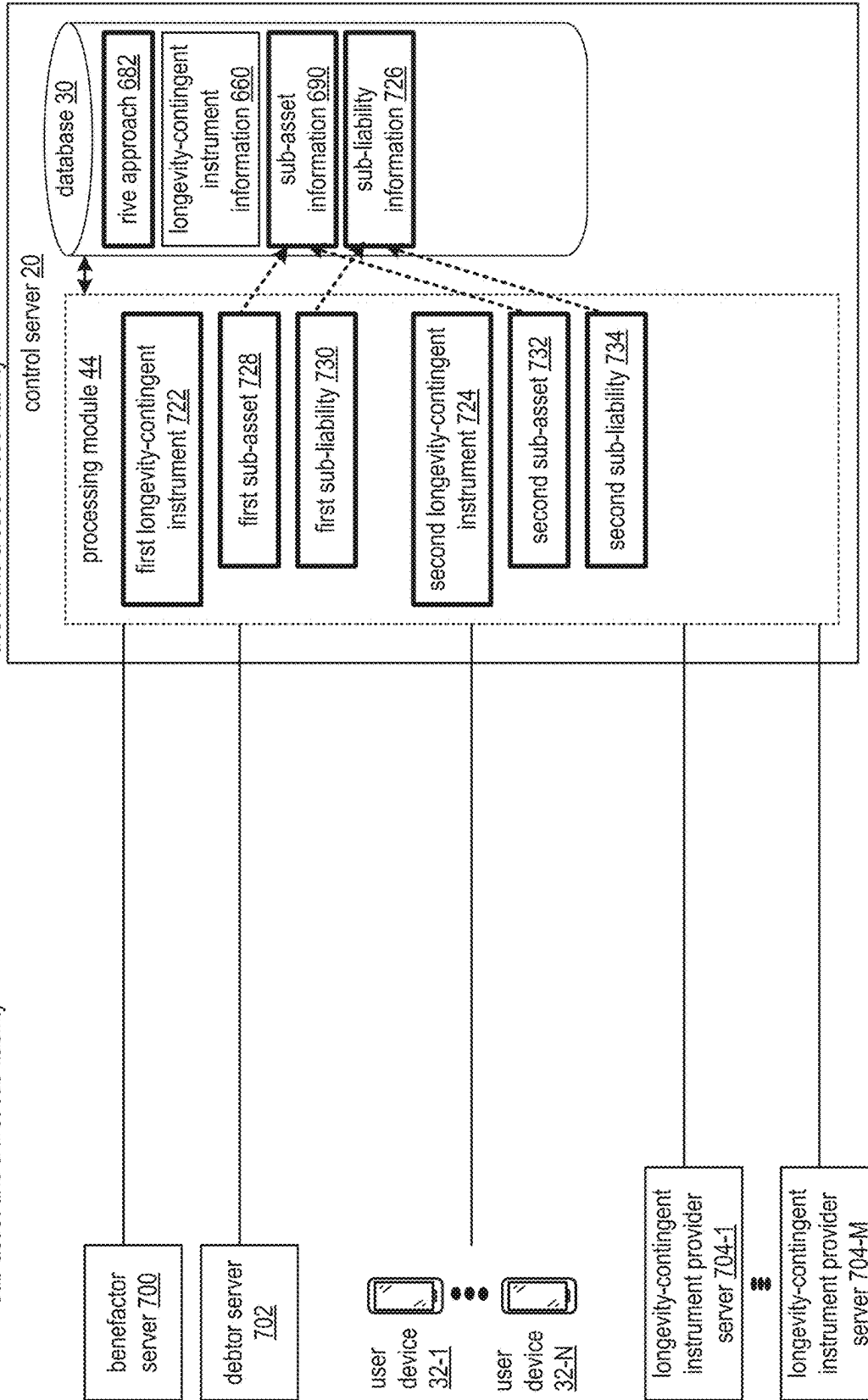


FIG. 5C

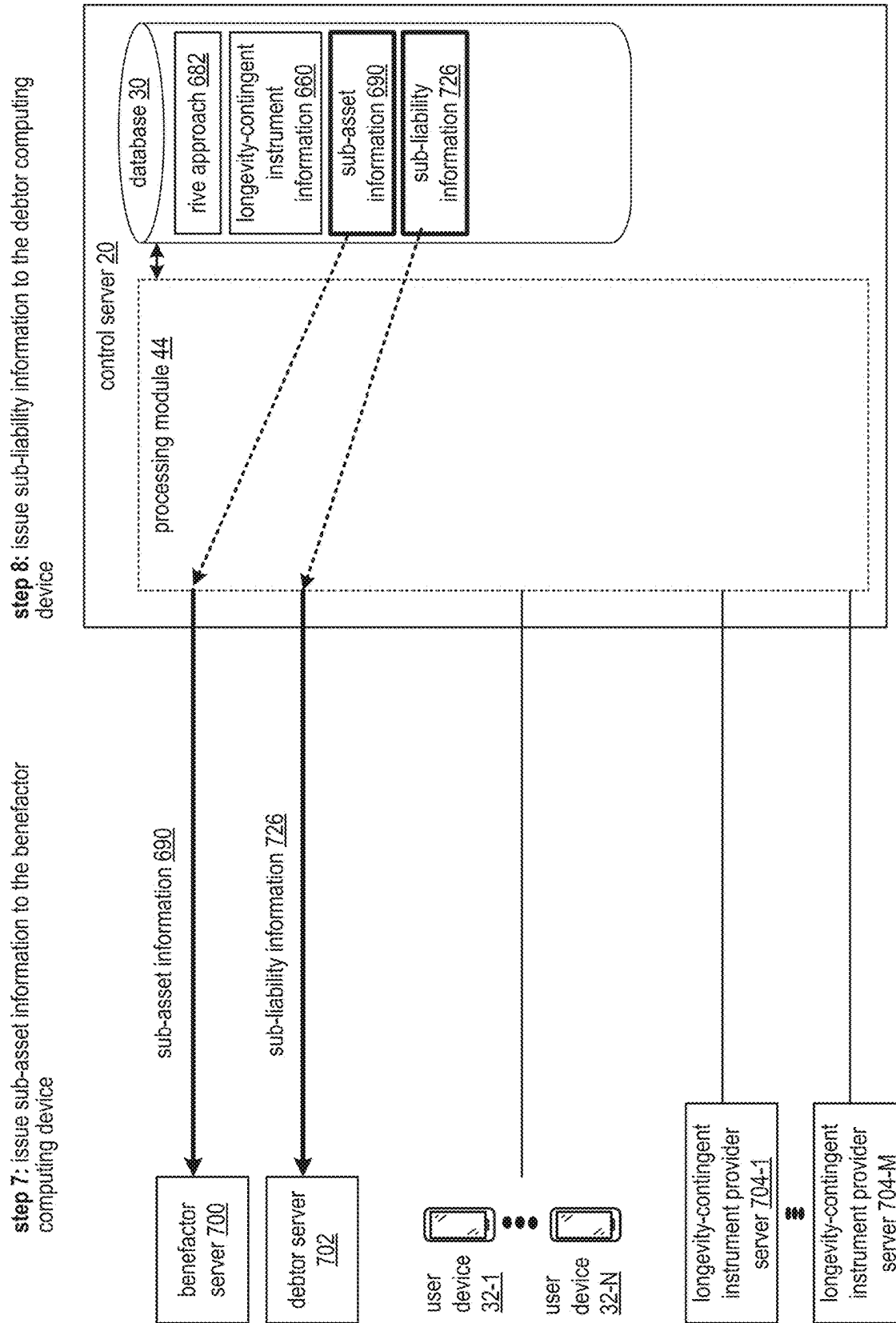


FIG. 5D

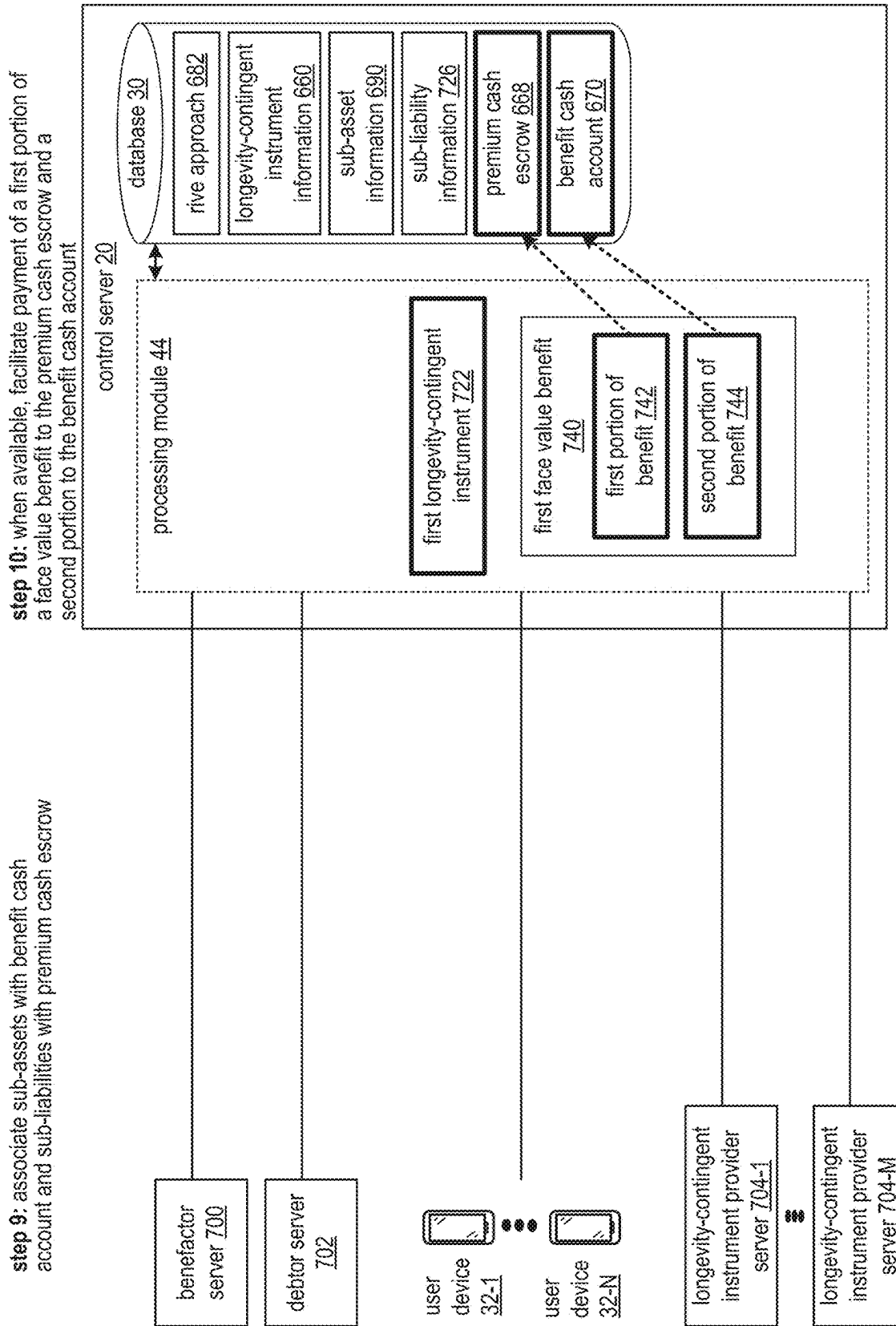


FIG. 5E

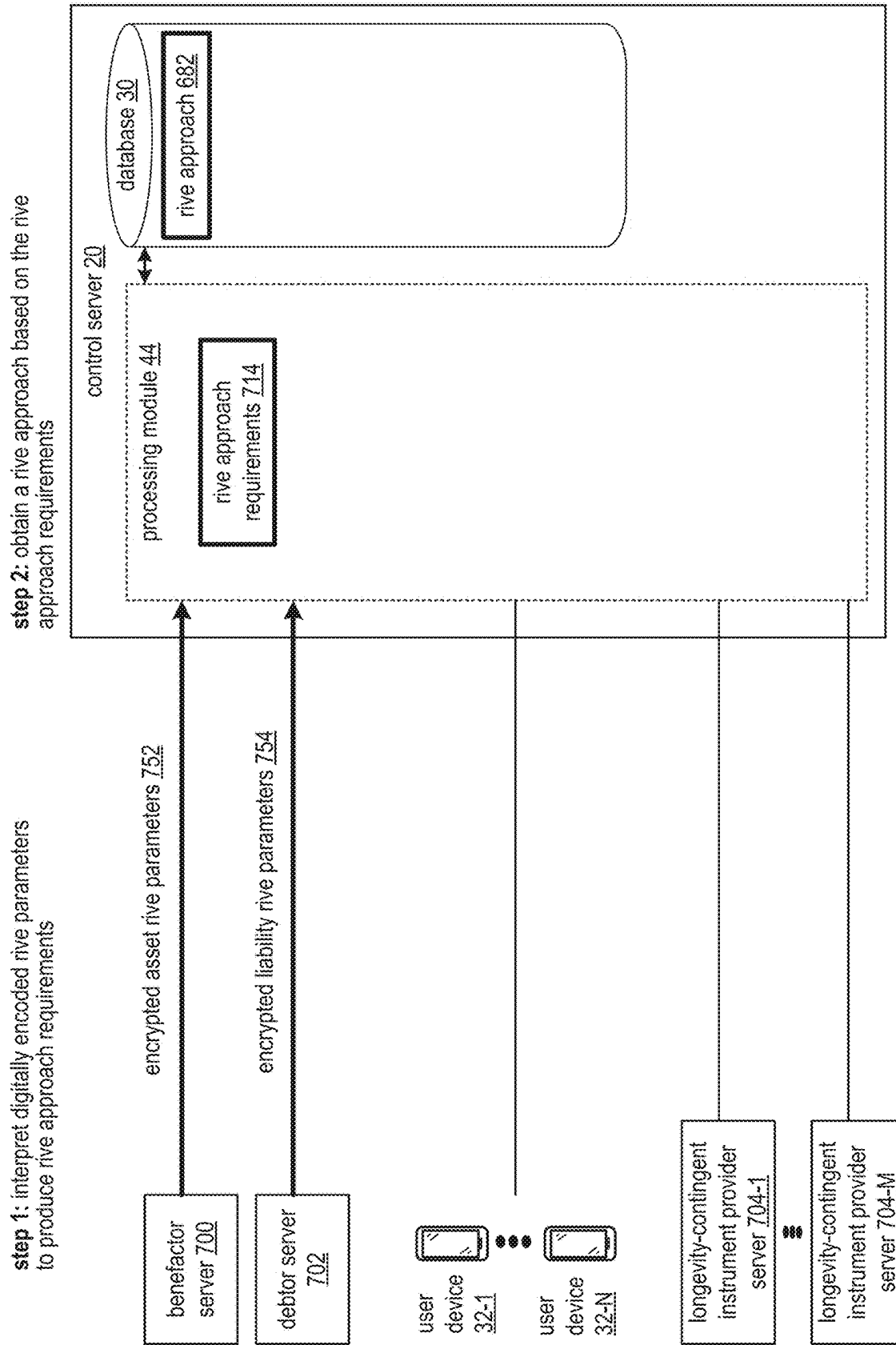


FIG. 6A

step 3: verify authenticity of blockchain-encoded records to produce an authenticity indicator

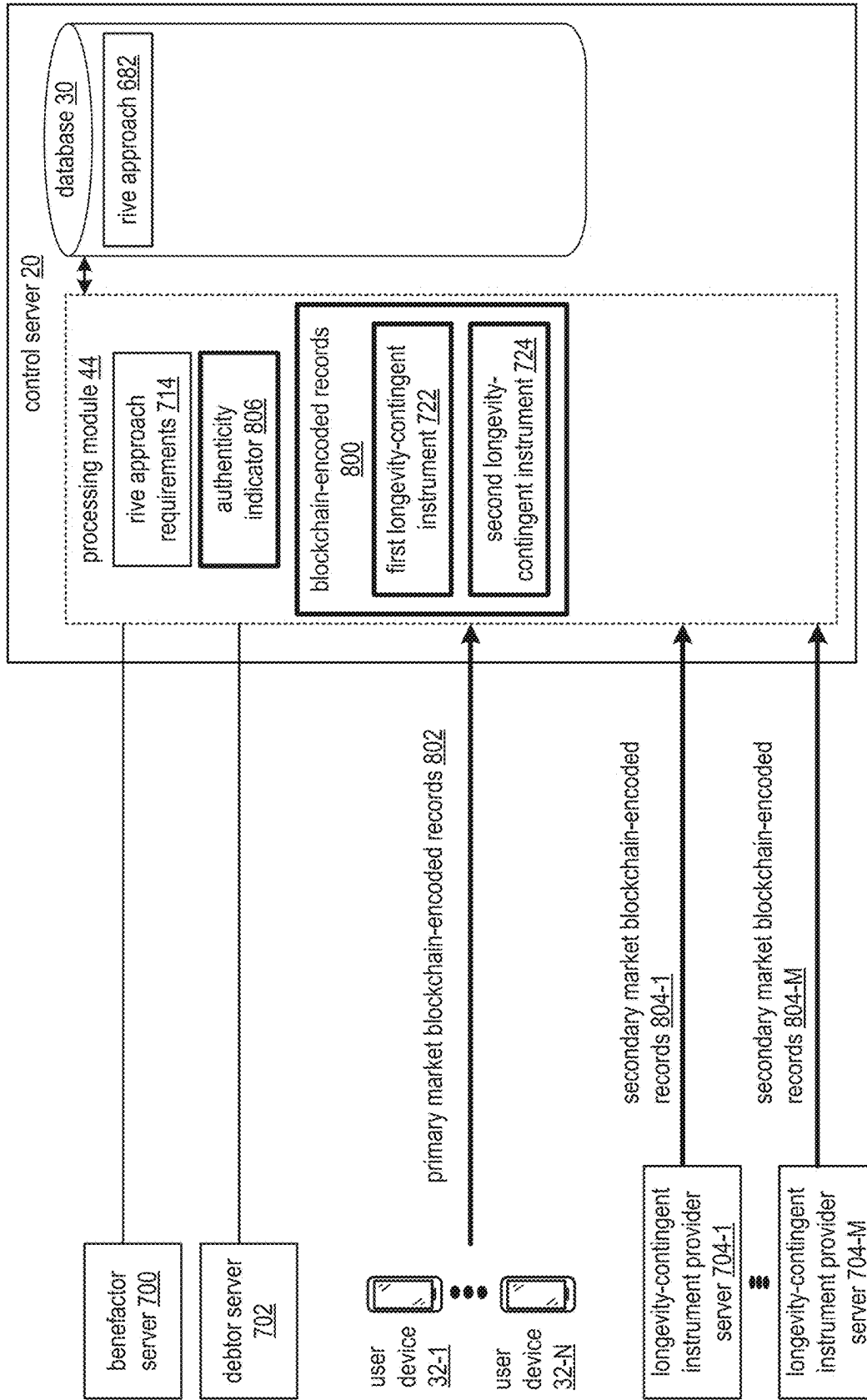


FIG. 6B

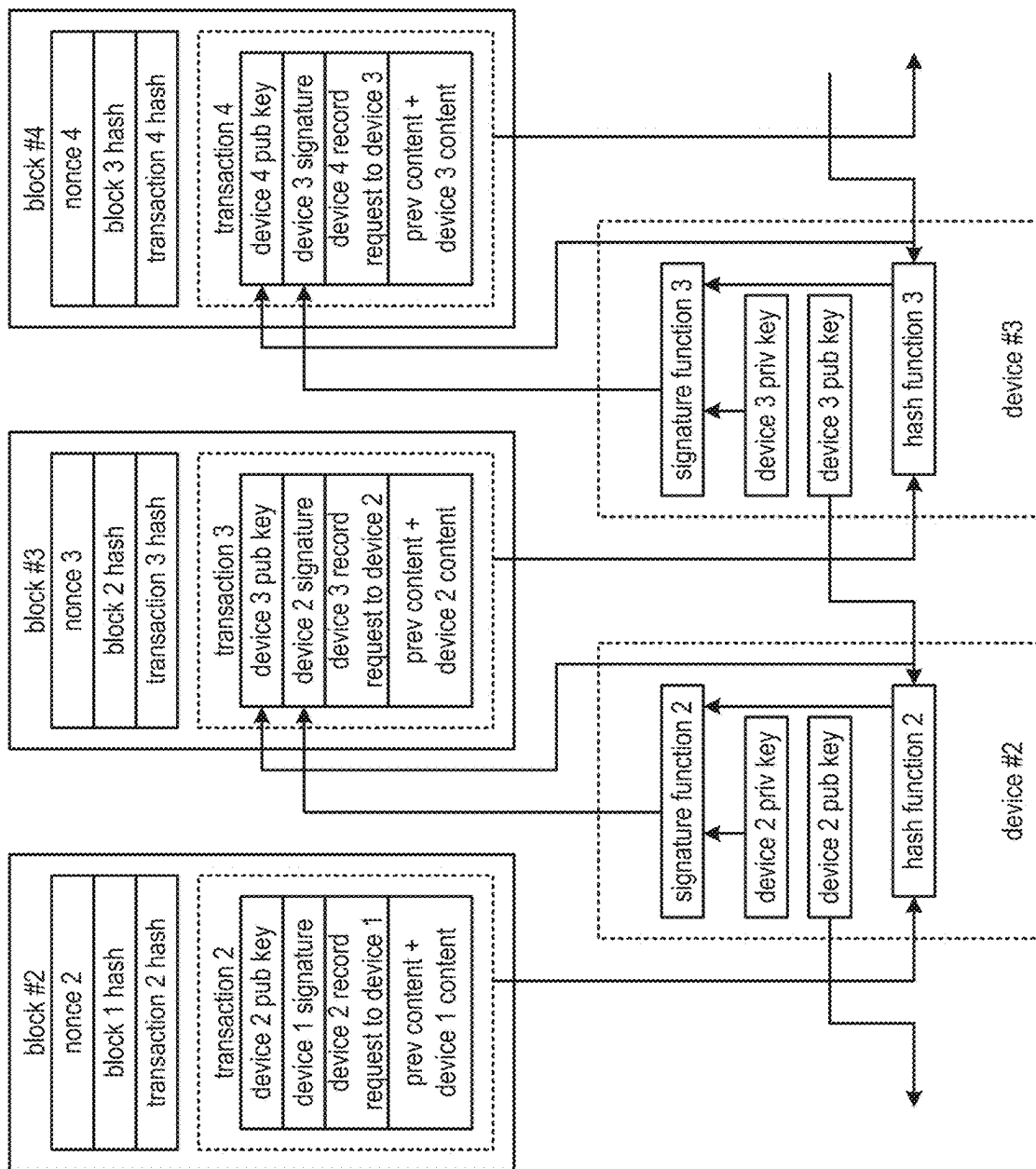


FIG. 6C

step 4: when the authenticity indicator is favorable, select the first longevity-contingent instrument and the second longevity-contingent instrument based on the rive approach to produce a set of longevity-contingent instruments

step 5: generate selection information

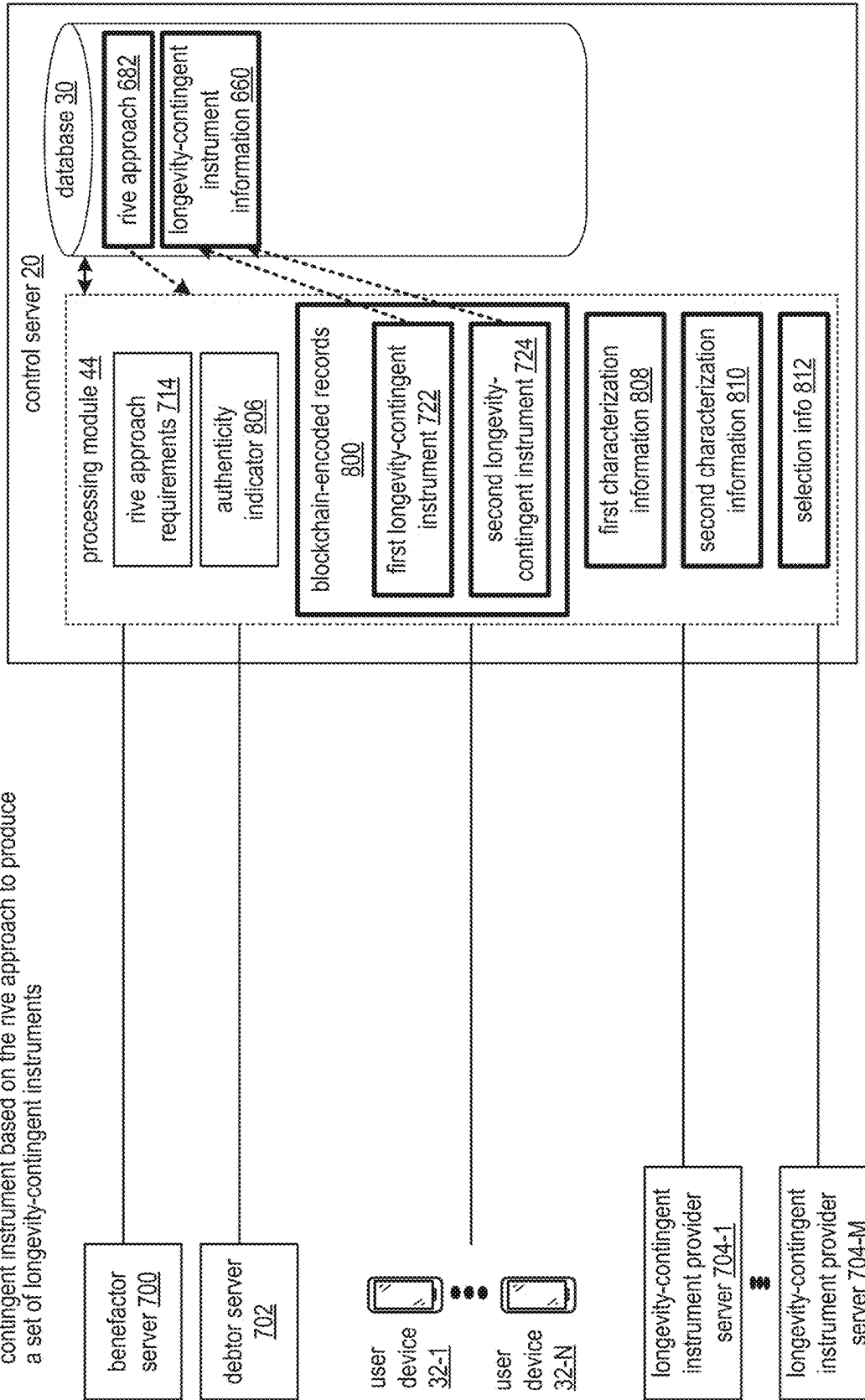


FIG. 6D

step 6: update blockchain-encoded records for the first and second longevity-contingent instruments to include the selection information

step 7: rive the first and second longevity-contingent instruments in accordance with the rive approach to produce sub-assets and sub-liabilities

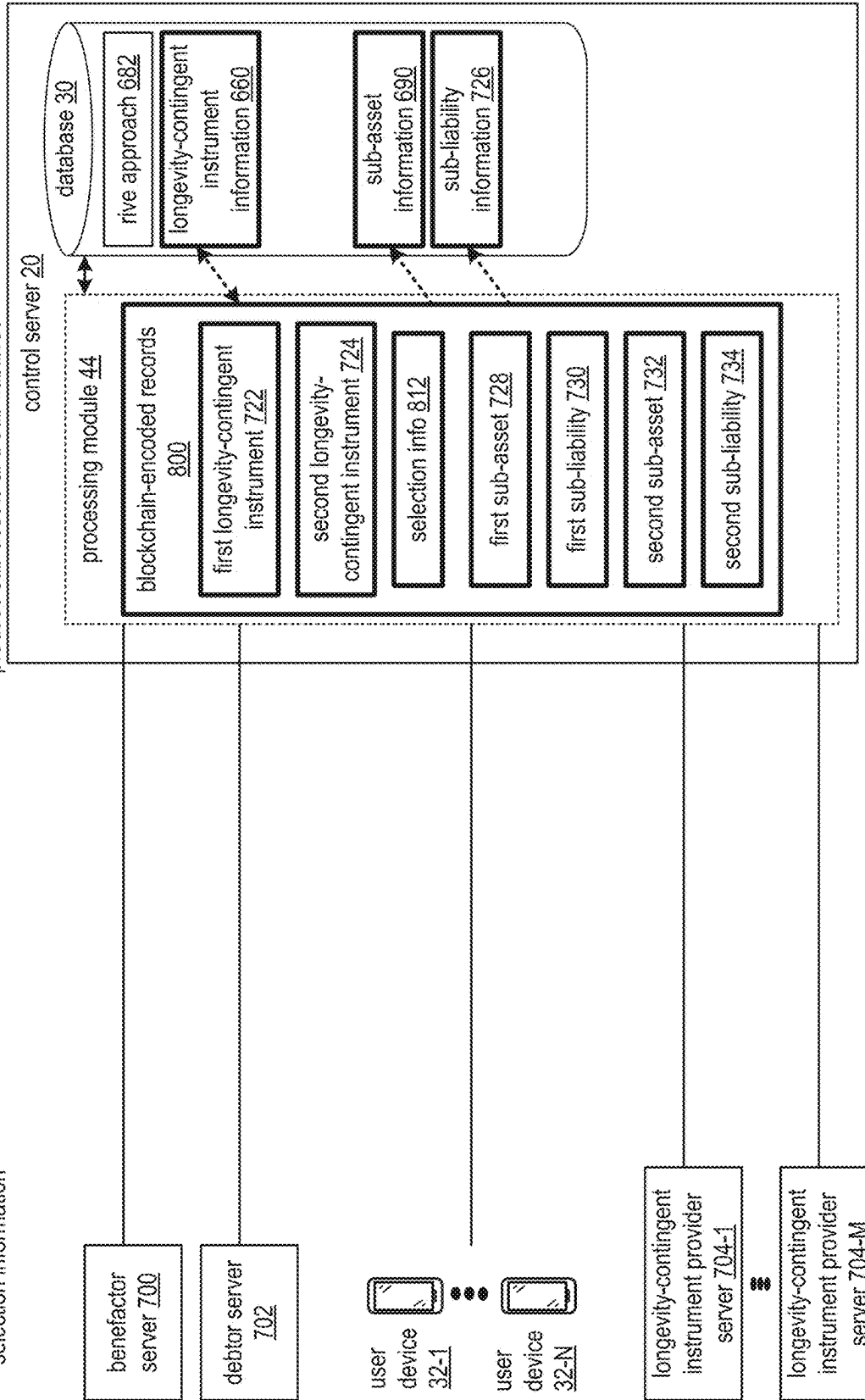


FIG. 6E

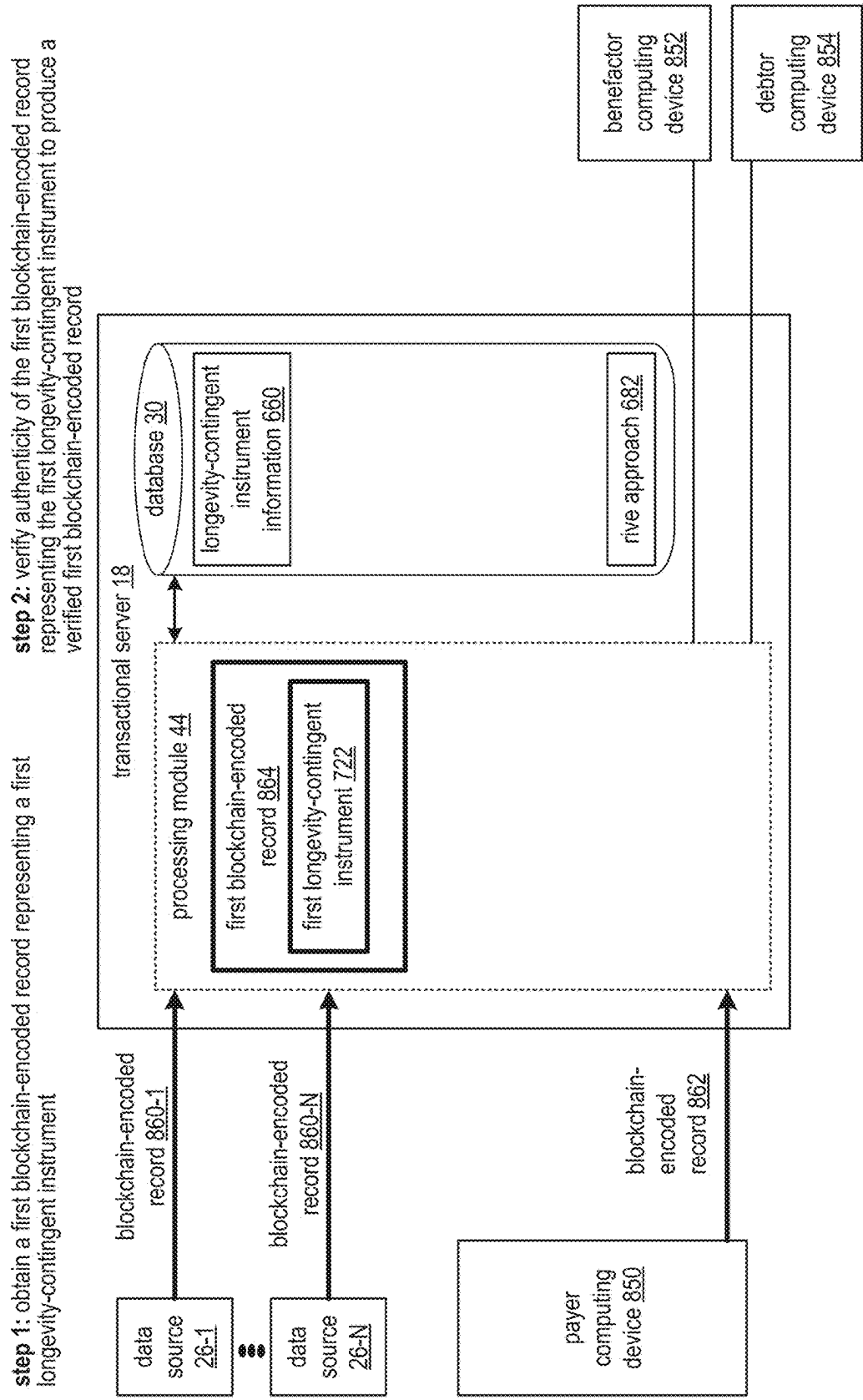


FIG. 7A

step 3: determine that the first longevity-contingent instrument is associated with an available and unfulfilled benefit status

step 4: when the first longevity-contingent instrument is associated with the available and unfulfilled benefit status, determine fulfillment information

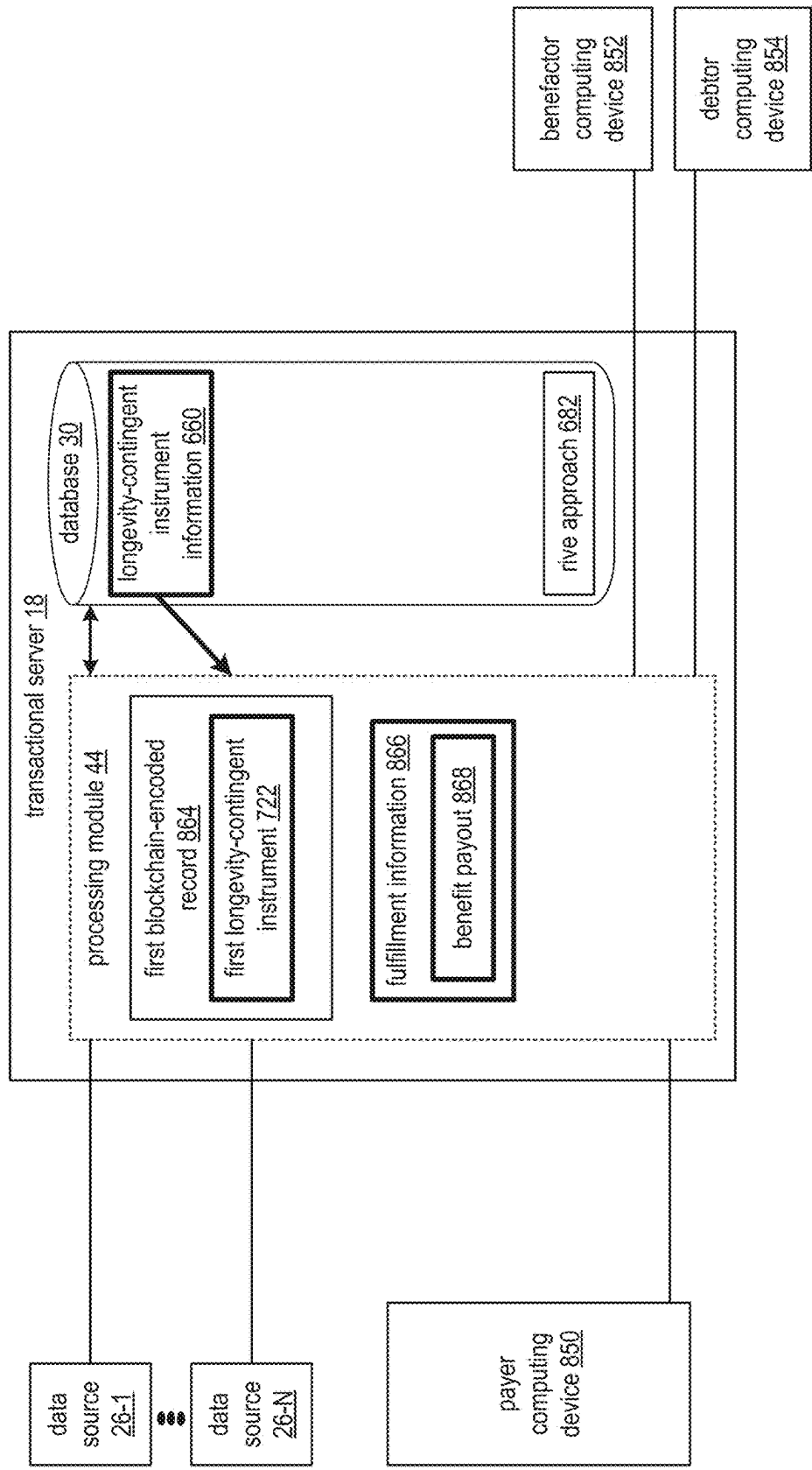


FIG. 7B

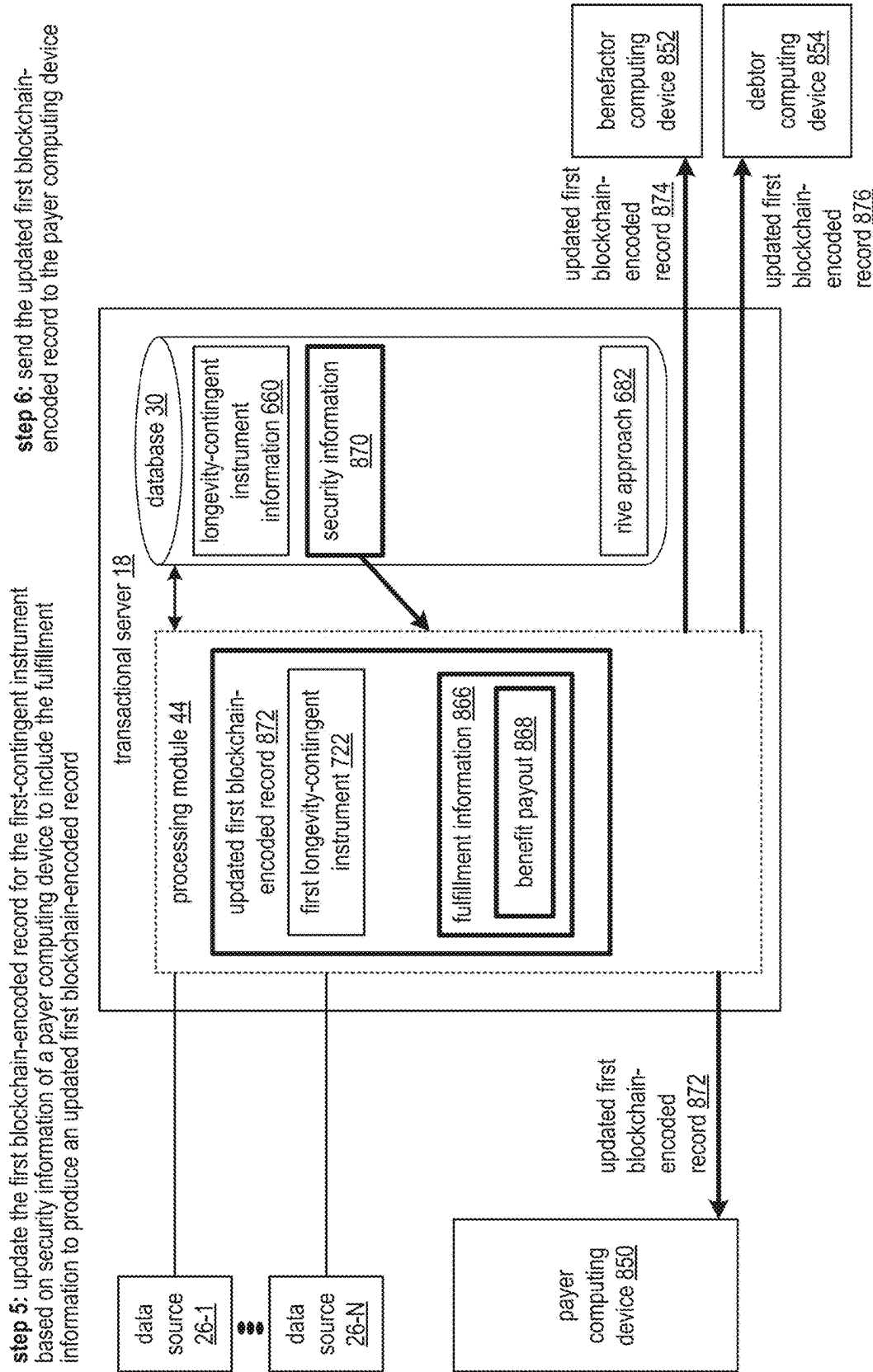


FIG. 7C

step 1: determine to update a set of longevity-contingent instruments

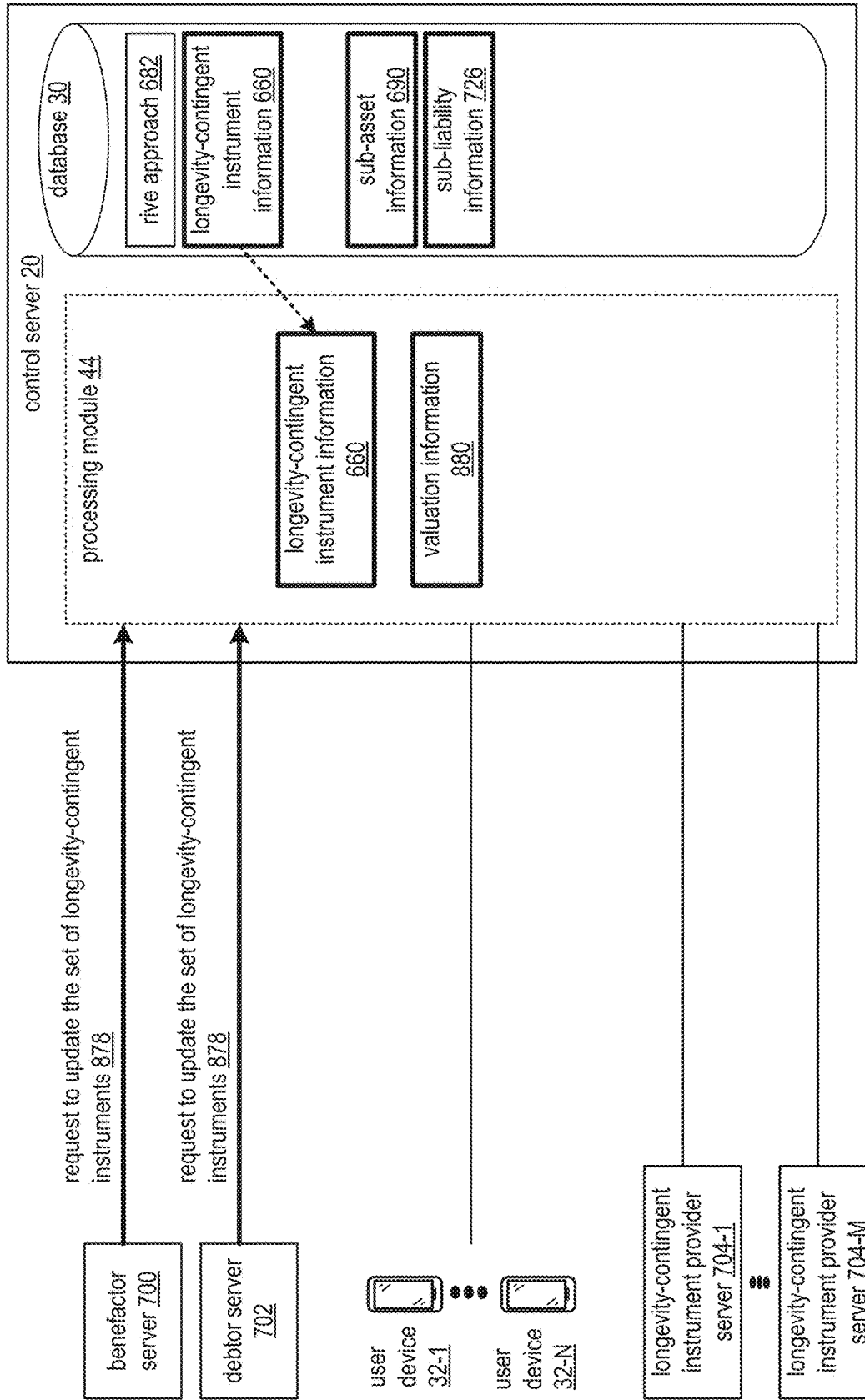


FIG. 8A

step 2: verify authenticity of a blockchain-encoded record to produce an authenticity indicator

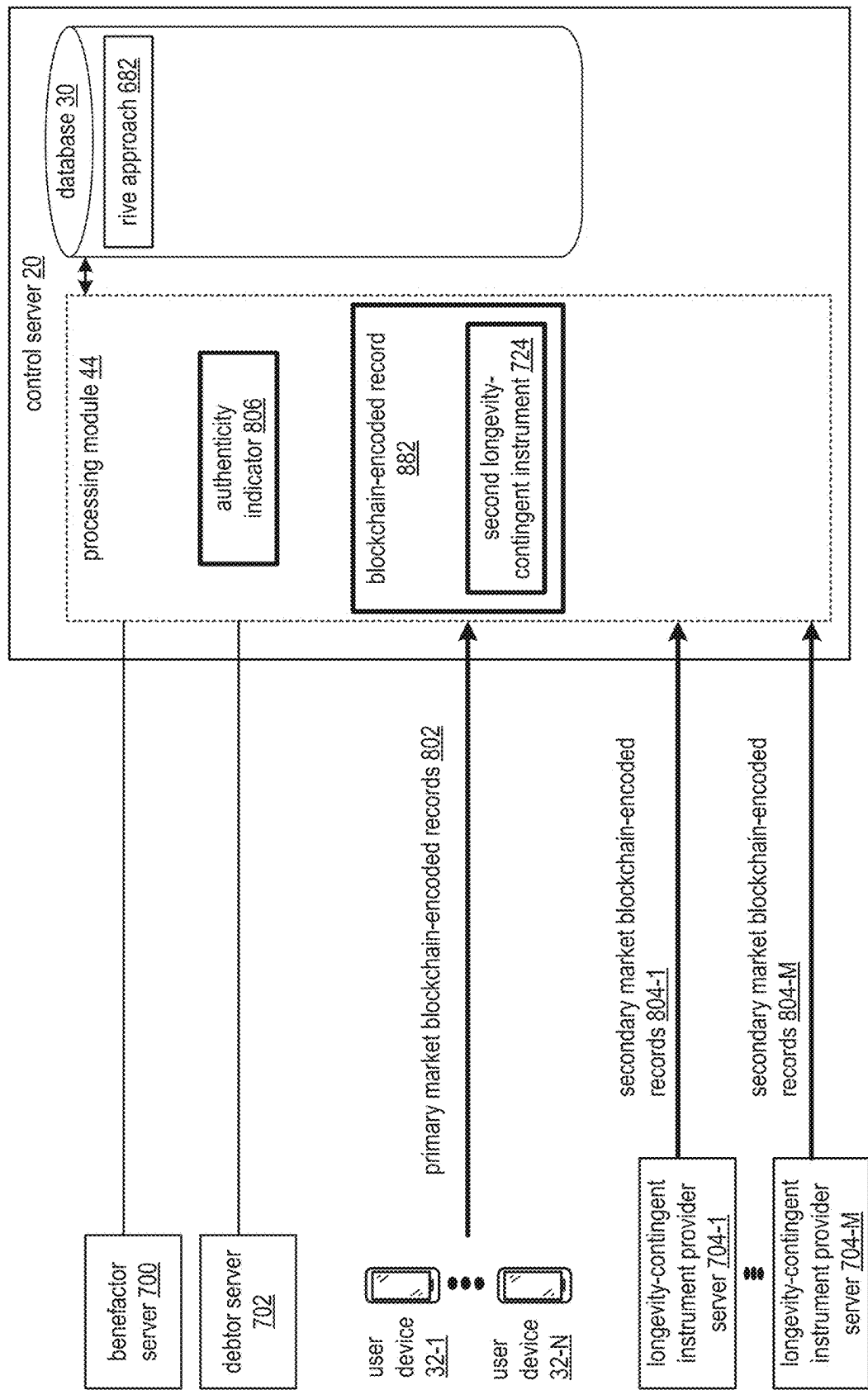


FIG. 8B

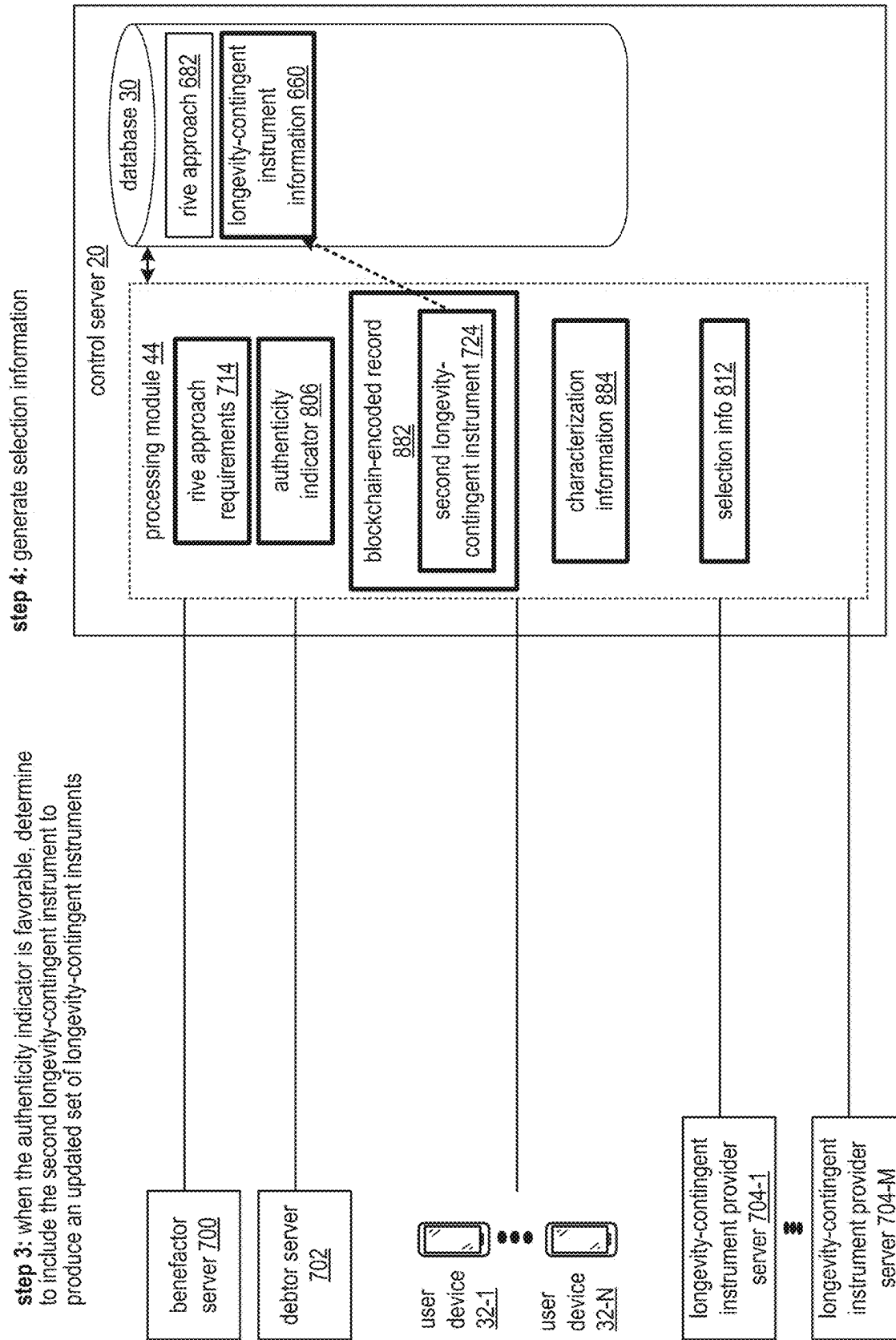


FIG. 8C

step 5: update the blockchain-encoded record for the second longevity-contingent instrument to include the selection information

step 6: rive the second longevity-contingent instrument in accordance with the rive approach to produce a second sub-asset and a second sub-liability

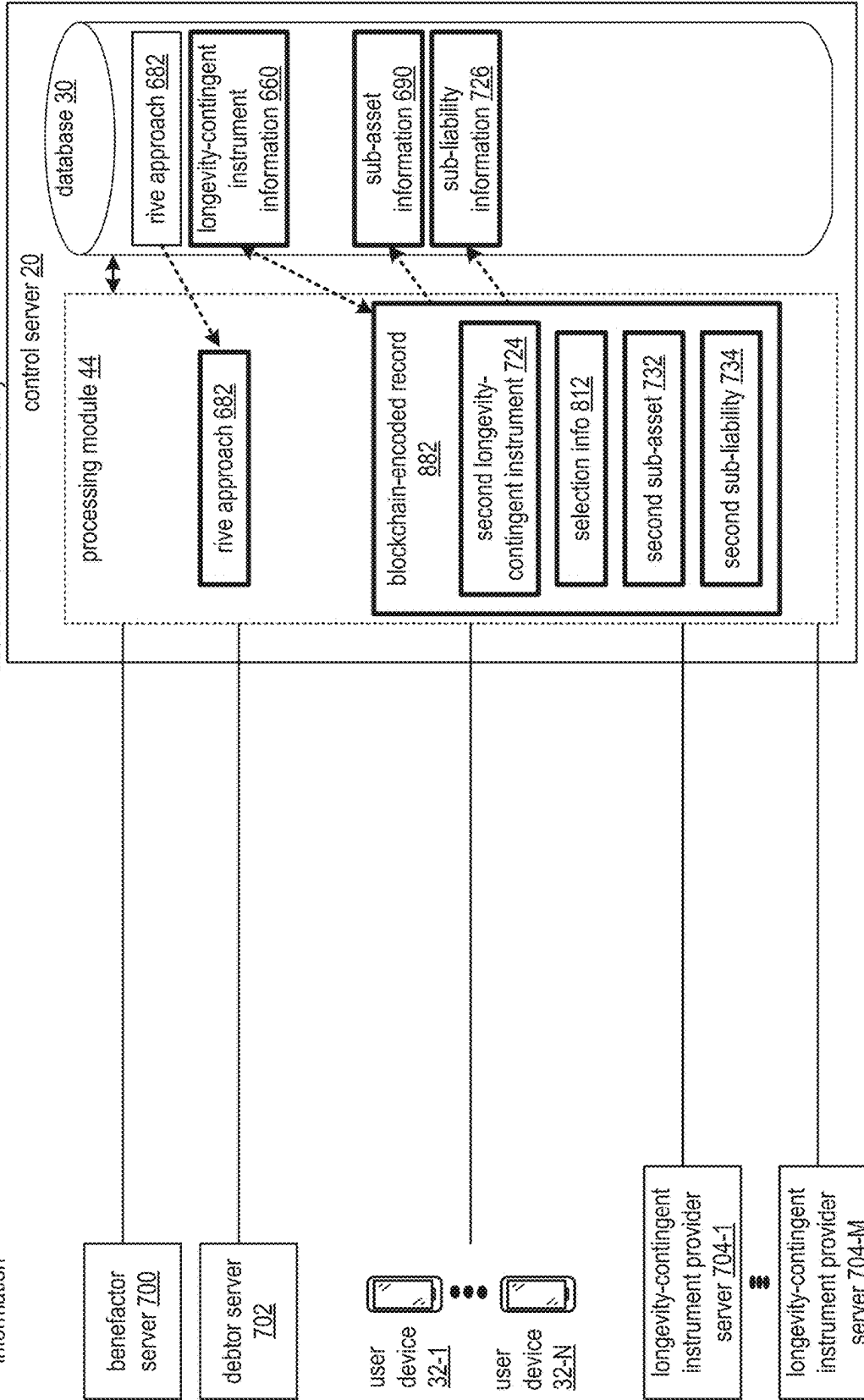
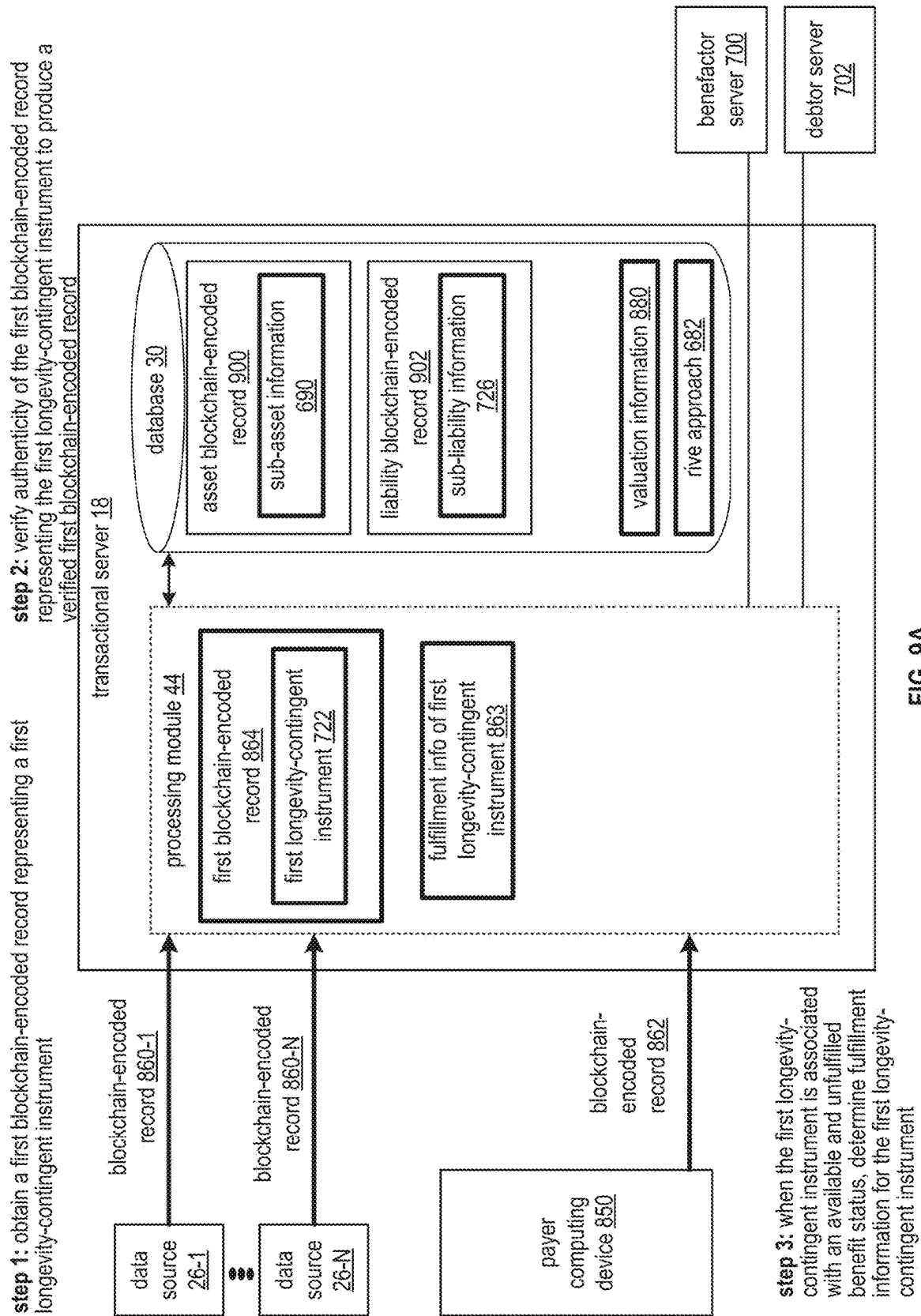


FIG. 8D



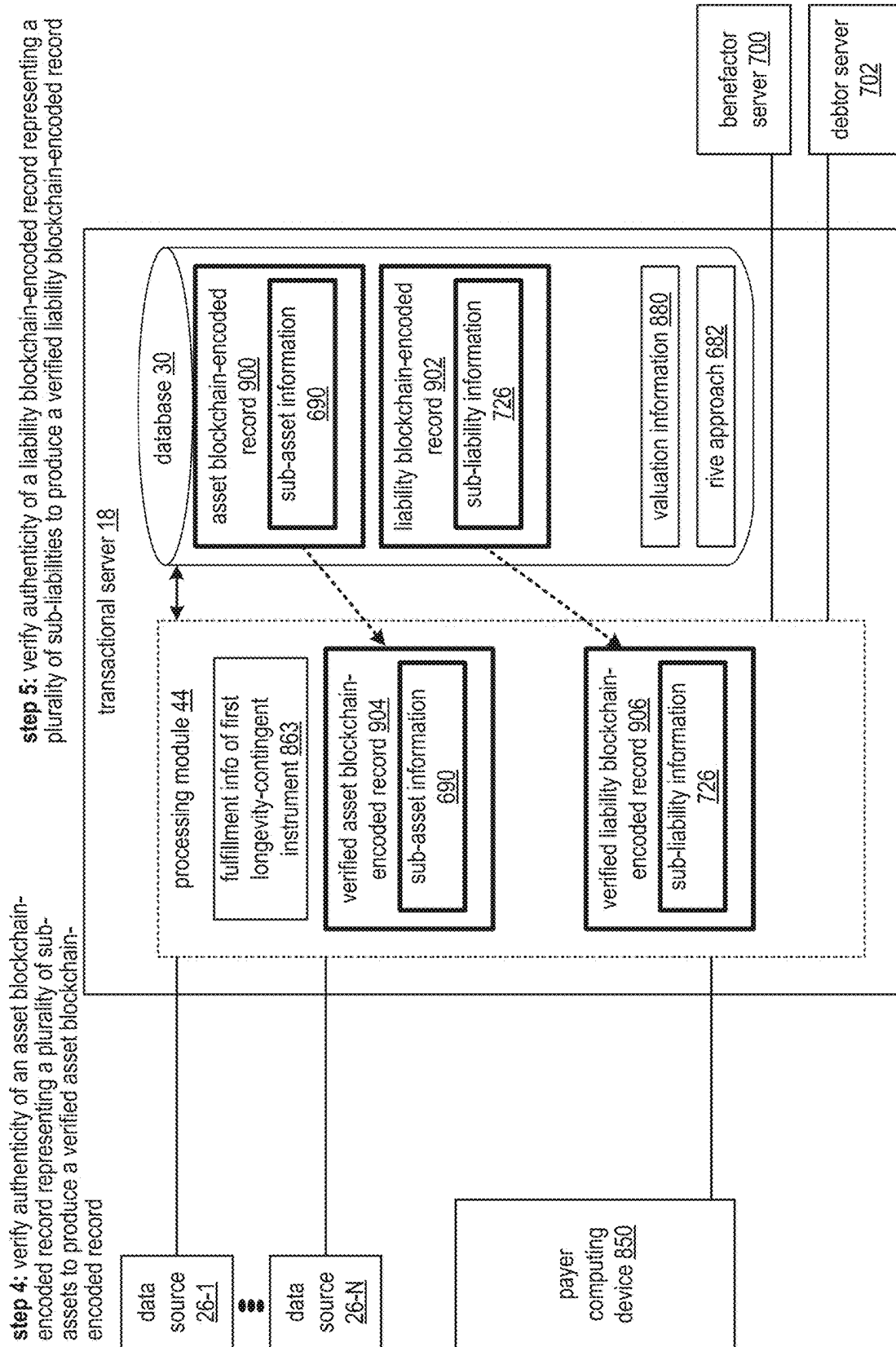


FIG. 9B

step 6: facilitate exclusion of the first longevity-contingent instrument from the set of longevity-contingent instruments

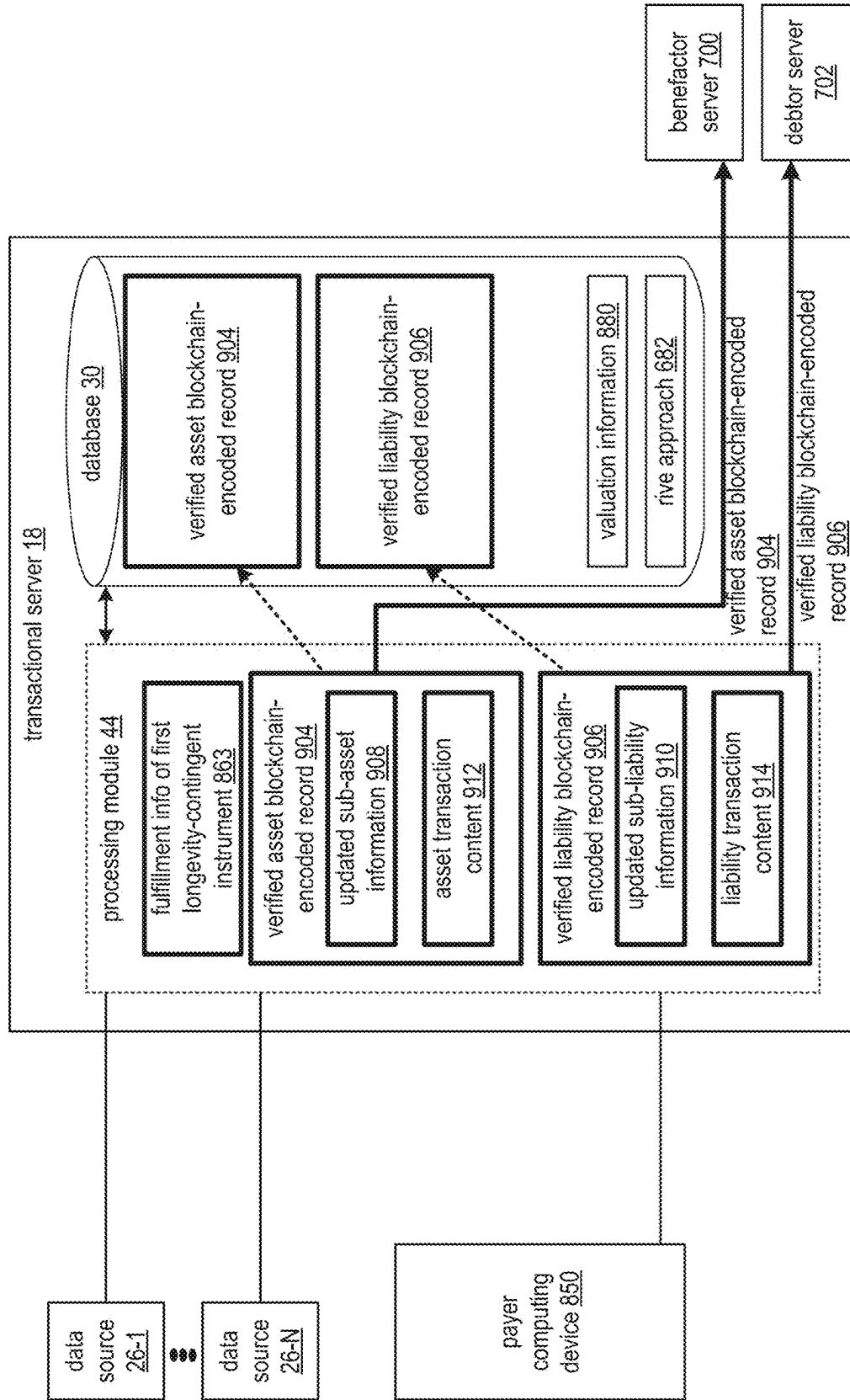


FIG. 9C

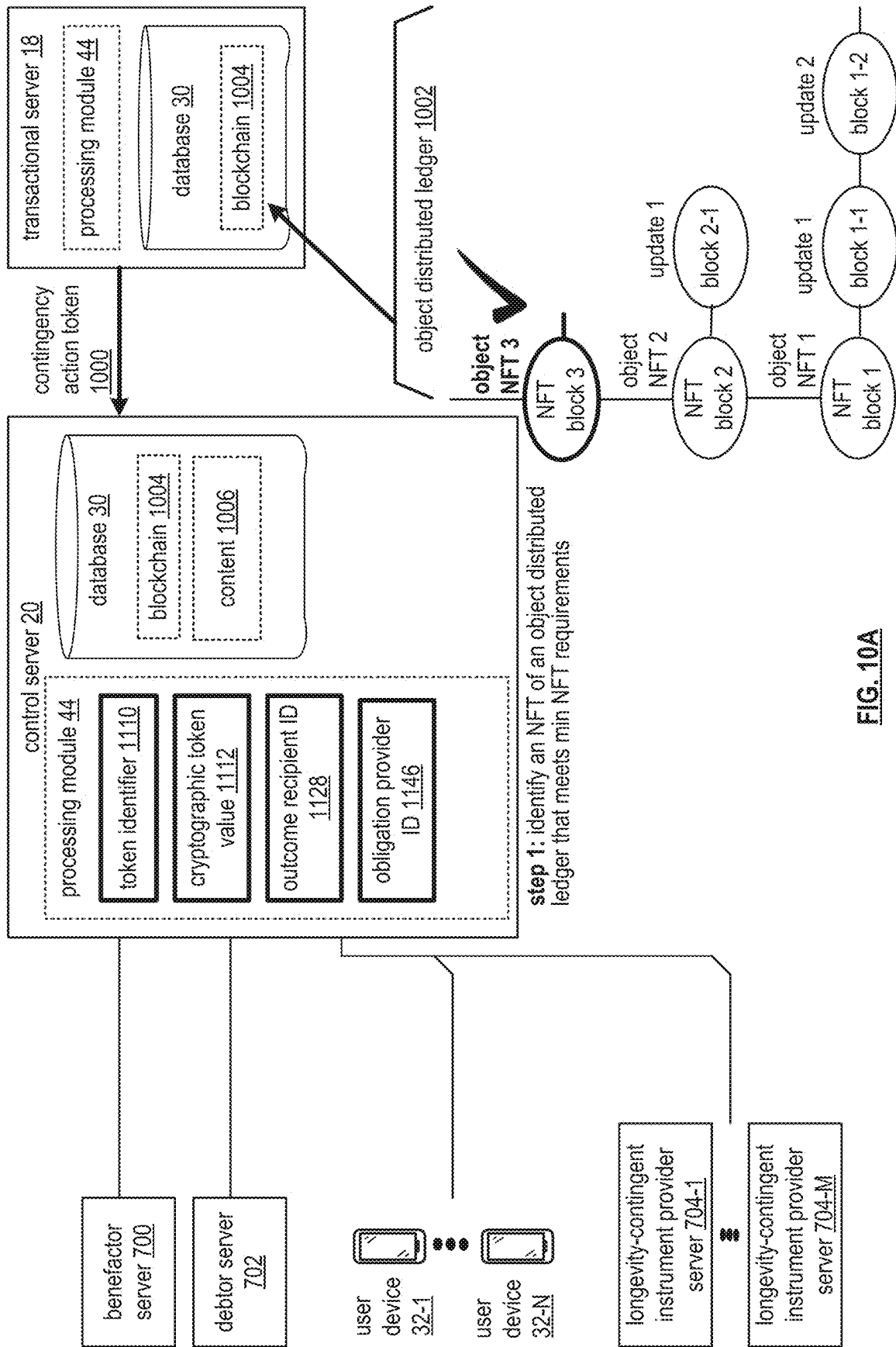


FIG. 10A

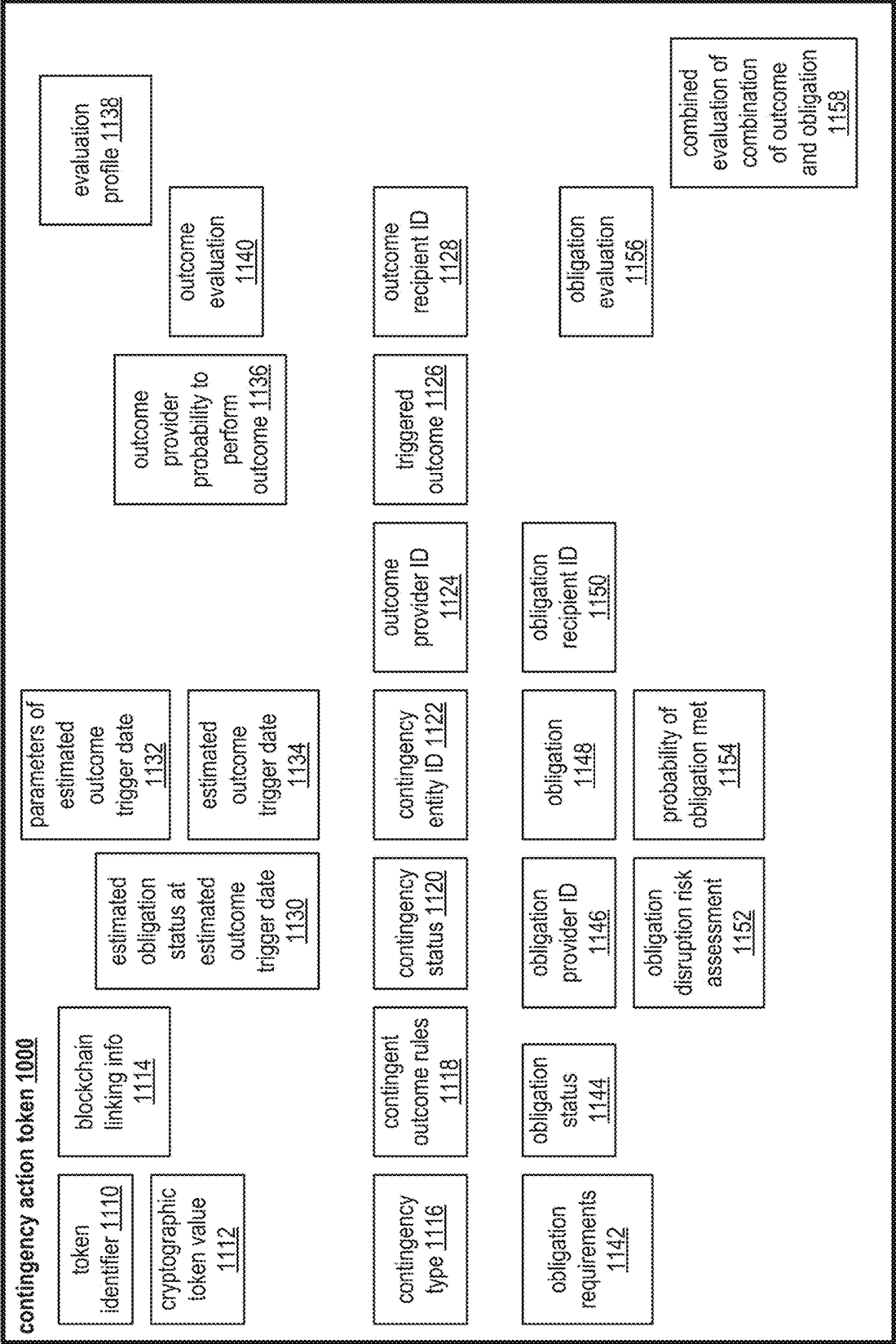


FIG. 10B

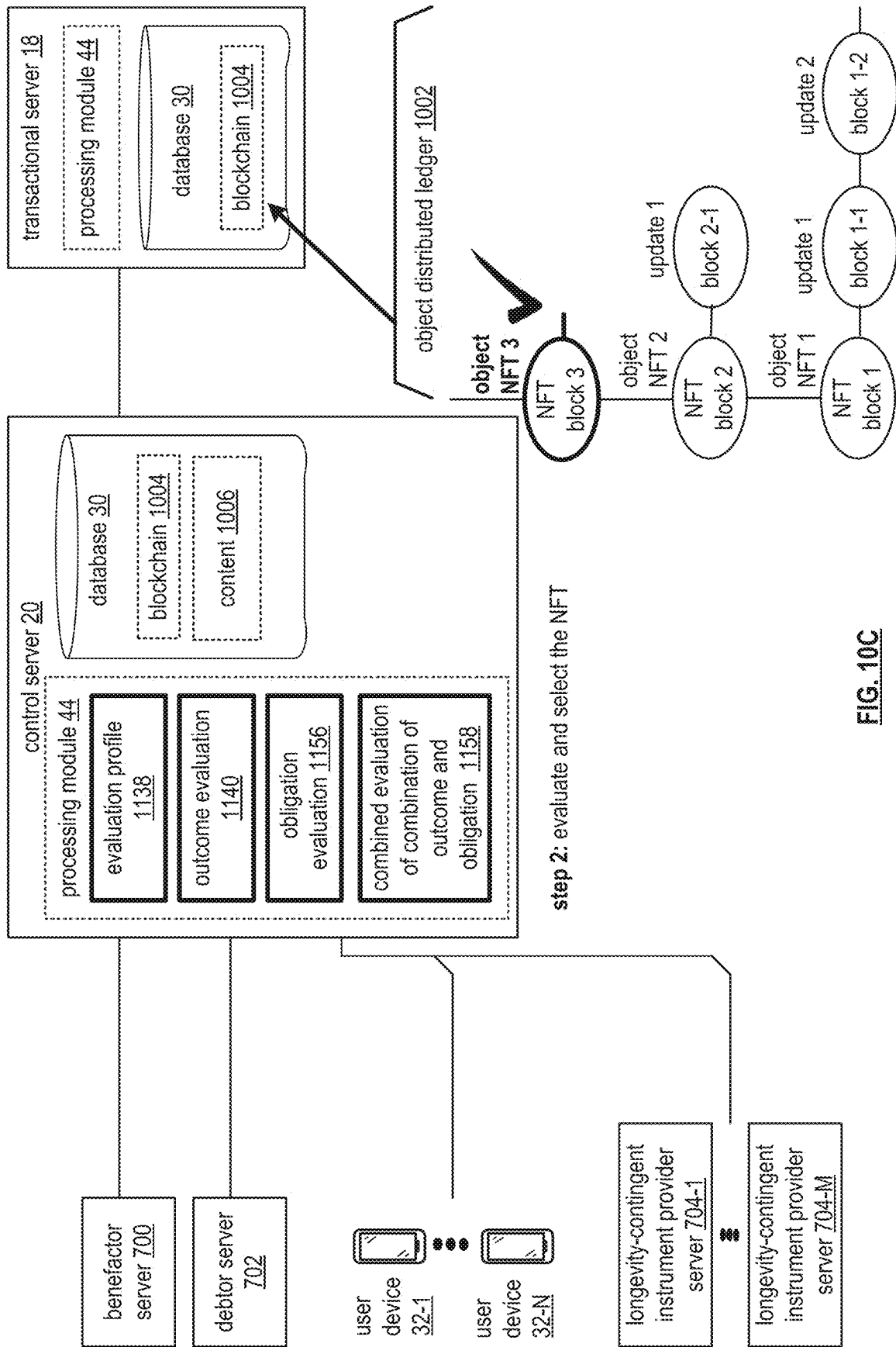


FIG. 10C

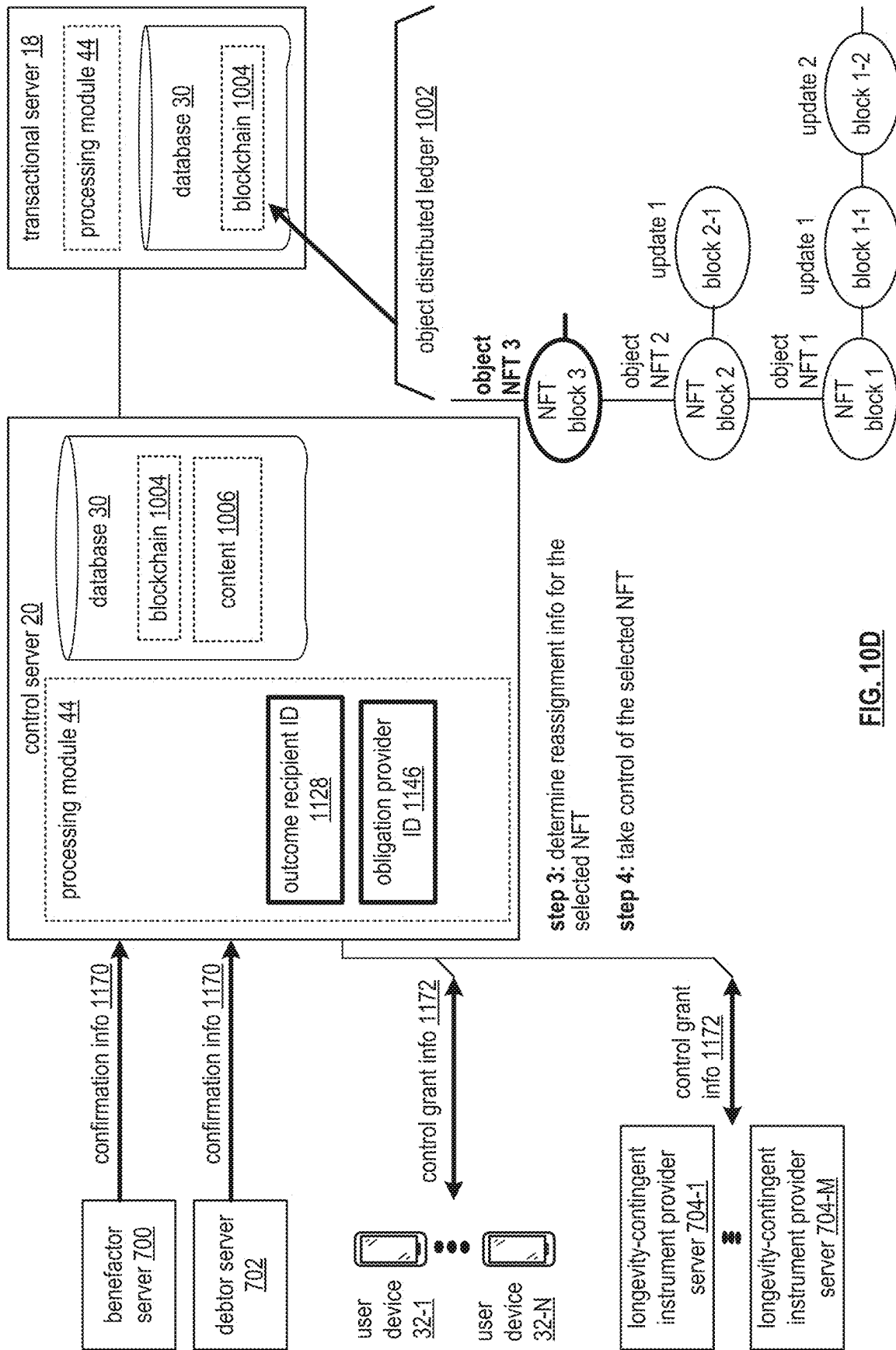


FIG. 10D

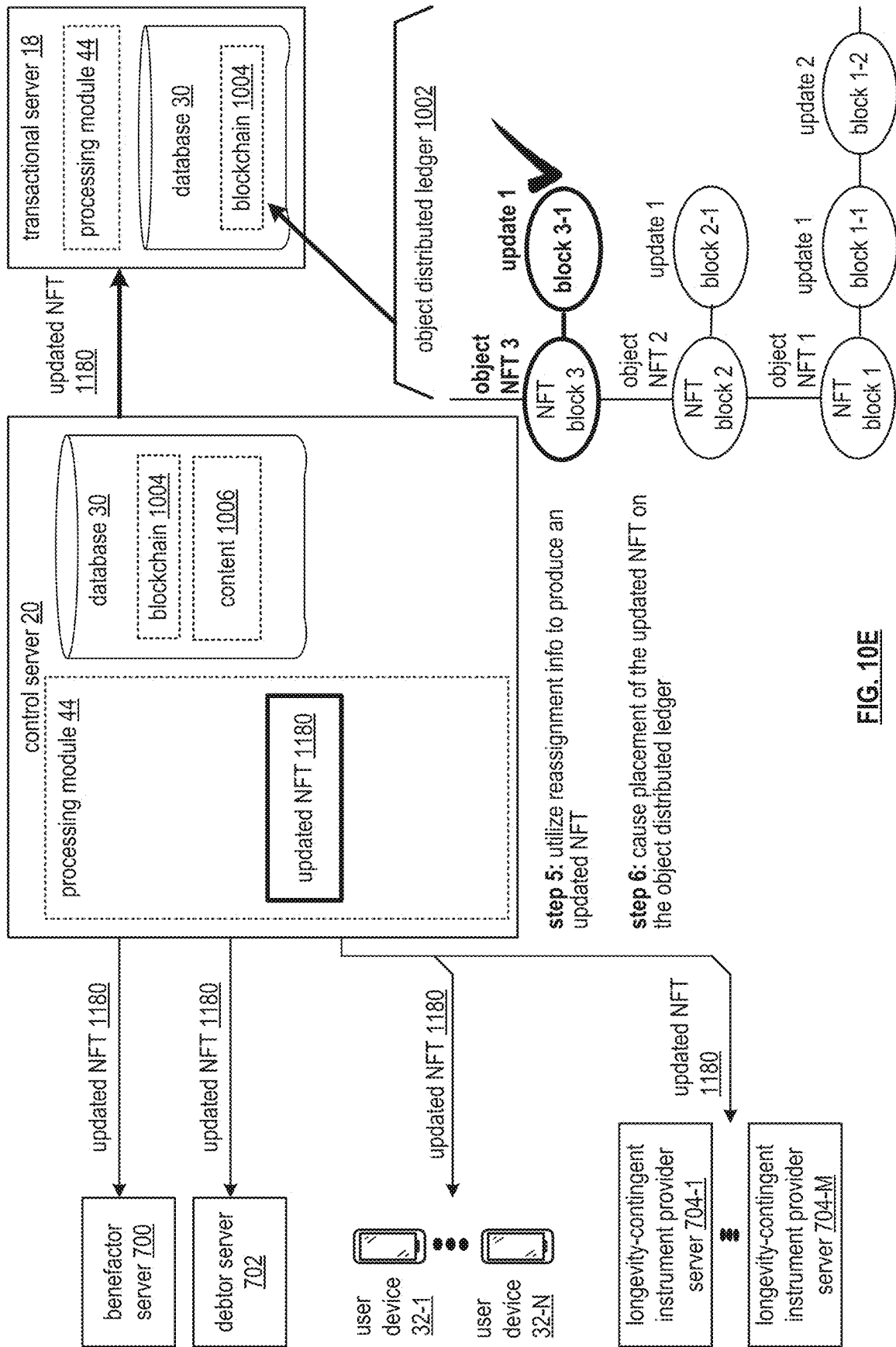


FIG. 10E

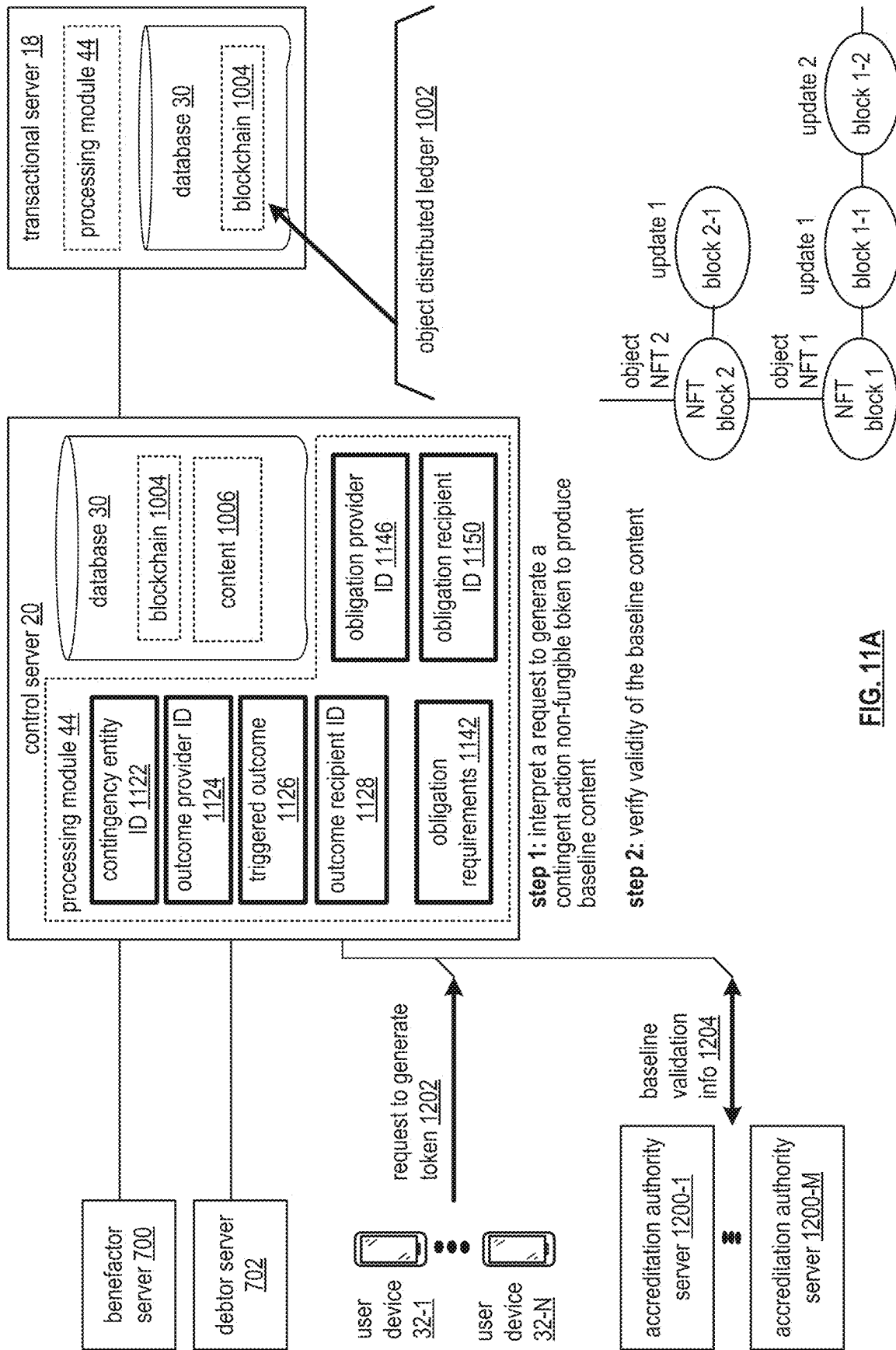


FIG. 11A

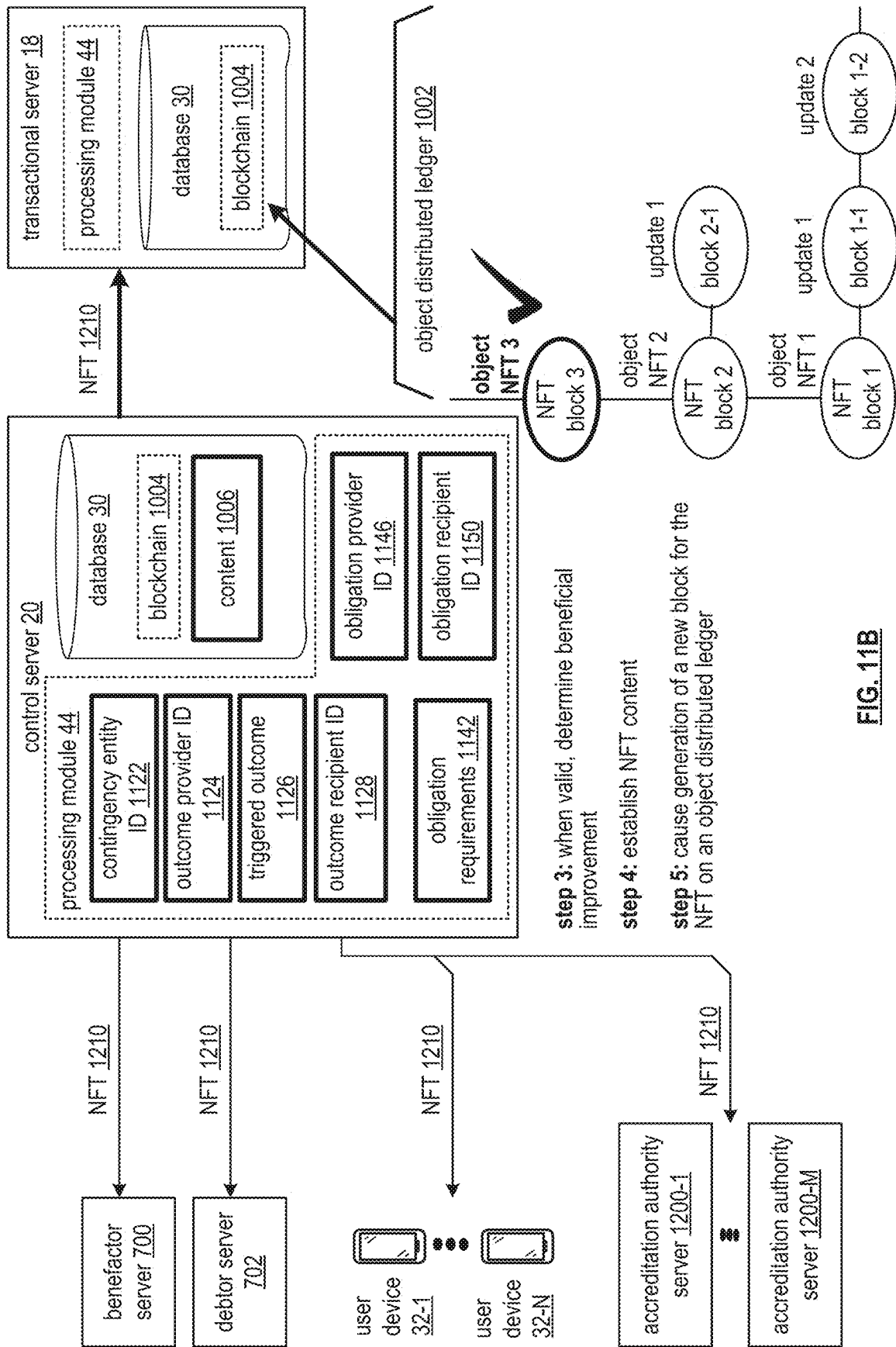


FIG. 11B

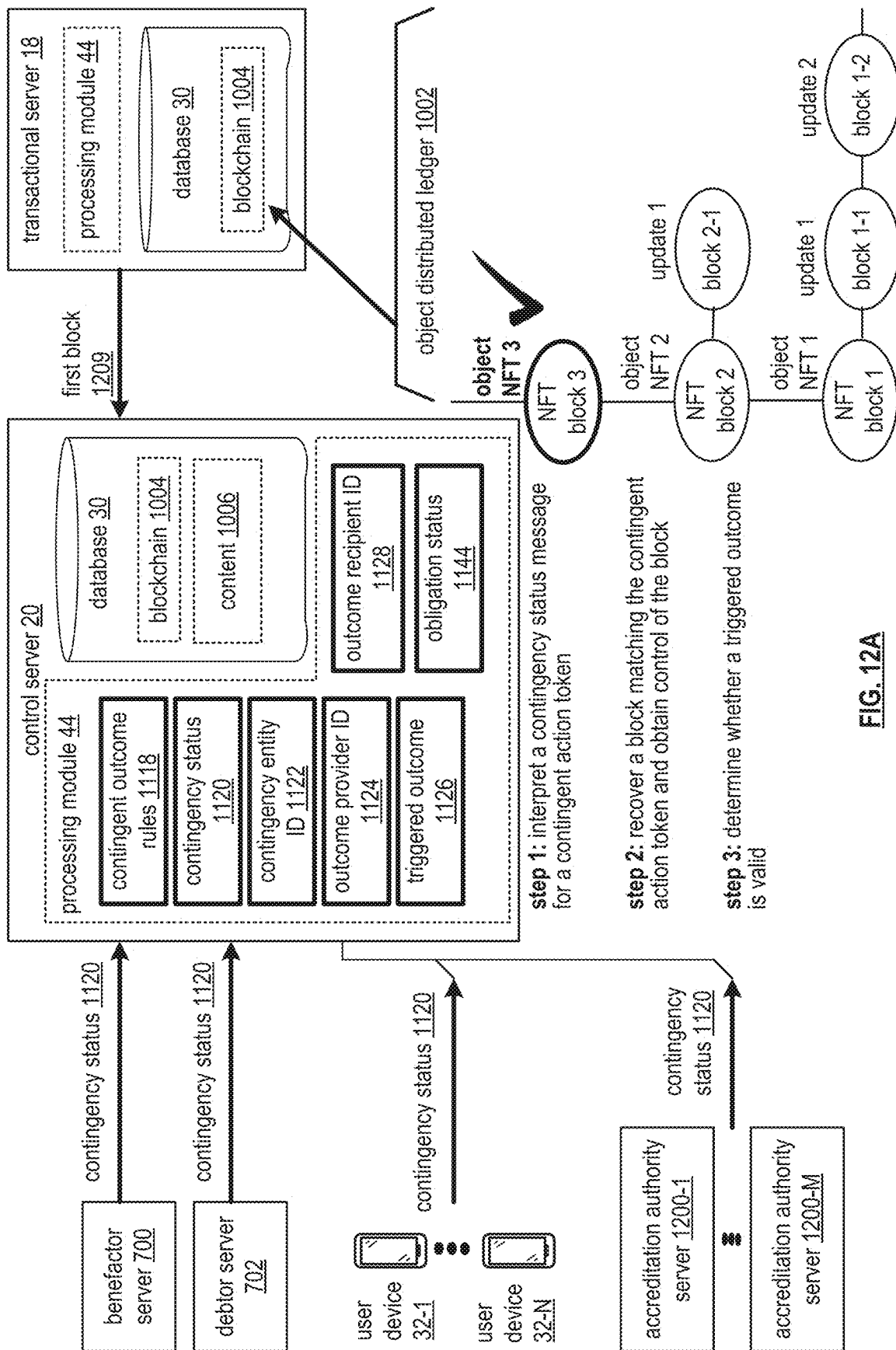


FIG. 12A

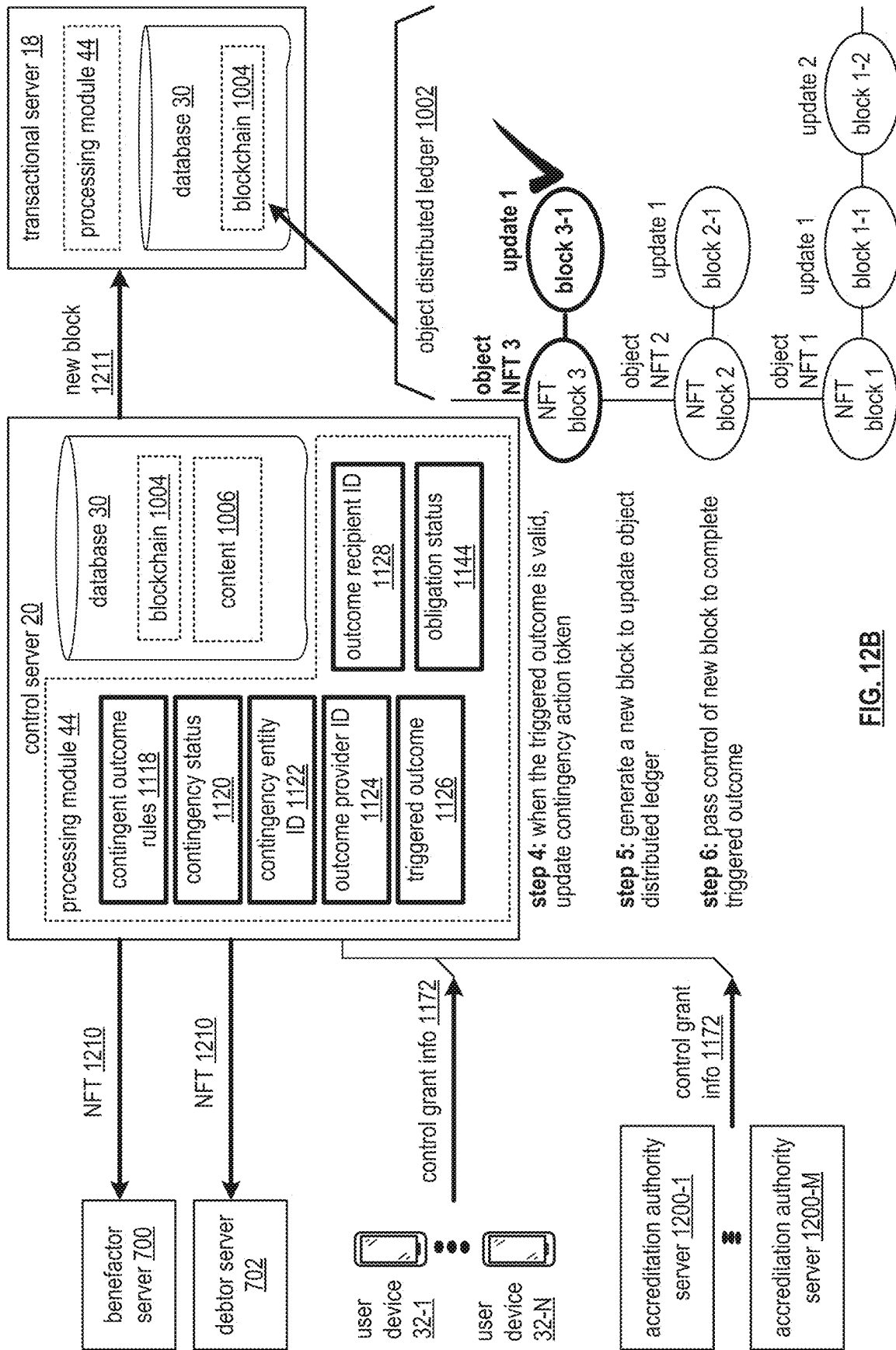


FIG. 12B

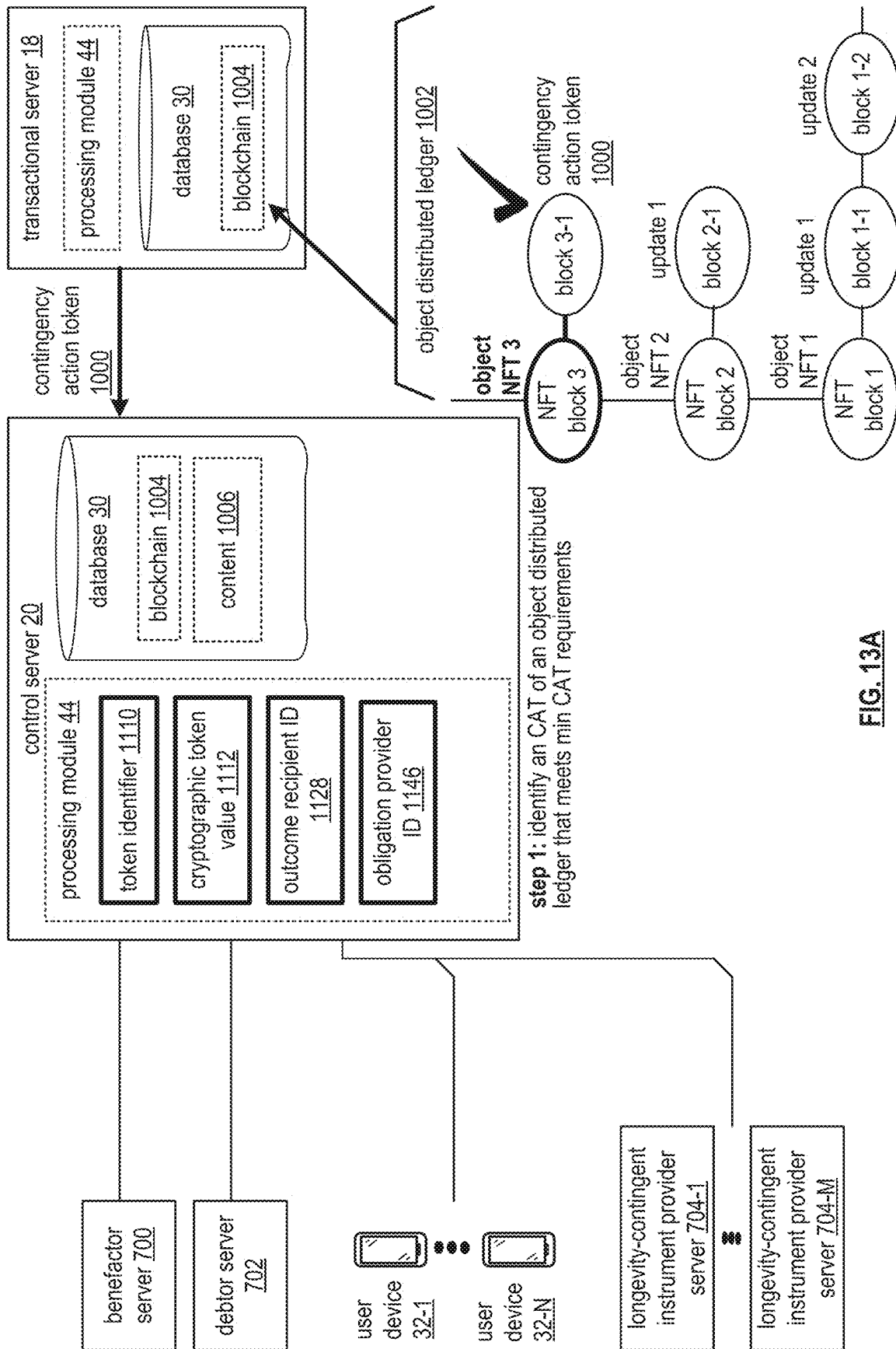


FIG. 13A

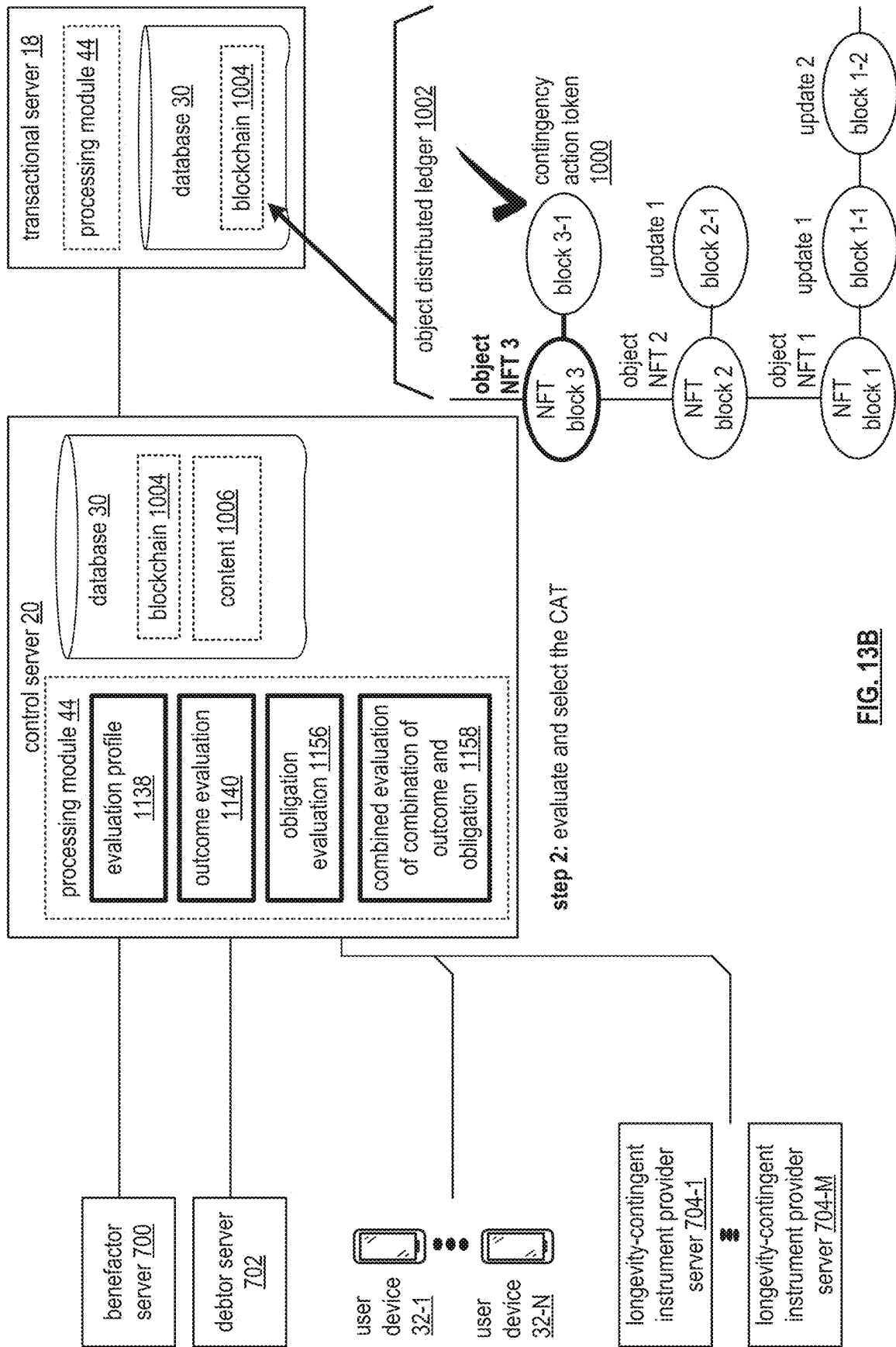


FIG. 13B

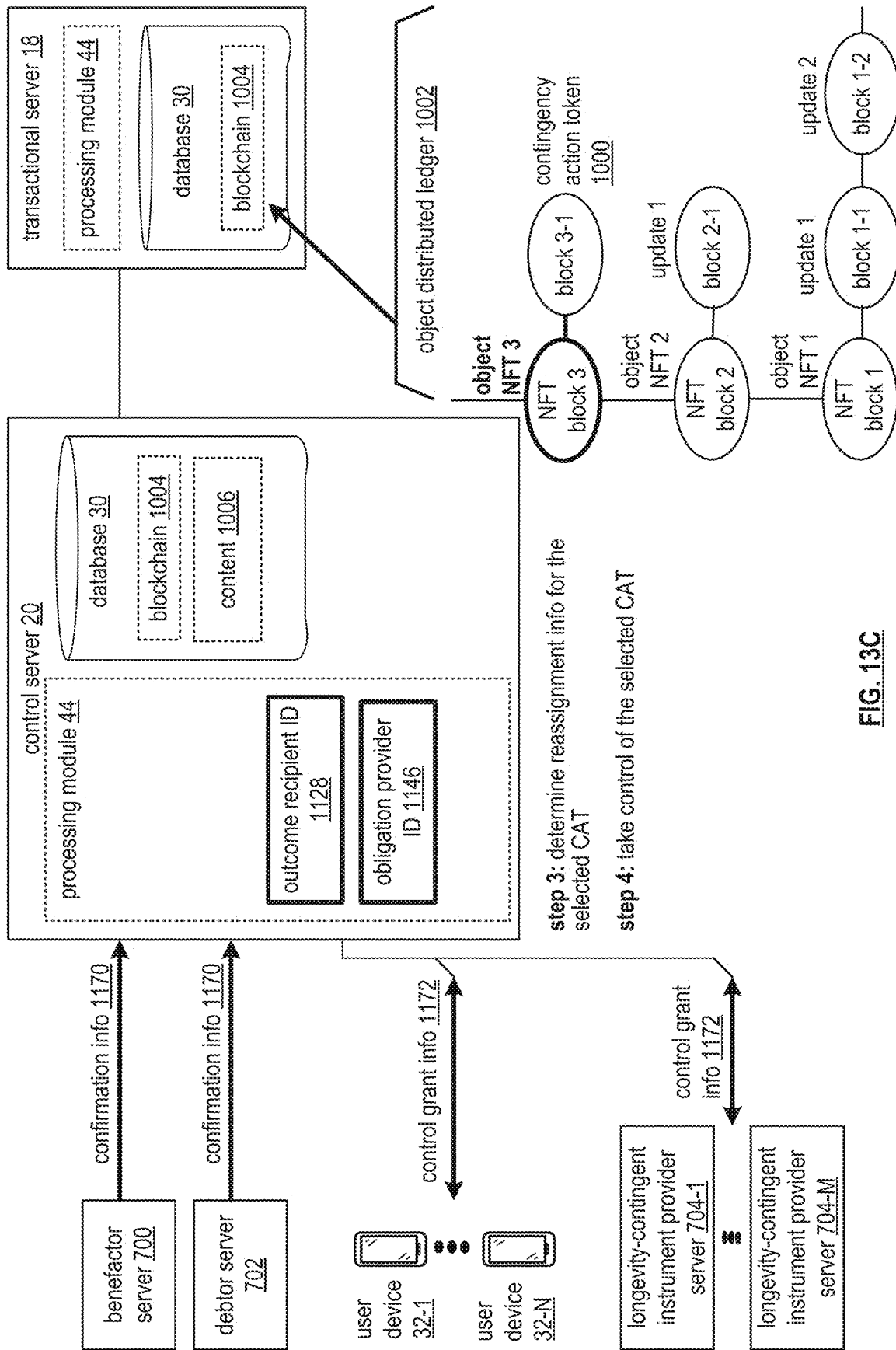


FIG. 13C

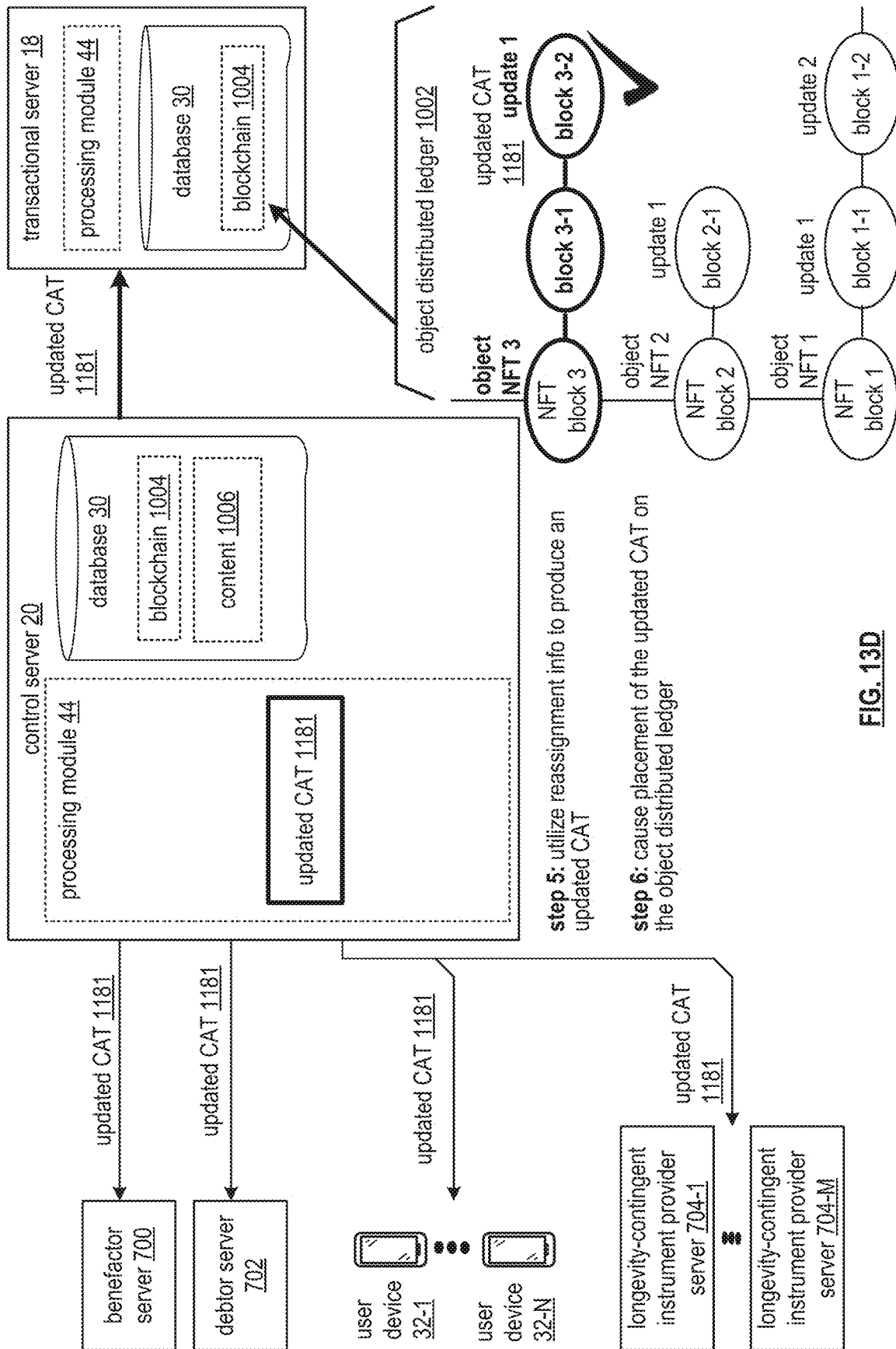


FIG. 13D

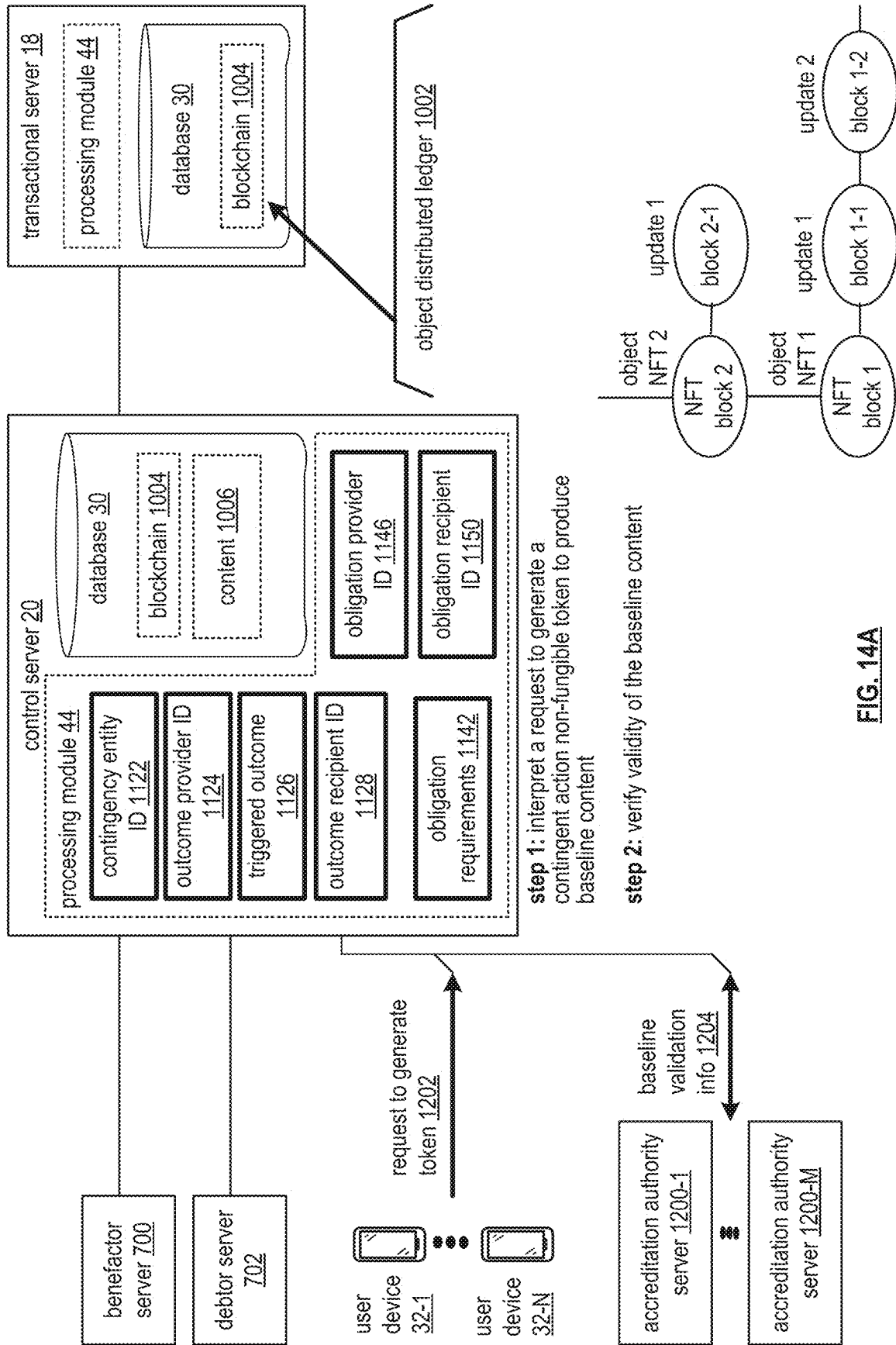
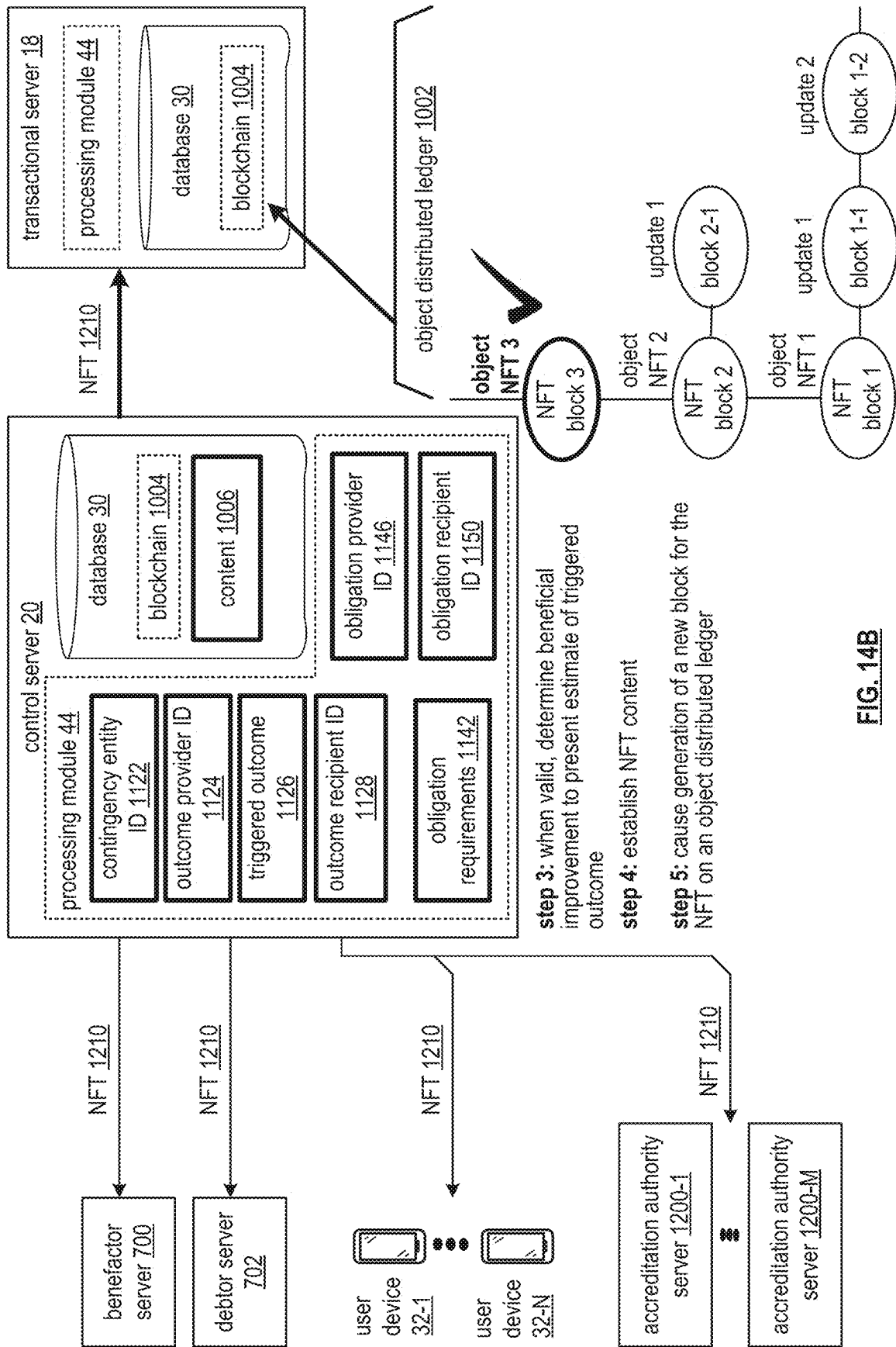
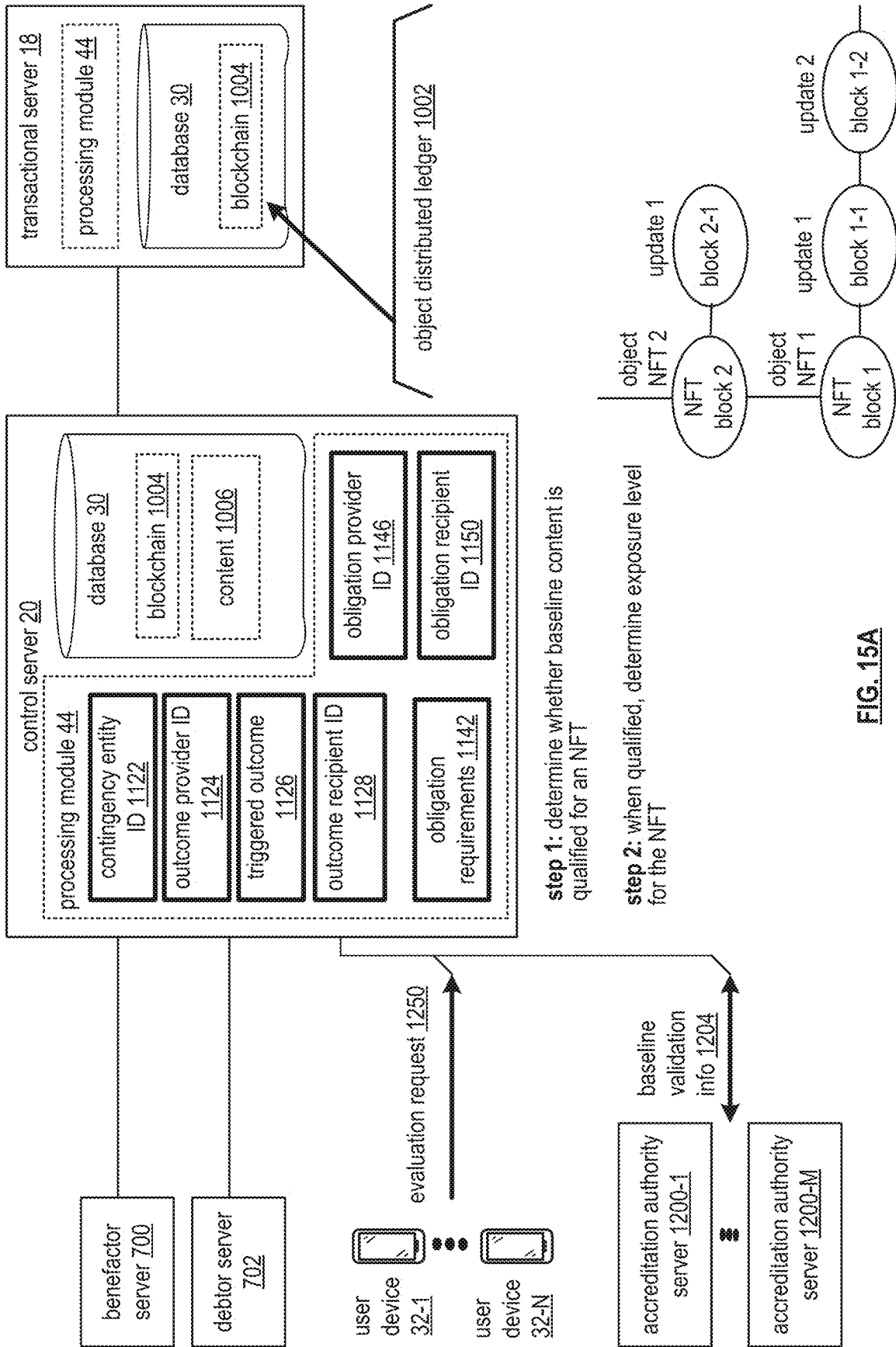
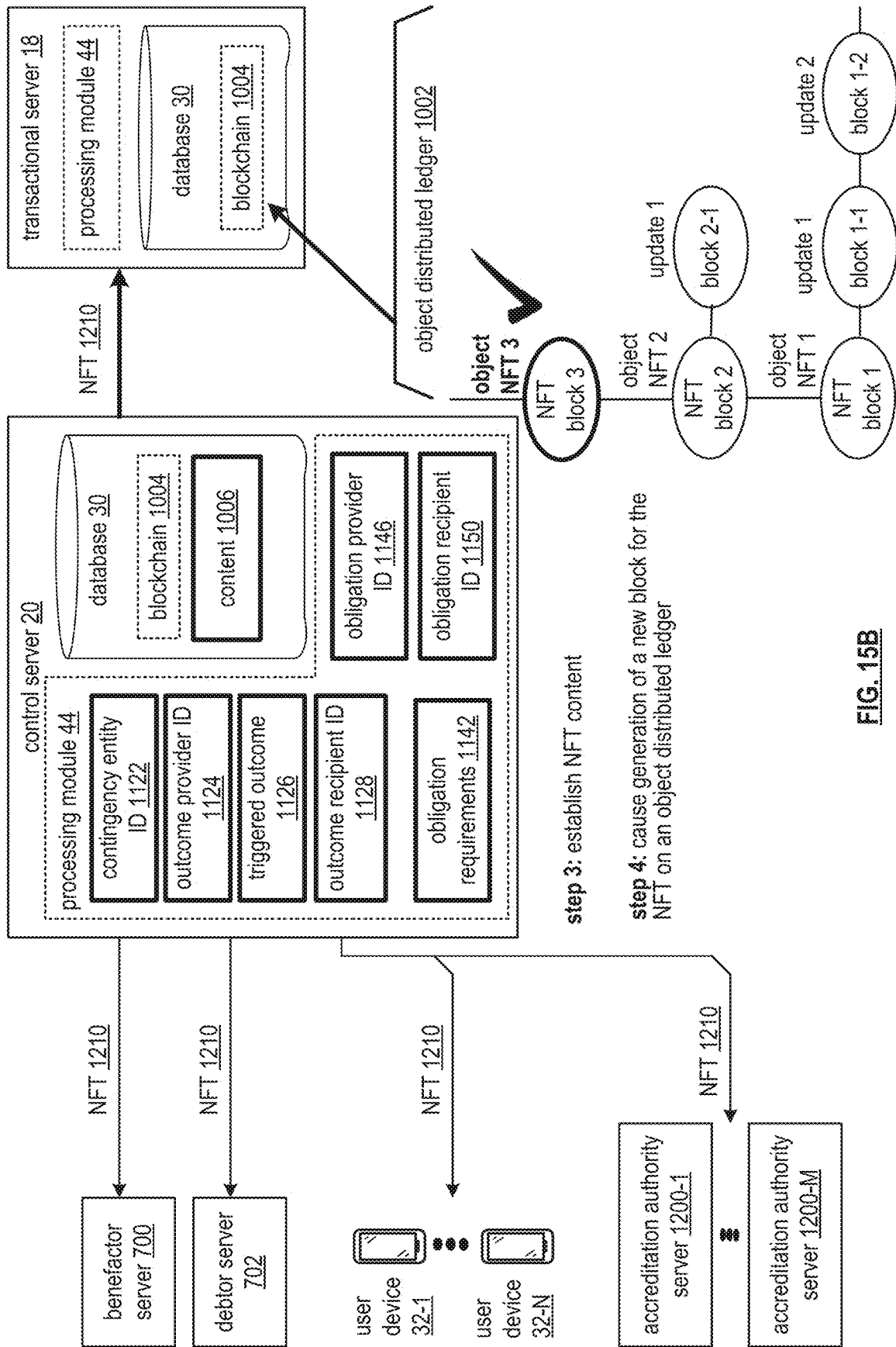


FIG. 14A







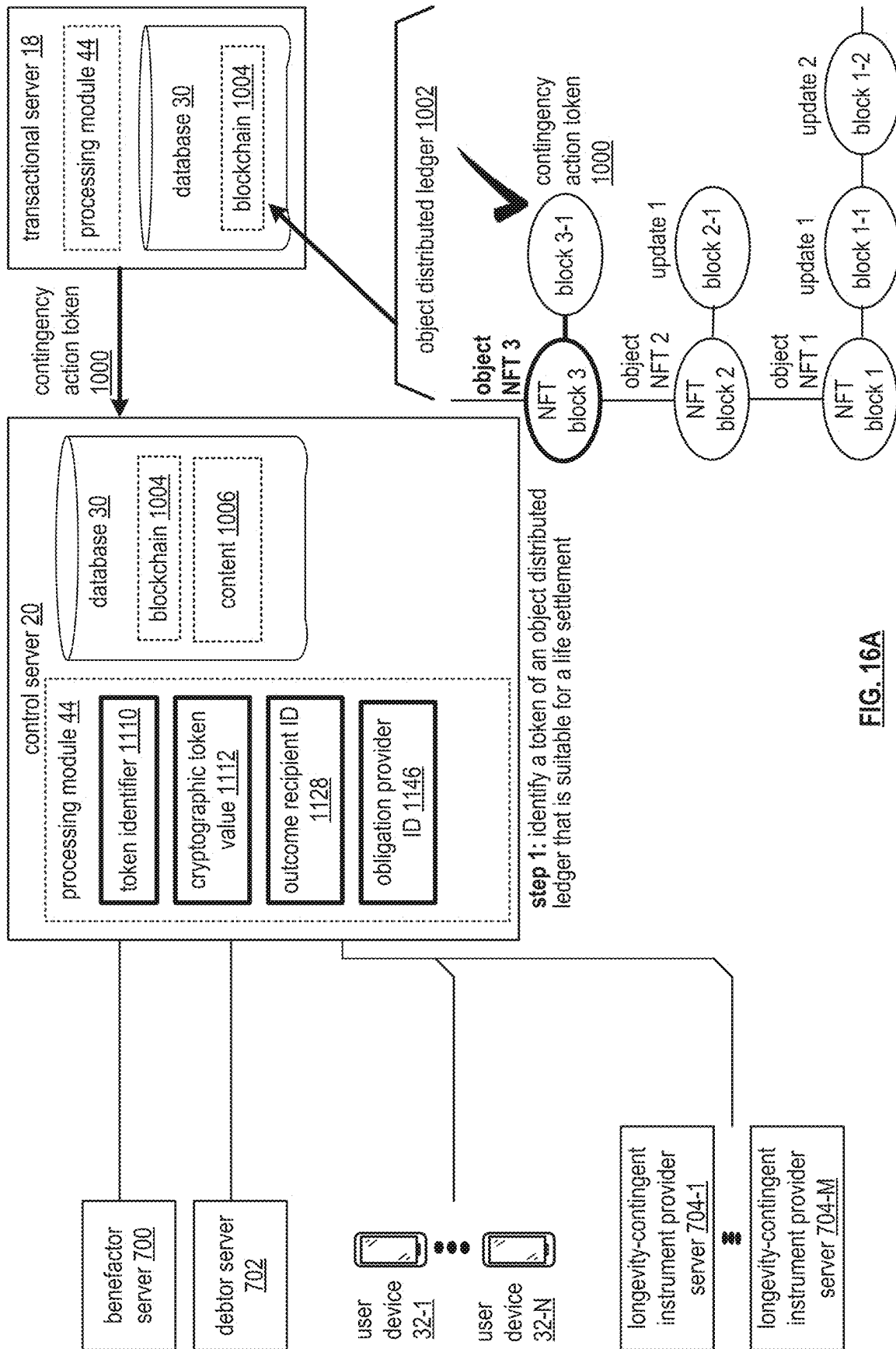


FIG. 16A

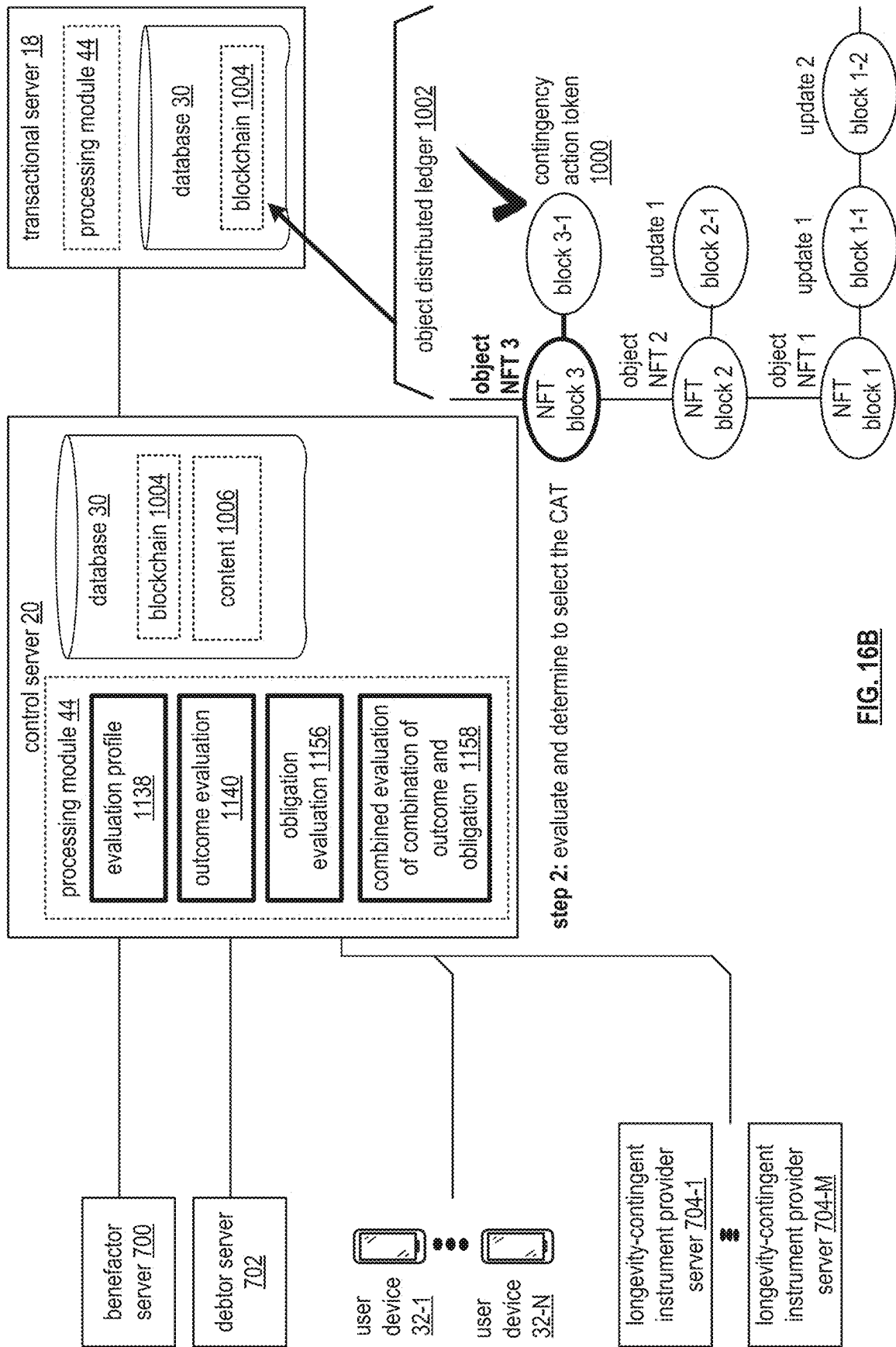


FIG. 16B

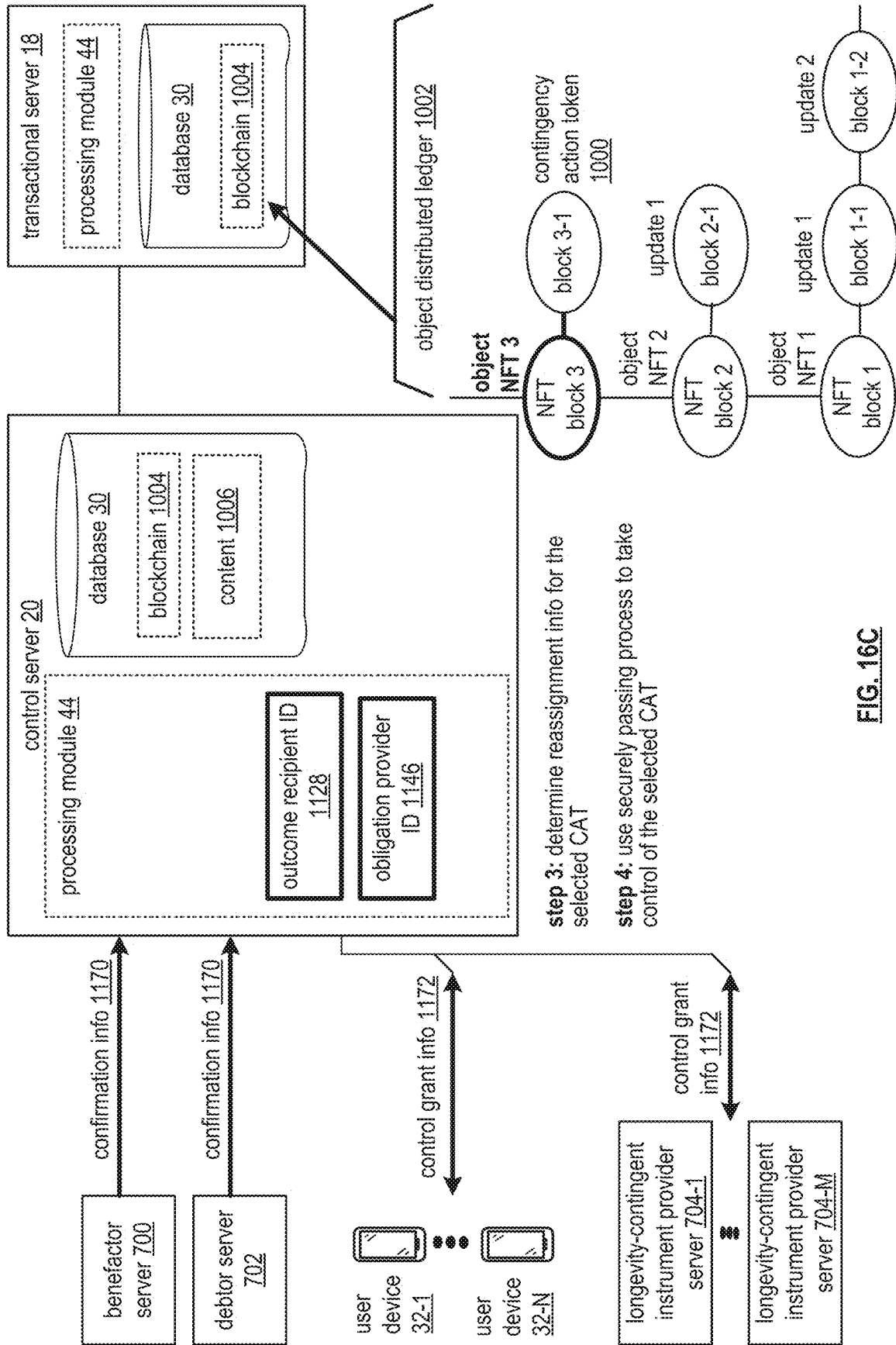


FIG. 16C

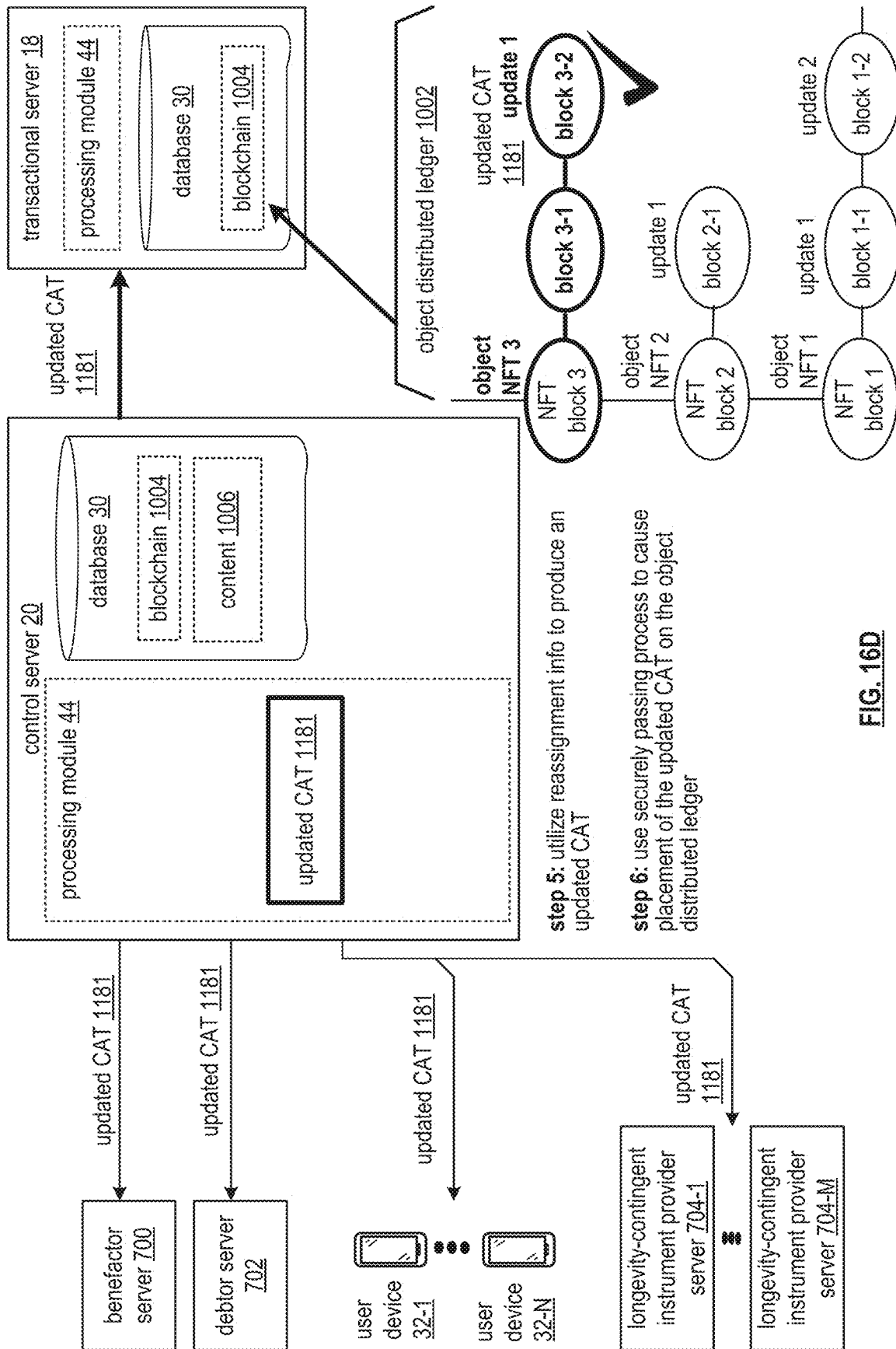


FIG. 16D

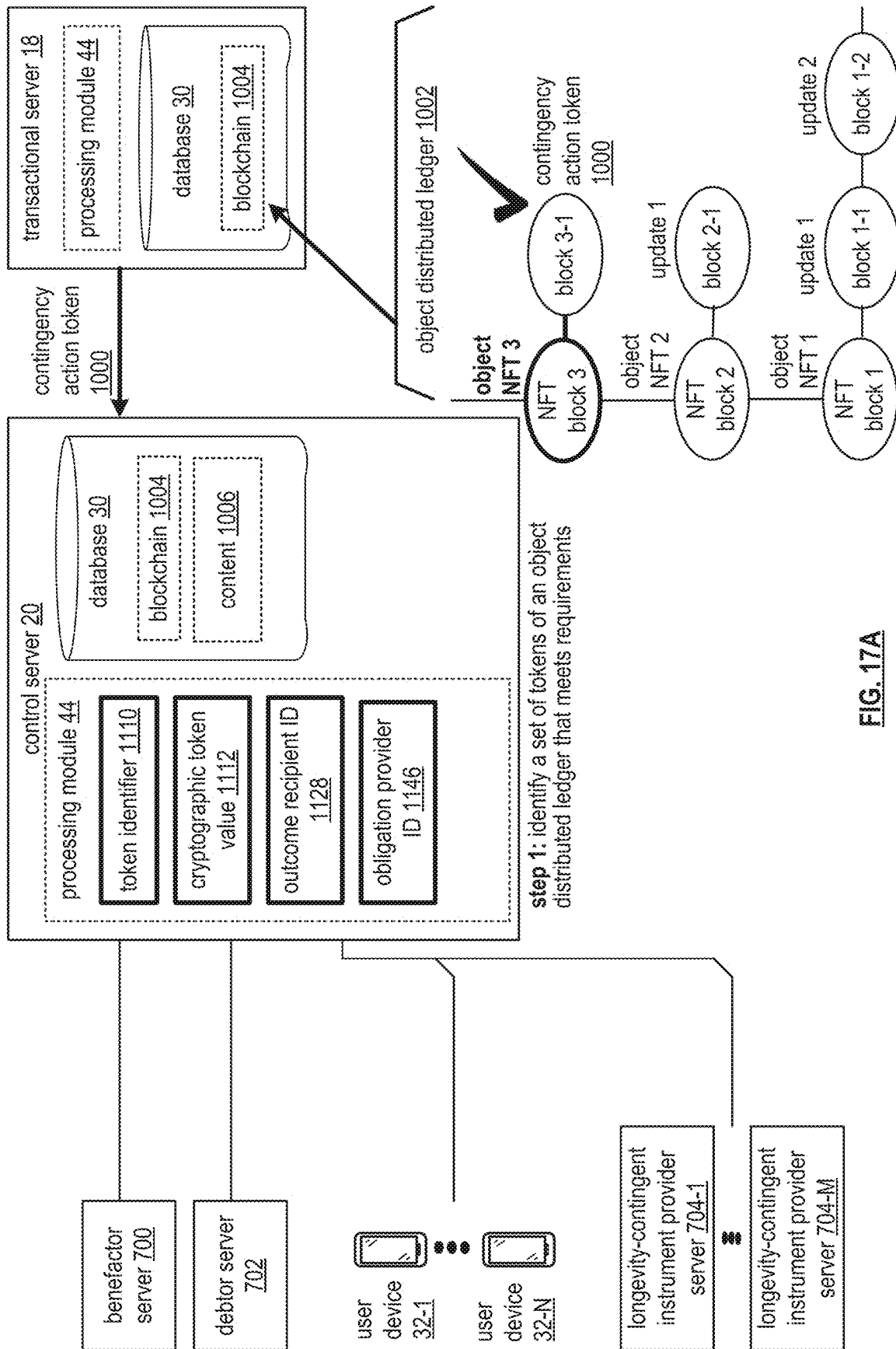


FIG. 17A

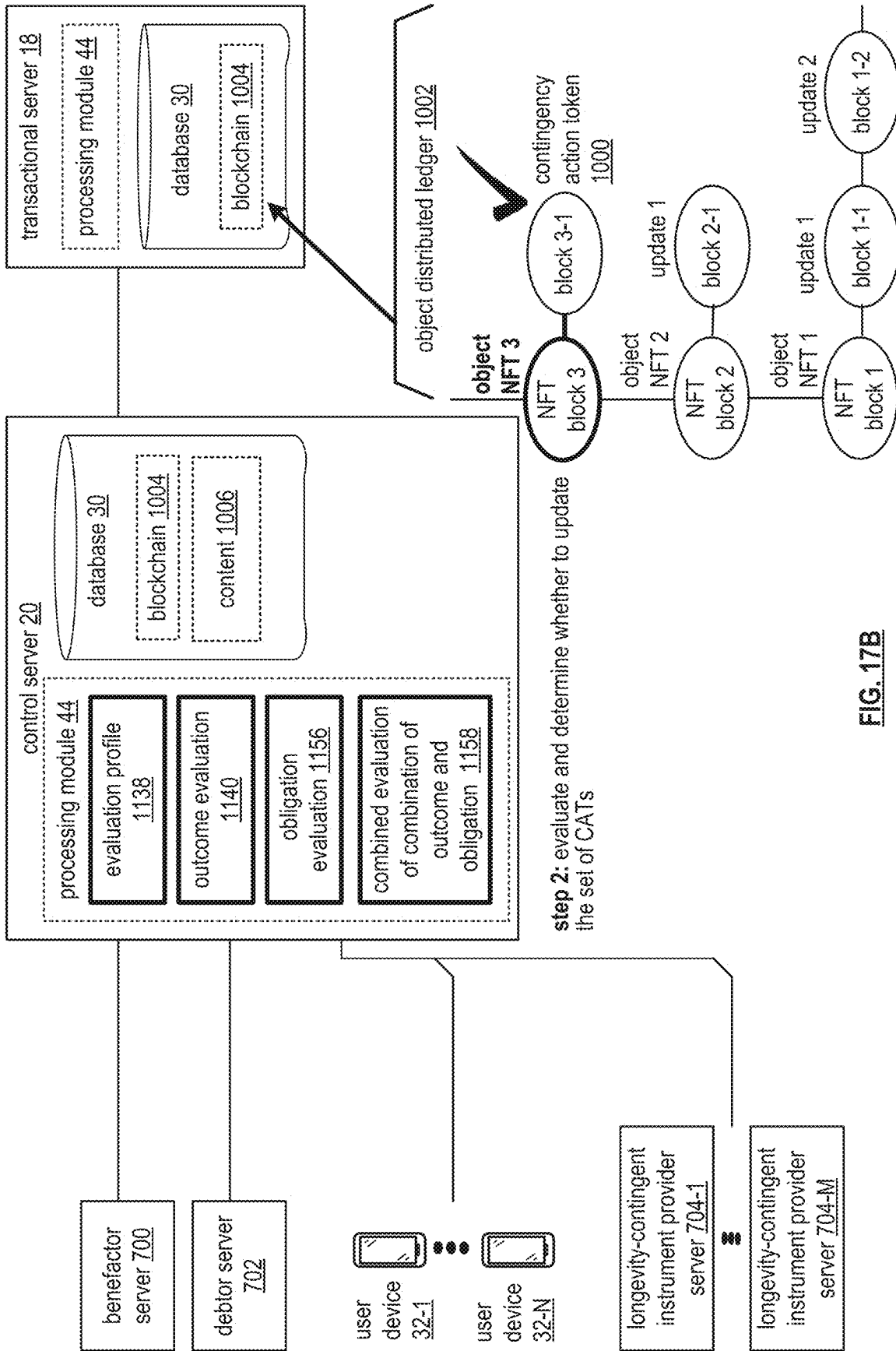
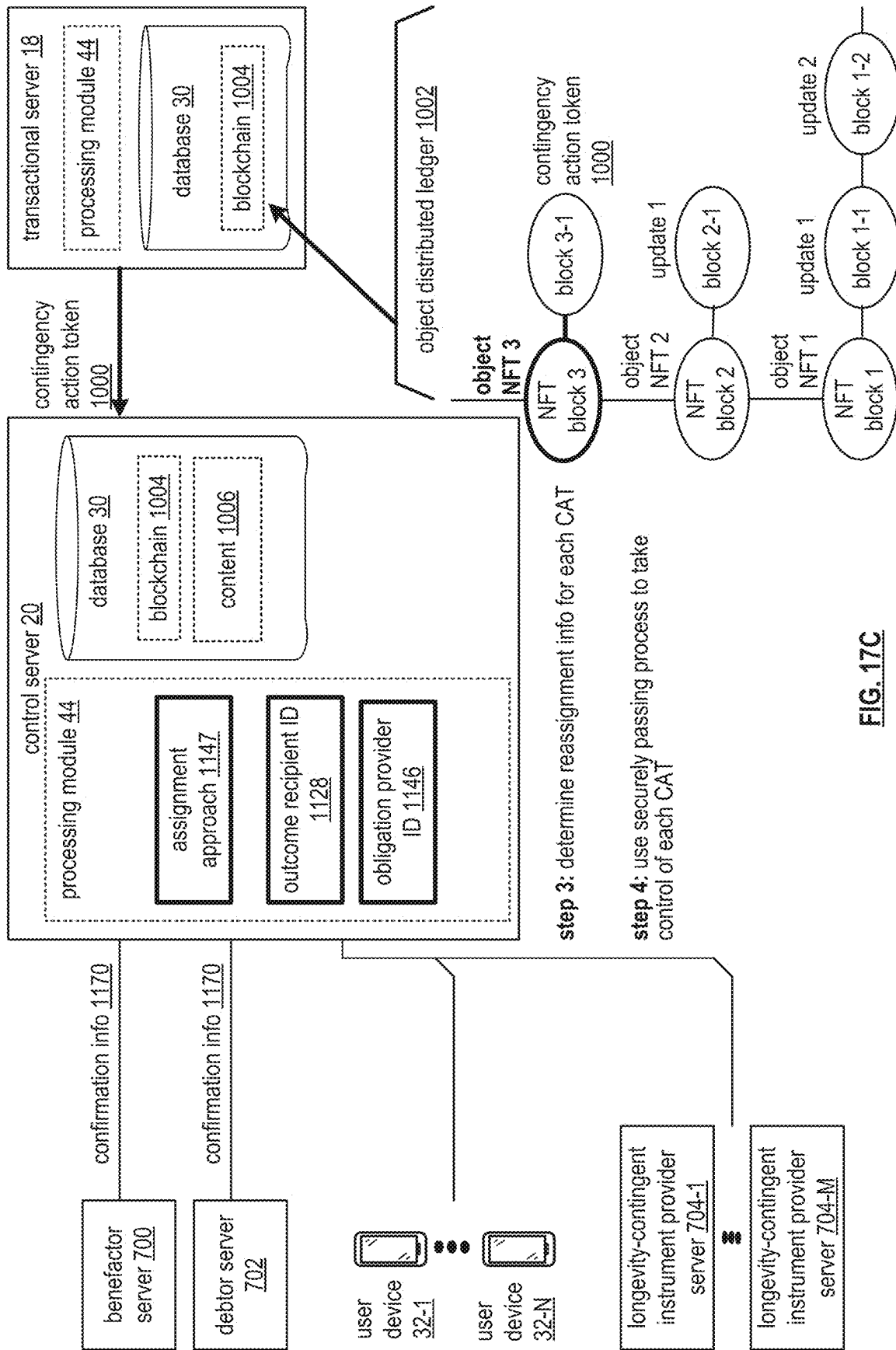


FIG. 17B



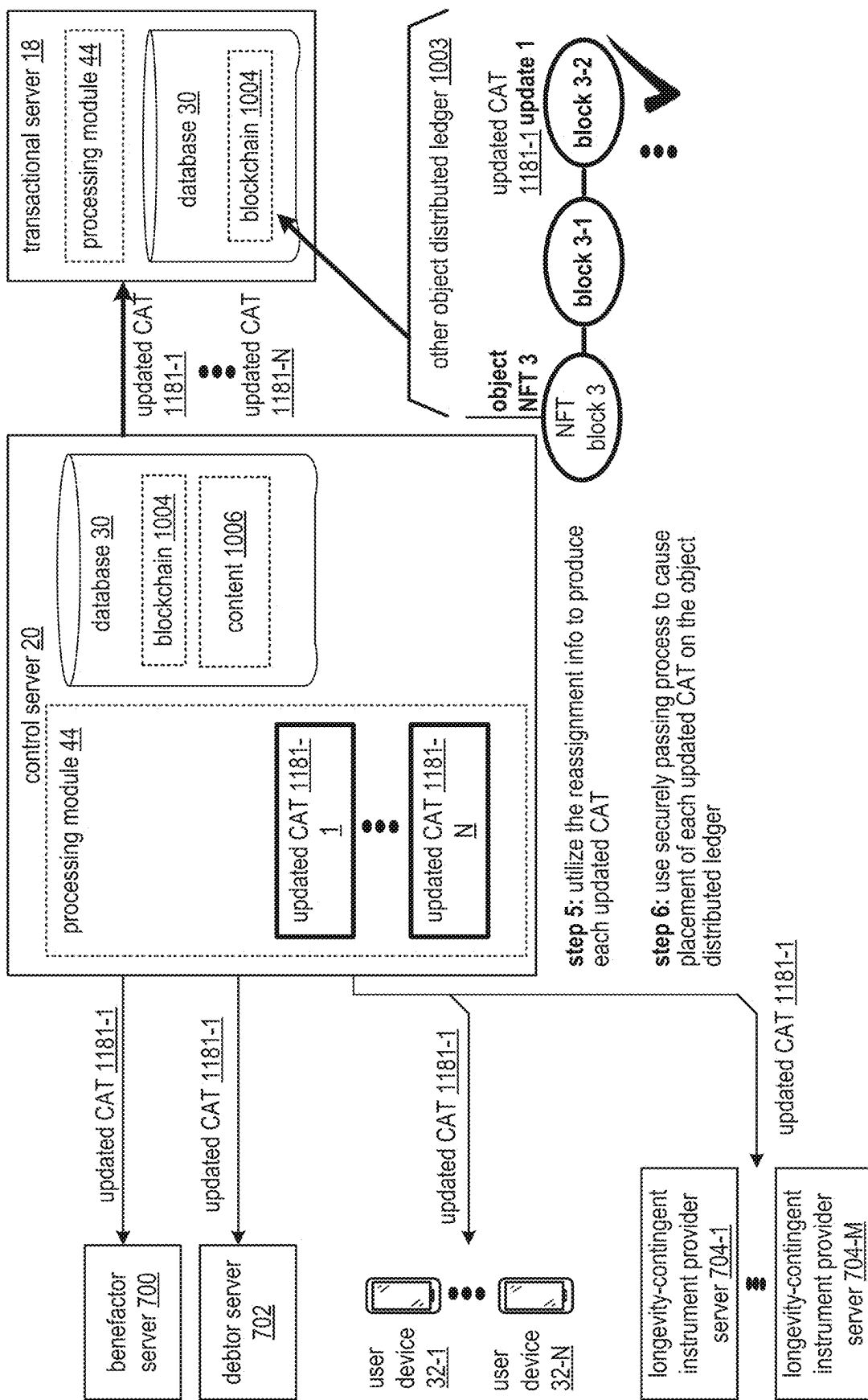


FIG. 17D

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SECURELY PROCESSING A CONTINGENT ACTION TOKEN**CROSS REFERENCE TO RELATED PATENTS**

The present U.S. Utility patent application claims priority pursuant to 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/414,532, entitled "SECURELY PROCESSING A CONTINGENT ACTION TOKEN," filed Oct. 9, 2022, which is hereby incorporated herein by reference in its entirety and made part of the present U.S. Utility patent application for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC

Not Applicable

BACKGROUND OF THE INVENTION**Technical Field of the Invention**

This invention relates generally to security aspects of communication systems and more particularly to security approaches to support digital asset representation reconfiguration and reassignment within the communication system.

Description of Related Art

Communication systems are known to communicate data between communication devices of the communication system. The data may be communicated in one or more of an unaltered form (e.g., raw data from a first communication device), in an altered form to provide enhanced transmission reliability (e.g., error encoded), in an altered form to provide enhanced security of access (e.g., credentialed access, encryption), and in an altered form to enhance communication resource utilization (e.g., compression). The data may represent a wide variety of data types including one or more of video, audio, text, graphics, and images. Text data is widely known to represent text character documentation, financial documents of numerical nature, and/or a combination thereof.

Global enterprise operations are increasingly utilizing communication systems to communicate representations of financial affairs. Financial documents associated with the financial affairs may include advertisements, solicitations, asset pricing information, purchase orders, invoices, payment transactions, asset distribution information, complex settlement information, financing information, financial market information, asset titling information, transaction guarantee information, global finance trend analysis information, and other information associated with the increasingly complex world of electronic commerce.

The global velocity of data communication and massive volume of data representing financial documents is ever-increasing and as a result it is a growing challenge to communicate, manipulate, and enhance the data related to financial affairs. Such challenges include refreshing an asset base of the financial system (e.g., including detecting growing issues with regards to desired funding levels of the

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financial system), unlocking untapped asset value (e.g., conversion of one asset type to another), and rapidly retitling new or re-spun assets (e.g., assigning new assets, reassigning converted assets).

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

FIG. 1 is a schematic block diagram of an embodiment of a communication system in accordance with the present invention;

FIG. 2 is a schematic block diagram of an embodiment of a device of a communication system in accordance with the present invention;

FIG. 3 is a schematic block diagram of an embodiment of a server of a communication system in accordance with the present invention;

FIGS. 4A-4E are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for servicing a plurality of rived longevity-contingent instruments within a computing system in accordance with the present invention;

FIGS. 5A-5E are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for riving longevity-contingent instruments within a computing system in accordance with the present invention;

FIGS. 6A-6E are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for generating a portfolio of blockchain-encoded rived longevity-contingent instruments within a computing system in accordance with the present invention;

FIGS. 7A-7C are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for utilizing a portfolio of blockchain-encoded rived longevity-contingent instruments within a computing system in accordance with the present invention;

FIGS. 8A-8D are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for updating a portfolio of blockchain-encoded rived longevity-contingent instruments within a computing system in accordance with the present invention;

FIGS. 9A-9C are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for utilizing blockchain-encoded records for rived longevity-contingent instruments within a computing system in accordance with the present invention;

FIGS. 10A-10E are schematic block diagrams of an embodiment of a computing system illustrating an embodiment of a method for utilizing a contingent action token within the computing system in accordance with the present invention;

FIGS. 11A-11B are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for generating a contingent action token within the computing system in accordance with the present invention;

FIGS. 12A-12B are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for utilizing a contingent action token within the computing system in accordance with the present invention;

FIGS. 13A-13D are schematic block diagrams of an embodiment of a computing system illustrating an embodi-

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ment of a method for selecting a contingent action token within the computing system in accordance with the present invention;

FIGS. 14A-14B are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for establishing a contingent action token within the computing system in accordance with the present invention; and

FIGS. 15A-15B are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for generating a contingent action token within the computing system in accordance with the present invention;

FIGS. 16A-16D are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for securely transitioning purpose of a contingent action token within the computing system in accordance with the present invention; and

FIGS. 17A-17D are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for securely processing a contingent action token within the computing system in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

FIG. 1 is a schematic block diagram of an embodiment of a communication system 10 that includes a legacy system 12, a plurality of N augmentation systems 14, a conversion server 16, a transactional server 18, a control server 20, one or more data sources 26, and a network 28. Alternatively, the communication system 10 may include any number of legacy systems 12 and any number of servers 16-20.

The legacy system 12 includes a plurality of user devices 32, a plurality of subscriber devices 34, a portion of the network 28, and a legacy server 22. Each user device 32 may be implemented utilizing one or more portable communication devices. Examples of portable communication devices include a smart phone, a basic cell phone, a Wi-Fi communication device, a satellite phone, and/or any other device that includes a computing core (e.g., providing processing module functionality), one or more wireless modems, sensors, and one or more user interfaces, and is capable of operating in a portable mode untethered from a fixed and/or wired network. For example, a particular user device 32 is implemented utilizing the smart phone, where the smart phone is utilized by a user associated with the legacy system 12. At least some of the user devices 32 are capable to communicate data encoded as wireless communication signals and/or wireless location signals with the portion of the network 28 associated with the legacy system 12 and/or directly or indirectly to other user devices 32 and/or to at least some of the user devices 34.

Each subscriber device 34 may be implemented utilizing one or more computing devices. Examples of portable computing devices includes a laptop computer, a tablet computer, a handheld computer, a desktop computer, a cable television set-top box, an application processor, an internet television user interface, and/or any other device that includes a computing core (e.g., providing the processing module functionality), one or more modems, sensors, and one or more user interfaces. For example, a particular user subscriber device 34 is implemented utilizing the laptop computer, where the laptop computer is utilized by a subscriber associated with the legacy system 12. The subscriber devices 34 are capable to communicate data that is encoded

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into wireless and/or wired communication signals via the portion of the network 28 associated with the legacy system 12 and/or directly or indirectly to other subscriber devices 34 and/or to at least some of the user devices 32.

The components of the communication system 10 are coupled via the network 28, which may include one or more of wireless and/or wireline communications networks, one or more wireless location networks, one or more private communications systems, a public Internet system, one or more local area networks (LAN), and one or more wide area networks (WAN). For example, the network 28 is implemented utilizing the Internet to provide connectivity between the legacy system 12, the plurality of augmentation systems 14, the one or more data source 26, and the servers 16-20. The wireless location networks communicate wireless location signals with the user devices 32. Each wireless location network may be implemented utilizing one or more of a portion of a global positioning satellite (GPS) satellite constellation, a portion of a private location service, a wireless local area network (WLAN) access point, a Bluetooth (BT) beacon and/or communication unit, and a radiofrequency identifier (RFID) tag and/or transceiver. Each wireless location network generates and transmits the wireless location signals in accordance with one or more wireless location industry standards (e.g., including synchronize timing information (i.e., GPS), and a geographic reference identifier (ID) (i.e., a beacon ID, a MAC address, an access point ID such as a wireless local area network SSID)).

The wireless communication networks of the network 28 include one or more of a public wireless communication network and a private wireless communication network and may operate in accordance with one or more wireless industry standards including 5G, 4G, universal mobile telecommunications system (UMTS), global system for mobile communications (GSM), long term evolution (LTE), wideband code division multiplexing (WCDMA), and IEEE 802.11. For example, a first user device 32 communicates data encoded as wireless communication signals with a 4G public wireless communication network of the network 28 and a second user device 32 communicates data encoded as wireless communication signals with a Wi-Fi wireless communication network of the network 28.

The legacy server 22 includes at least one processing module 44 and at least one database 30. The processing module 44 processes control messages 36 and data messages 38 via the network 28 with one or more of the user devices 32, the subscriber devices 34, the augmentation systems 14, the data sources 26, the conversion server 16, a transactional server 18, and the control server 20. The processing module 44 further stores and retrieves data in the database 30. The processing module 44 is discussed in greater detail with respect to FIGS. 2-3 and the database 30 is discussed in greater detail with reference to FIG. 3.

Each augmentation system 14 includes another plurality of user devices 32, another plurality of subscriber devices 34, another portion of the network 28, and an augmentation server 24. The augmentation server 24 includes another processing module 44 and another database 30. Each of the conversion server 16, the transactional server 18, and the control server 20 includes another processing module 44 and another database 30.

Each data source 26 may be implemented utilizing one or more of a server, a subscription service, a website data feed, or any other portal to data messages 38 that provide utility for operation of the communication system 10. Further examples of the data source 26 includes one or more of a

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financial market server, a census server, a government record server, another transactional server, another control server, another conversion server, another legacy server, a weather service, a screen scraping algorithm, a website, another database, a schedule server, a live traffic information feed, an information server, a service provider, and a data aggregator. The data messages **38** includes one or more of live financial market information, historical financial market information, weather information, a user daily activity schedule (e.g., a school schedule, a work schedule, a delivery schedule, a public transportation schedule), real-time traffic conditions, a road construction schedule, a community event schedule, address of residence information, user lifestyle information (e.g., smoker, non-smoker, physical activities, etc.), user death records, mortality tables, and other information associated with a user.

In general, and with respect to the asset reconfiguration and reassignment within the communication system **10**, the communication system **10** provides a secure records processing approach to support three primary functions. The three primary functions include: 1) determining desired financial attributes of a financial system (e.g., supported by an underperforming legacy asset base), 2) facilitating acquisition of an augmenting asset bundle to enhance the financial system (e.g., enhancing and/or replacing the legacy asset base, and 3) facilitating the enhancement of the financial system utilizing the augmenting asset bundle such that the financial system substantially achieves the desired financial attributes. The communication system **10** may perform one or more of the three primary functions to provide the asset reconfiguration and reassignment.

The financial system is associated with the legacy system **12** where a plurality of users of the user devices **32** and the subscriber devices **34** are investors/beneficiaries of the legacy asset base supporting the financial system. The plurality of users may include thousands, hundreds of thousands, or even millions of users. The financial system includes any system to derive value for the plurality of users (e.g., balance sheet value and/or cash flow value) from the legacy asset base. Examples of the financial system includes a money market, a bond fund, a hedge fund, a pension system, and a stock fund. The desired financial attributes include one or more of present and future values of the legacy asset base, cash flows enabled by the legacy asset base, ongoing costs associated with the financial system, and return on investment levels for the legacy asset base. The legacy asset base may include thousands, hundreds of thousands, or even millions of individual assets, where assets may include tangible hard assets (e.g., property title, precious metals, commodities, etc.) and monetary assets (e.g., bonds, stocks, life insurance policies,

The augmenting asset bundle includes a bundle of selected assets acquired from one or more of the augmentation systems **14**, where candidate assets associated with the augmentation systems **14** includes thousands, hundreds of thousands, and even millions of assets. The assets are selected such that when combined or replacing assets of the legacy assets, the desired financial attributes of the financial system can substantially be reached. The facilitating of the enhancement of the financial system utilizing the augmenting asset bundle manipulates (e.g., splits, un-bundles, transforms, re-bundles, retitles, etc.) the selected assets for combination with or the replacement of assets of the legacy asset base.

The first primary function includes the communication system **10** determining desired financial attributes of a financial system. In an example of operation where the

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financial system of the legacy system **12** is a pension system for over 100,000 pensioners, the legacy asset base includes assets that are a combination of cash and bonds, and the augmentation systems **14** lists millions of available life insurance policies, the processing module **44** of the control server **20** determines to evaluate the financial system. For example, the control server **20** receives, via the network **28**, a control message **36** from the conversion server **16**, where the control message **36** includes a request to address underperformance of the legacy asset base associated with the legacy system **12**. Having determined to evaluate the financial system, the control server **20** characterizes the financial system to produce a desired cash flow and desired valuation improvement or left for the legacy asset base. For example, the control server **20** receives, via the network **28**, another control message **36** from the legacy server **22** that includes information associated with the financial system, and evaluates the information associated with the financial system to determine the desired cash flow and desired valuation lift.

The second primary function includes the communication system **10** facilitating acquisition of an augmenting asset bundle to enhance the financial system. In an example of operation, the processing module **44** of the control server **20** accesses augmenting asset information to extract candidate asset characteristics and down selects candidate assets that compare favorably to augmenting asset preferences. The candidate asset characteristics includes one or more of asset identifier (ID), asset type (e.g., stock, bond, life insurance policy, tangible asset), estimated fair market value (FMV) of the asset, purchase price of the asset, a risk level associated with the asset, a risk level associated with the particular augmentation system tied to the asset, associated liabilities (e.g., premium payments), associated payouts (e.g., a death benefit of an insurance policy), estimated payout timing (e.g., estimated year of a life insurance death benefit payout), an estimated return on investment (ROI) level, and demographics of entities associated with the asset (e.g., age and other characteristics of an insured person associated with an insurance policy). The augmenting asset preferences includes one or more of a maximum desired risk level associated with the asset, a maximum desired risk level associated with the augmentation system tied to the asset, a maximum liability level, a minimum payout level, a minimum ROI level, and one or more preferred demographics of the entities associated with the asset. For example, the control server **20** receives control messages **36** from one or more of the augmentation servers **24**, where the control messages **36** includes the candidate asset characteristics, and receives further control messages **36** from the conversion server **16**, where the further control messages **36** includes the augmenting asset preferences.

Having obtained the candidate asset characteristics and the augmenting asset preferences, the control server **20** searches through available assets of the one or more augmentation systems **14** to down select the candidate assets that compare favorably to the augmenting asset preferences. For example, the control server **20** exchanges control messages **36** with the augmentation server of each of the one or more augmentation systems **14** to identify each available asset, compares the asset characteristics of the available asset to the augmenting asset preferences, and identify assets where the comparison is favorable (e.g., estimated ROI greater than minimum desired ROI, estimated risk level lower than maximum desired risk level, etc.) to produce the down selected candidate assets.

Having identified the down selected candidate assets, the control server **20** determines a financial contribution of each

of the down selected candidate assets. For example, the control server **20** estimates a balance sheet contribution (e.g., a portion of the desired lift) and a cash flow contribution (e.g., a portion of the desired cash flow) for each down selected candidate asset based on the candidate asset characteristics. The control server **20** may produce the estimates based on the down selected candidate assets in an un-altered form and may produce further estimates based on altered forms of the down selected candidate assets, where each of the altered down selected candidate assets are reconfigured. The reconfiguring of a plurality of assets (e.g., selected candidate assets) includes the deconstruction of each of the assets into deconstructed asset elements of two or more element types in accordance with a deconstruction approach and re-bundling pluralities of deconstructed asset elements into two or more new asset bundles in accordance with a re-bundling approach to substantially satisfied the desired cash flow and desired valuation lift of the financial system, where each new asset bundle is generally titled to a different entity. For instance, the control server **20** utilizes a default deconstruction approach and default re-bundling approach to produce financial contributions of the down selected candidate assets when reconfigured (e.g., deconstructed and re-bundled in accordance with the default deconstruction approach and default re-bundling approach).

Having determined the financial contributions of each of the down selected candidate assets, the control server **20** selects assets from the down selected candidate assets to produce the augmenting asset bundle. The selecting includes choosing an asset selection approach to make the selections and completing the selecting utilizing the identified selection approach. The selection approaches include one or more of selecting assets that individually produce a highest level of ROI, selecting assets that produce a highest level of cash flow, selecting assets that produce a highest level of lift, selecting assets associated with highest levels of favorable financial contributions weighted by risk (e.g., asset risk, augmenting system risk, and transactional server entity risk), a random selection approach, and any other approach to optimize selection of the assets when considering utilization of deconstructed elements of the assets. The choosing of the asset selection approach may be based on one or more of a predetermination, a request, a correlation of historically utilized selection approaches and financial results, and a weighting factor that considers multiple desired outcomes.

Having chosen the asset selection approach, the control server **20** utilizes the asset selection approach to select assets from the down selected candidate assets based on the financial contributions to produce the augmenting asset bundle revealing characteristics of the selected assets (e.g., asset ID, asset type, etc.). For example, the control server **20** exchanges further control messages **36** with the one or more augmentation servers **24** to complete acquisition of the selected assets of the augmenting asset bundle based on the financial contributions of the selected assets.

The third primary function includes the communication system **10** facilitating the enhancement of the financial system utilizing the augmenting asset bundle such that the financial system substantially achieves the desired financial attributes. In an example of operation, the control server **20** selects a server to perform the reconfiguring of the acquired assets. The selection may be based on one or more of a predetermination, a request, and historical reconfiguring results. For example, the control server **20** selects the conversion server **16** to perform the reconfiguring of the acquired assets

Having selected the conversion server **16** to perform the reconfiguring of the acquired assets, the control server **20** facilitates the reconfiguring of the assets of the augmenting asset bundle. The facilitating includes selecting the deconstruction approach, selecting the re-bundling approach, and initiating the reconfiguring utilizing the selected approaches. The selecting may be based on one or more of a predetermination, a request, information extracted from data messages **38** of one or more of the data sources **26** (e.g., current market conditions), and historical financial results based on various approaches. The initiating of the reconfiguring includes performing the reconfiguring by the control server **20** and/or issuing a control message **36** to the conversion server **16**, where the control message **36** includes a request to perform the reconfiguring of the assets of the augmenting asset bundle in accordance with the selected deconstruction approach and the selected re-bundling approach. The control message **36** may further include the characteristics of the selected assets of the augmenting asset bundle. For example, the conversion server **16** deconstructs each asset of the augmenting asset bundle in accordance with the deconstruction approach to produce two or more deconstructed asset elements (e.g., of two or more element types) and re-bundles pluralities of the deconstructed asset elements in accordance with the re-bundling approach to produce the two or more asset bundles.

Having facilitated the reconfiguring of the assets, the control server **20** facilitates the reassignment of the reconfigured assets where the two or more asset bundles are to be titled to two or more entities of the communication system **10** to substantially satisfied the desired cash flow and desired valuation lift of the financial system. The facilitating includes issuing titling information to the conversion server **16** such that the conversion server **16** titles the two or more asset bundles in accordance with the titling information. Having received the titling information, the conversion server **16** produces two asset bundles and issues the titling information via a control message **36** to the legacy server **22** to associate a first asset bundle with the legacy system **12** and issues the titling information via another control message **36** to the transactional server **18** to associate a second asset bundle with the transactional server **18**.

Having facilitated the titling of the two or more asset bundles, the control server **20** identifies the transactional server **18** to facilitate subsequent financial transactions utilizing the new asset bundles produced from the re-bundling of the deconstructed elements of the acquired assets. For example, the control server **20** issues a control message **36**, via the network **28**, to the transactional server **18**, where the control message **36** includes subsequent financial transaction information (e.g., how to utilize the new asset bundles). For instance, the transactional server **18** exchanges control messages **36** with an augmentation server **24** associated with a particular asset to settle a periodic liability (e.g., the transactional server **18** facilitates a liability payment to the augmentation server **24** such as a life insurance premium payment) and to collect a cash flow (e.g., a life insurance policy death benefit payment). As another instance, the transactional server **18** partitions the cash flow from the augmentation server **24** into a first portion and a second portion, where the first portion is associated with the legacy server **22** (e.g., a portion of the life insurance policy death benefit payment flows to the pension system associated with the financial system of the legacy server **22**) and the second portion is associated with the transactional server **18** (e.g., a holdback if any). Such financial transactions may include

one or more of electronic money wire transfers and block-chain encoded secure funds transfer.

In various embodiments, a non-transitory computer readable storage medium includes at least one memory section that stores operational instructions that, when executed by one or more processing modules of one or more computing devices that each include a processor and a memory, causes each processing module to perform operations including the above-described asset reconfiguration and reassignment within the communication system.

FIG. 2 is a schematic block diagram of an embodiment of the user device 32 and the subscriber device 34 of the communication system 10 that includes a computing core 50, a visual output device 74 (e.g., a display screen, a light-emitting diode), a user input device 76 (e.g., keypad, keyboard, touchscreen, voice to text, etc.), an audio output device 78 (e.g., a speaker, a transducer, a motor), a visual input device 80 (e.g., a photocell, a camera), a sensor 82 (e.g., an accelerometer, a velocity detector, electronic compass, a motion detector, electronic gyroscope, a temperature device, a pressure device, an altitude device, a humidity detector, a moisture detector, an image recognition detector, a biometric reader, an infrared detector, a radar detector, an ultrasonic detector, a proximity detector, a magnetic field detector, a biological material detector, a radiation detector, a mass and/or weight detector, a density detector, a chemical detector, a gas detector, a smoke detector, a fluid flow volume detector, a DNA detector, a wind speed detector, a wind direction detector, a medical condition detector, a human activity detector, a motion recognition detector, and a battery level detector), one or more universal serial bus (USB) devices 1-U, one or more peripheral devices, one or more memory devices (e.g., a local memory, a flash memory device 92, one or more hard drives 94, one or more solid state (SS) memory devices 96, and/or cloud memory 98), an energy source 100 (e.g., a battery, a generator, a solar cell, and a fuel cell), one or more wireless location modems 84 (e.g., a GPS receiver, a Wi-Fi transceiver, a Bluetooth transceiver, etc.), one or more wireless communication modems 86 (e.g., 4G, 5G cellular), a wired local area network (LAN) 88, and a wired wide area network (WAN) 90.

The computing core 50 includes a video graphics processing module 52, one or more processing modules 44, a memory controller 56, one or more main memories 58 (e.g., RAM), one or more input/output (I/O) device interface modules 62 (e.g., interfaces), an input/output (I/O) controller 60, a peripheral interface 64, one or more USB interface modules 66, one or more network interface modules 72, one or more memory interface modules 70, and/or one or more peripheral device interface modules 68. Each of the interface modules 62, 66, 68, 70, and 72 includes a combination of hardware (e.g., connectors, wiring, etc.) and operational instructions stored on memory (e.g., driver software) that is executed by the processing module 44 and/or a processing circuit within the interface module. Each of the interface modules couples to one or more components of the user device 32. For example, one of the IO device interface modules 62 couples to an audio output device 78. As another example, one of the memory interface modules 70 couples to flash memory 92 and another one of the memory interface modules 70 couples to cloud memory 98 (e.g., an on-line storage system and/or on-line backup system).

The main memory 58 and the one or more memory devices include a computer readable storage medium that stores operational instructions that are executed by one or more processing modules 44 of one or more computing

devices (e.g., the user device 32) causing the one or more computing devices to perform functions of the communication system 10. For example, the processing module 44 retrieves the stored operational instructions from the HD memory 94 for execution.

FIG. 3 is a schematic block diagram of an embodiment of the servers 16-24 of the communication system 10 that includes a computing core 110 and elements of the user device 32 (e.g., FIG. 2), including one or more of the visual output device 74, the user input device 76, the audio output device 78, the memories 92-98 to provide the database 30 of FIG. 1, the wired LAN 88, and the wired WAN 90. The computing core 110 includes elements of the computing core 50 of FIG. 2, including the video graphics module 52, the plurality of processing modules 44, the memory controller 56, the plurality of main memories 58, the input-output controller 60, the input-output device interface module 62, the peripheral interface 64, the memory interface module 70, and the network interface modules 72.

FIGS. 4A-4E are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for servicing a plurality of rived longevity-contingent instruments within a computing system. The computing system includes data sources 26-1 through 26-N, the augmentation server 24 of FIG. 1, the transactional server 18 of FIG. 1, and legacy servers 22-1 through 22-2. In an embodiment, the data sources 26-1 through 26-N are implemented utilizing the data source 26 of FIG. 1. In an embodiment, the legacy servers 22-1 through 22-2 are implemented utilizing the legacy server 22 of FIG. 1, where legacy server 22-1 is associated with a pension system and legacy server 22-2 is associated with one or more sponsors associated with the pension system. The transactional server 18 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1.

The plurality of rived longevity-contingent instruments includes a pool of life insurance policies (e.g., the instruments), where the policies have been rived (e.g., split of benefit ownership from premium liability responsibility). Each longevity-contingent instrument is associated with a premium payment stream (e.g., series of premium payments). For example, an insurance company of a first life insurance policy requires a monthly premium payment to maintain the first life insurance policy in force. Together, the pool of life insurance policies is associated with a plurality of premium payment streams.

A financial offering that includes the pool of life insurance policies requires an aggregated payment of the plurality of premium payment streams associated with the pool of life insurance policies. In an embodiment, the one or more sponsors associated with the legacy servers 22-1 through 22-2 are liable for the aggregated payment of the plurality of periodic premium payments in accordance with a rive approach 682. The rive approach 682 is discussed in greater detail with regards to FIG. 4C.

Each longevity-contingent instrument is further associated with a payout (e.g., death benefit) when a longevity status changes, e.g., a death of an insured person associated with the life insurance policy of the longevity-contingent instrument. For example, when the insured person passes, the life insurance company of the first life insurance policy provides payment of the payout to an entity associated with ownership of the first life insurance policy.

Riving of the policies splits the policy to associate liability of periodic premium payments with one or more debtors (e.g., sponsors) and to associate the policy payout with one or more benefactors (e.g., a pension and a sponsor). For

example, the riving results in associating multiple sponsors of a common union pension with the liability of periodic premium payments. As another example, the riving results in associating the multiple sponsors of the common union pension and the common union pension with the policy payout.

The servicing of the plurality of longevity-contingent instrument includes steps associated with both the payouts upon longevity status change and the payment of the premium payment streams. The method of the servicing is discussed in greater detail with reference to FIGS. 7A-7C.

FIG. 4A illustrates an example of operation of steps of a method for the servicing of the plurality of longevity-contingent instruments where, in a first step, the processing module 44 interprets a digitally encoded data packet from another computing device to produce a first longevity indicator of a first longevity-contingent instrument of a plurality of longevity-contingent instruments. The first longevity-contingent instrument is rived in accordance with the rive approach 682 to produce a first sub-asset of a plurality of sub-assets and a first sub-liability of a plurality of sub-liabilities. The first sub-liability is associated with a first premium payment stream of a plurality of premium payment streams of the plurality of sub-liabilities.

A first death-notification of a multitude of death-notifications is encoded to produce the digitally encoded data packet. For example, the processing module 44 receives a multitude of death-notifications 662-1 through 662-N from data sources 26-1 through 26-N. The processing module 44 decodes the multitude of death-notifications to produce death-notification information. The processing module 44 accesses the database 30 to extract a plurality of insured person identifiers of the plurality of longevity-contingent instruments from longevity-contingent instrument information 660. A first insured person identifier of the plurality of insured person identifiers is associated with the first longevity-contingent instrument. The processing module 44 generates the first longevity indicator 664 to indicate a deceased status when the death-notification information includes a deceased person identifier that substantially matches the first insured person identifier of the first longevity-contingent instrument.

In another example, the processing module 44 interprets asset settlement information 144 to produce an indication of payment of the payout 674. The processing module 44 generates the first longevity indicator 664 when the payment of the payout 674 includes the deceased person identifier that substantially matches the first insured person identifier of the first longevity-contingent instrument.

In yet another example, the processing module 44 interprets either of the asset settlement information 144 and a corresponding death-notification 662-1 to produce a longevity status change 676. The processing module 44 generates the first longevity indicator 664 when the longevity status change 676 includes the deceased person identifier that substantially matches the first insured person identifier of the first longevity-contingent instrument.

FIG. 4B further illustrates the example of the servicing of the plurality of longevity-contingent instruments where, having produced the first longevity indicator 664, in a second step, the processing module 44 updates a first longevity status indicator 666 for the first longevity-contingent instrument within the database 30 utilizing the first longevity indicator to produce an updated first longevity status indicator. For example, the processing module 44 produces the updated first longevity status indicator to indicate a benefit

status when the first longevity indicator 664 indicates that the insured person has deceased.

Having updated the first longevity status indicator 666, when the updated first longevity status indicator is associated with the benefit status, in a third step, the processing module 44 determines a payout 678 associated with the first sub-asset. The determining the payout 678 includes a variety of approaches. A first approach includes interpreting a payment notification message 672. For example, the processing module 44 interprets the asset settlement information 144 to produce the payment notification message 672, where the payment notification message 672 includes the payout 678. In another example, the processing module 44 interprets the asset settlement information 144 to produce the indication of payment of the payout 674, where the indication of payment of the payout 674 includes the payout 678.

A second approach to determine the payout 678 includes accessing the database 30 to extract a face value of the first longevity-contingent instrument. For example, the processing module 44 accesses the longevity-contingent instrument information 660 to extract the face value (e.g., a stated value of an associated life insurance policy).

A third approach to determine the payout 678 includes accessing the database 30 to extract a benefit value (e.g., an agreed to value) of the first sub-asset. For example, the processing module 44 accesses sub-asset information 690 to extract the benefit value.

Alternatively, or in addition to, the processing module 44 indicates that the first sub-asset has matured. For example, the processing module updates the sub-asset information 690 to indicate that the sub-asset has matured (e.g., to benefit payout).

FIG. 4C further illustrates the example of the servicing of the plurality of longevity-contingent instruments where the processing module 44, having identified the payout 678, in a fourth step determines a first portion of the payout 680 to associate with a premium cash escrow 668 in accordance with the rive approach 682. The association enables subsequent utilization of the premium cash escrow 668 to fund the aggregated payment of the plurality of premium payment streams on behalf of the one or more debtors.

The rive approach includes a variety of approaches. The approaches include a surplus approach where a balance associated with the premium cash escrow 668 is maintained at a level that is more than enough to make the aggregated premium payment streams. The approaches further include a deficit approach where the balance associated with the premium cash escrow 668 is maintained at a level that is less than enough to make the aggregated premium payment streams (e.g., another party such as a pension sponsor is liable to make up differences).

The approaches further include a breakeven approach where the balance associated with the premium cash escrow 668 is maintained at a level that is just enough to make the aggregated premium payment streams. The approaches further include a pro rata approach where the first portion is in accordance with a negotiated percentage of the payout (e.g., always 50% or even 40%). The approaches further include a consistency approach where the balance associated with the premium cash escrow 668 receives a stream of constant inflows to support the aggregated premium payment streams.

When the rive approach 682 includes the surplus approach, the determining of the first portion of the payout 680 includes calculating the first portion of the payout such that a sum of a plurality of first portion payouts within a first

time frame is greater than a sum of a subset of the plurality of premium payment streams for the first time frame. When the rive approach 682 includes the deficit approach, the determining of the first portion of the payout 680 includes calculating the first portion of the payout such that the sum of the plurality of first portion payouts within the first time frame is less than the sum of the subset of the plurality of premium payment streams for the first time frame.

When the rive approach 682 includes the break-even approach, the determining of the first portion of the payout 680 includes calculating the first portion of the payout such that the sum of the plurality of first portion payouts within the first time frame is substantially the same as the sum of the subset of the plurality of premium payment streams for the first time frame. When the rive approach 682 includes the pro rata approach, the determining of the first portion of the payout 680 includes establishing the first portion of the payout in accordance with a pre-determined percentage of the payout. When the rive approach 682 includes the consistency approach, the determining of the first portion of the payout 680 includes establishing the first portion of the payout in accordance with a pre-determined first portion level (e.g., a default constant amount).

Having determined the first portion of the payout 680, the processing module 44, in a fifth step determines a second portion of the payout 686 to associate with a benefit cash account 670 based on the first portion of the payout 680 and in accordance with the rive approach 682. The benefit cash account 670 is associated with the one or more benefactors. The determining of the second portion of the payout 686 includes a variety of approaches. The approaches include the pro rata approach, the consistency approach, and a difference approach.

When the rive approach includes the pro rata approach, the determining of the second portion of the payout 686 includes establishing the second portion of the payout 686 in accordance with a pre-determined percentage of the payout. For example, the processing module 44 multiplies the pre-determined percentage by the payout 678 to produce the second portion of the payout 686 (e.g., 60% of the payout).

When the rive approach includes the consistency approach, the determining of the second portion of the payout 686 includes establishing the second portion of the payout 686 in accordance with a pre-determined second portion level (e.g., a constant amount). For example, the processing module 44 sets the second portion of the payout 686 to be a fixed number based on the predetermined second portion level (e.g., a flat \$100,000).

When the rive approach includes the difference approach, the determining of the second portion of the payout 686 includes establishing the second portion of the payout in accordance with a difference between the payout and the first portion of the payout (e.g., what's leftover). For example, the processing module 44 subtracts the first portion of the payout 680 from the payout 678 to produce the second portion of the payout 686 (e.g., \$1 million payout minus \$480,000 first portion equals \$520,000).

Alternatively, or in addition to, the processing module 44 determines a third portion of the payout. For instance, the payout 678 equals the sum of the first through third portions, where the third portion is a service fee. In yet another alternative, the processing module determines further portions of the payout when more than one benefactor directly receives a portion of the payout 678 (e.g., multiple pensions associated with the plurality of longevity-contingent assets).

FIG. 4D further illustrates the example of the servicing of the plurality of longevity-contingent instruments where the

processing module 44, in sixth step, facilitates reconciling of the first portion of the payout 680 to the premium cash escrow 668 and the second portion of the payout 686 to the benefit cash account 670. For example, the processing module 44 increments the premium cash escrow 668 of the database 30 by an amount of the first portion of the payout 680. Alternatively, or in addition to, the processing module 44 issues a payment message to another server associated with the premium cash escrow 668 (e.g., a debtor). As another example, the processing module 44 increments the benefit cash account 670 of the database 30 by an amount of the second portion of the payout 686. Alternatively, or in addition to, the processing module 44 issues a payment message to another server associated with the benefit cash account 670 (e.g., a benefactor).

Having facilitated the reconciling of the first portion of the payout 680 and the second portion of the payout 686, in a seventh step the processing module 44 facilitates the aggregated payment of the plurality of premium payment streams utilizing the premium cash escrow 668 and one or more premium offsets 688-1 and 688-2 from the one or more debtors (e.g., via their legacy servers 22-1 and 22-2). For example, the processing module 44 accrues premium payments 684 utilizing a portion of the premium cash escrow 668, determines a level of a required payment of the premium payment streams, calculates a difference between the accrued premium payment 684 and the level of required payment to produce a supplementing level, and obtains the supplementing level of funds from the legacy servers 22-1 and 22-2 via premium offsets 688-1 and 688-2.

Having obtained the portion of the premium cash escrow 668, the premium offsets 688-1, and the premium offsets 688-2, the processing module 44 sums the portion of the premium cash escrow 668, the premium offset 688-1, and the premium offset 688-2 to produce the premium payments 684. Having produced the premium payments 684, the processing module 44 issues liability settlement information 142 to the augmentation server 24, where the liability settlement information 142 pertains to the premium payments 684.

FIG. 4E further illustrates the example of the servicing of the plurality of longevity-contingent instruments where, in an eight step the processing module 44 facilitates payment from the benefit cash account 670 to the one or more benefactors. For example, the processing module 44 issues sub-asset settlement information 146 to the legacy server 22-1 that is associated with the pension system, where the sub-asset settlement information 146 includes a portion of the benefit cash account 670 (e.g., the second portion of the payout 686). Alternatively, or in addition to, the processing module 44 issues the second portion of the payout 686 to another server associated with one or more other benefactors.

Having facilitated the payment of the benefit cash account 670, the processing module 44, from time to time in a ninth step, adjusts the rive approach 682 to favor increasing the second portion of the payout when a first sum of a first plurality of second portion payouts within a first time frame is less than a first sum of a first subset of the plurality of premium payment streams for the first time frame. For example, the processing module 44 increases the percentage of the second portion of the payout to bolster the premium payments.

Alternatively, the processing module 44, from time to time in the ninth step, adjusts the rive approach to favor decreasing the second portion of the payout when a second sum of a second plurality of second portion payouts within

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a second time frame is greater than a second sum of a second subset of the plurality of premium payment streams for the second time frame. For example, the processing module 44 decreases the percentage of the payout 686 to not overfund the premium payments.

The method described above module can alternatively be performed by various modules of the communication system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the communication system 10, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 5A-5E are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for riving longevity-contingent instruments within a computing system. The computing system includes a benefactor server 700, a debtor server 702, user devices 32-1 through 32-N, longevity-contingent instrument provider servers 704-1 through 704-M, and the control server 20 of FIG. 1. In an embodiment, the benefactor server 700 and the debtor server 702 are implemented utilizing the legacy server 22 of FIG. 1, where the benefactor server 700 is associated with at least one pension system and the debtor server 702 is associated with at least one sponsor associated with the at least one pension system. In an embodiment, the user devices 32-1 through 32-N are implemented utilizing the user devices 32 of FIG. 1. In an embodiment, the longevity-contingent instrument provider servers 704-1 through 704-M are implemented utilizing the augmentation server 24 of FIG. 1. The control server 20 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1.

FIG. 5A illustrates an example of operation of steps of a method for the riving of the longevity-contingent instruments where, in a first step, the processing module 44 interprets digitally encoded rive parameters from one or more of a benefactor computing device (e.g., the benefactor server 700) and a debtor computing device (e.g., the debtor server 702) to produce rive approach requirements 714. The interpreting includes a series of operations. A first operation includes decoding a first subset of the digitally encoded rive parameters received from the benefactor computing device to produce asset rive parameters. For example, the processing module 44 decodes digitally encoded rive parameters from the benefactor server 700 to produce asset rive parameters 710. The asset rive parameter 710 includes one or more of a required net cash flow pattern, a target investment yield rate, and a maximum initial benefactor contribution level.

A second operation includes decoding a second subset of the digitally encoded rive parameters received from the debtor computing device to produce liability rive parameters. For example, the processing module 44 decodes digitally encoded rive parameters from the debtor server 702 to produce liability rive parameter 712. The liability rive parameters 712 includes one or more of a maximum contribution cash flow pattern and a maximum initial debtor contribution level. A third operation includes aggregating the asset rive parameters 710 and the liability rive parameters 712 to produce the rive approach requirements 714.

Having produced the rive approach requirements 714, in a second step, the processing module 44 determines a rive

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approach 682 for riving a set of longevity-contingent instruments of a multitude of available longevity-contingent instruments based on the rive approach requirements 714. A first longevity-contingent instrument of the set of longevity-contingent instruments includes a first face value benefit (e.g., death benefit) and a first premium payment stream. A second longevity-contingent instrument of the set of longevity-contingent instruments includes a second face value benefit and a second premium payment stream. When available (e.g., when an insured person passes and the death benefit is provided), a first portion of the first face value benefit is utilized to fund at least some of the second premium payment stream in accordance with the rive approach 682. The premium payment stream includes series of time-certain obligated payments to maintain the corresponding longevity-contingent instrument (e.g., with a corresponding provider, i.e. insurance company).

The determining of the rive approach 682 includes one of a variety of ways. A first way, when the rive approach requirements indicate that a first allocated portion of the plurality of sub-assets is to be greater than the plurality of sub-liabilities, includes establishing the rive approach as a surplus approach. A second way, when the rive approach requirements indicate that the first allocated portion of the plurality of sub-assets is to be less than the plurality of sub-liabilities includes establishing the rive approach as a deficit approach. A third way, when the rive approach requirements indicate that the first allocated portion of the plurality of sub-assets is to be substantially the same as the plurality of sub-liabilities includes establishing the rive approach as a break-even approach.

A fourth way of determining the rive approach 682, when the rive approach requirements indicate that the first allocated portion of the plurality of sub-assets is to be a pre-determined percentage of the plurality of sub-assets includes establishing the rive approach as a pro rata approach. A fifth way, when the rive approach requirements indicate that the first allocated portion of the plurality of sub-assets is to be a pre-determined first portion level includes establishing the rive approach as a consistency approach.

FIG. 5B further illustrates the example of the riving of the longevity-contingent instruments where, having determined the rive approach 682, in a third step, the processing module 44 analyzes a subset of the multitude of available longevity-contingent instruments to produce characterization information 720. The subset of the multitude of available longevity-contingent instruments includes the first longevity-contingent instrument 722 and the second longevity-contingent instrument 724. The characterization information 720 includes first characterization information for the first longevity-contingent instrument 722 and second characterization information for the second longevity-contingent instrument 724.

The multitude of available longevity-contingent instruments are generally available from one or both of a primary market and a secondary market. Accessing the primary market includes obtaining the longevity-contingent instruments directly from initial policyholders (e.g., the originally insured). Accessing the secondary market includes obtaining the longevity-contingent instruments from brokers and providers, where the longevity-contingent instruments have changed hands from the initial policyholders to one or more intermediaries (e.g., the brokers, etc.). The analyzing of the subset of the multitude of available longevity-contingent instruments to produce the characterization information includes several sub-steps. A first sub-step includes access-

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ing the multitude of available longevity-contingent instruments. For example, the processing module 44 receives primary market longevity-contingent instrument information 716 from one or more of the user devices 32-1 through 32-N. A first instance includes the user device 32-1 issuing the primary market longevity-contingent instrument information 716 to the control server 20 in an unsolicited fashion when desiring to offer a life insurance policy for sale. A second instance includes the control server 20 receiving the primary market longevity-contingent instrument information 716 from the user device 32-2 in response to a solicitation message from the control server 20.

As another example of accessing a multitude of available longevity-contingent instruments, the processing module 44 receives one or more of secondary market longevity-contingent instrument information 718-1 through 718-M from one or more of the longevity-contingent instrument provider servers 704-1 through 704-M. The receiving includes receiving the information in an unsolicited fashion and receiving the information in response to the control server 20 issuing a solicitation.

Having accessed the multitude of available longevity-contingent instruments, a second sub-step to analyze the subsets of the multitude of available longevity-contingent instruments includes determining the first characterization information to include one or more elements. A first element includes a first estimated timeframe for payout of the first face value benefit (e.g., generate a life expectancy based on one or more of insured age, gender, smoker, health impairments, historical life expectancy data, etc.). A second element includes a present value of the first face value benefit utilizing the first estimated timeframe (e.g., generate a present value range for a range of discounted cash flow analysis interest rates and for a range around the first estimate timeframe, i.e., dither the life expectancy). A third element includes a present value of the first premium payment stream.

A third sub-step to analyze the subsets of the multitude of available longevity-contingent instruments includes determining the second characterization information to include one or more further elements. A first further element includes a second estimated timeframe for payout of the second face value benefit. A second further element includes a present value of the second face value benefit utilizing the second estimated timeframe. A third further element includes a present value of the second premium payment stream.

A fourth sub-step to analyze the subsets of the multitude of available longevity-contingent instruments includes aggregating the first characterization information and the second characterization information to produce the characterization information 720. The characterization information 720 further includes insured age, gender, smoker, insured health record, historical life expectancy data, a requested purchase price, an offered purchase price, etc.).

Having analyzed the multitude of available longevity-contingent instruments to produce the characterization information 720, in a fourth step, when the first characterization information and the second characterization information compare favorably to the rive approach requirements 714, the processing module 44 selects the first longevity-contingent instrument 722 and the second longevity-contingent instrument 724 to include in the set of longevity-contingent instruments. For example, the processing module 44 identifies the first and second longevity-contingent instruments, causes title transfer (e.g., purchase via a transaction with the user device 32-1 and/or longevity-contingent instrument

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provider servers 704-1), and lists the first and second longevity-contingent instruments in the longevity-contingent instrument information 660 of the database 30.

FIG. 5C further illustrates the example of the riving of the longevity-contingent instruments where, having selected the longevity-contingent instruments, in a fifth step, the processing module 44 rives the first longevity-contingent instrument 722 based on the first face value benefit, the first premium payment stream and in accordance with the rive approach 682 to produce a first sub-asset 728 of a plurality of sub-assets of the set of longevity-contingent instruments and a first sub-liability 730 of a plurality of sub-liabilities of the set of longevity-contingent instruments. The first sub-liability 730 is associated with the first premium payment stream.

The riving of the first longevity-contingent instrument 722 includes generating beneficiary ownership of the first face value benefit to be associated with the first sub-asset 728. For example, the processing module 44 facilitates listing a legal entity of the first sub-asset as a partial beneficiary of the first longevity-contingent instrument and updates the sub-asset information 690 with the first sub-asset 728. As another example, the processing module 44 facilitates listing another legal entity of the first sub-liability as one of another partial beneficiary of the first longevity-contingent instrument and updates the sub-liability information 726 with the first sub-liability 730.

The riving of the first longevity-contingent instrument 722 further includes generating fiduciary responsibility of the first premium payment stream to be associated with the first sub-liability. For example, the processing module 44 facilitates listing the other legal entity of the first sub-liability as having fiduciary responsibility of the first premium payment stream of the first longevity-contingent instrument 722.

Having rived the first longevity-contingent instrument 722, in a sixth step, the processing module 44 rives the second longevity-contingent instrument 724 based on the second face value benefit, the second premium payment stream and in accordance with the rive approach 682 to produce a second sub-asset 732 of the plurality of sub-assets and a second sub-liability 734 of the plurality of sub-liabilities. The second sub-liability 734 is associated with the second premium payment stream. The processing module 44 further updates the sub-asset information 690 with the second sub-asset 732 and updates the sub-liability information 726 with the second sub-liability 734.

FIG. 5D further illustrates the example of the riving of the longevity-contingent instruments where, having rived the longevity-contingent instruments, in a seventh step, the processing module 44 issues sub-asset information 690 to the benefactor computing device (e.g., to the benefactor server 700). The sub-asset information 690 is based on the plurality of sub-assets and the rive approach 682. The issuing includes generating the sub-asset information 690 from all of the sub-assets and sending, via the network 28 of FIG. 1, the sub-asset information 690 to the benefactor server 700.

Having issued the sub-asset information, in an eighth step, the processing module 44 issues sub-liability information 726 to the debtor computing device (e.g., to the debtor server 702). The sub-liability information 726 is based on the plurality of sub-liabilities and the rive approach 682. The issuing includes generating the sub-liability information 726 from all of the sub-liabilities and sending, via the network 28 of FIG. 1, the sub-liability information 726 to the debtor server 702.

FIG. 5E further illustrates the example of the riving of the longevity-contingent instruments where, having issued the sub-liability information to the debtor computing device, in a ninth step, the processing module 44 associates the plurality of sub-assets with a benefit cash account 670 and associates the plurality of sub-liabilities with a premium cash escrow 668. The benefit cash account 670 is associated with the benefactor computing device and the premium cash escrow 668 is associated with the debtor computing device.

Having associated the sub-assets and the sub-liabilities, in a tenth step, the processing module 44, when available (e.g., upon payment of a death benefit), facilitates payment of a first portion of the first face value benefit 742 to the premium cash escrow 668 in accordance with the first sub-liability. The first portion of the first face value benefit is determined in accordance with the rive approach 682. The tenth step further includes the processing module 44, when available, facilitating payment of a second portion of the first face value benefit 744 to the benefit cash account 670 in accordance with the first sub-asset. The second portion of the first face value benefit is determined in accordance with the rive approach 682 and the first portion of the first face value benefit. Alternatively, or in addition to, the processing module 44 facilitates payment of a portion of the second premium payment stream utilizing one or more of the premium cash escrow 668 and a premium offset from the debtor computing device.

The method described above module can alternatively be performed by various modules of the communication system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the communication system 10, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 6A-6E are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for generating a portfolio of blockchain-encoded rived longevity-contingent instruments within a computing system. The computing system includes a benefactor server 700, a debtor server 702, user devices 32-1 through 32-N, longevity-contingent instrument provider servers 704-1 through 704-M, and the control server 20 of FIG. 1.

In an embodiment, the benefactor server 700 and the debtor server 702 are implemented utilizing the legacy server 22 of FIG. 1, where the benefactor server 700 is associated with at least one benefit entity (e.g., pension system) and the debtor server 702 is associated with at least one sponsor entity associated with the at least one benefit entity. In an embodiment, the user devices 32-1 through 32-N are implemented utilizing the user devices 32 of FIG. 1. In an embodiment, the longevity-contingent instrument provider servers 704-1 through 704-M are implemented utilizing the augmentation server 24 of FIG. 1. The control server 20 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1.

FIG. 6A illustrates an example of operation of steps of a method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, in a first step, the processing module 44 interprets digitally encoded rive parameters from one or more of a benefactor

computing device (e.g., the benefactor server 700) and a debtor computing device (e.g., the debtor server 702) to produce rive approach requirements 714. The interpreting includes a series of one or more operations. A first operation includes decrypting encrypted asset rive parameters 752 received from the benefactor server 700 to produce a first subset of the digitally encoded rive parameters. A second operation includes decoding the first subset of the digitally encoded rive parameters to produce asset rive parameters.

A third operation includes decrypting encrypted liability rive parameters 754 received from the debtor server 702 to produce a second subset of the digitally encoded rive parameters. A fourth operation includes decoding the second subset of the digitally encoded rive parameters to produce liability rive parameters. A fifth operation includes aggregating the asset rive parameters and the liability rive parameters to produce the rive approach requirements 714.

Having produced the rive approach requirements 714, in a second step of the method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments, the processing module 44 obtains a rive approach 682 for riving a set of longevity-contingent instruments of a multitude of available longevity-contingent instruments based on the rive approach requirements 714. A first longevity-contingent instrument of the set of longevity-contingent instruments includes a first face value benefit and a first premium payment stream. The first longevity-contingent instrument assigns the first face value benefit and the first premium payment stream to a first ownership entity (e.g., originally insured or a broker/holding entity).

A second longevity-contingent instrument of the set of longevity-contingent instruments includes a second face value benefit and a second premium payment stream. The second longevity-contingent instrument assigns the second face value benefit and the second premium payment stream to a second ownership entity (e.g., another originally insured or the broker/holding entity). In an embodiment, when an insured person passes and a death benefit is provided, availability of a first portion of the first face value benefit is utilized to fund at least some of the second premium payment stream in accordance with the rive approach 682.

The obtaining of the rive approach 682 includes determining, retrieving, and receiving. For example, the processing module 44 determines the rive approach 682 based on the rive approach requirements 714 as previously discussed. As another example, the processing module 44 retrieves the rive approach requirements 714 from the database 30. As yet another example, the processing module 44 receives the rive approach requirements 714 from another computing device.

FIG. 6B further illustrates the example of operation of steps of the method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, having obtained the rive approach 682, in a third step, the processing module 44 verifies authenticity of a group of blockchain-encoded records 800 representing a subset of the multitude of available longevity-contingent instruments to produce an authenticity indicator 806. The subset of the multitude of available longevity-contingent instruments includes the first longevity-contingent instrument 722 and the second longevity-contingent instrument 724.

The verifying of the authenticity includes obtaining the group of blockchain-encoded records 800 and analyzing the group of blockchain-encoded records 800 for authenticity. The obtaining of the group of blockchain-encoded records 800 includes accessing one or both of a primary market and a secondary market. Accessing the primary market includes

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obtaining blockchain-encoded records for longevity-contingent instruments directly from initial policyholders (e.g., originally insured individuals). Accessing the secondary market includes obtaining further blockchain-encoded records for further longevity-contingent instruments from brokers and providers, where the blockchain-encoded records of longevity-contingent instruments have changed hands from the initial policyholders to one or more intermediaries (e.g., the brokers, etc.).

The accessing of the blockchain-encoded records **800** includes a series of sub-steps. A first sub-step includes identifying the multitude of available longevity-contingent instruments by one or more of issuing a solicitation message for longevity-contingent instrument information and receiving the longevity-contingent instrument information. For example, the processing module **44** issues a solicitation message to one or more of the user devices **32-1** through **32-N**, and in response, receives primary market blockchain-encoded records **802**. As another example, the processing module **44** issues the solicitation message to one or more of the longevity-contingent instrument provider servers **704-1** through **704-M**, and in response, receives at least one of secondary market blockchain-encoded records **804-1** through **804-M**. Alternatively, the processing module **44** receives the blockchain-encoded records **800** in an unsolicited fashion.

The analyzing of the group of blockchain-encoded records **800** for authenticity includes utilizing a symmetric key signature approach or another approach including a straightforward signature verification. When utilizing the symmetric key signature approach, the processing module **44** decrypts a first signature of a first blockchain-encoded record of the blockchain-encoded records **800** utilizing a first public key of a first public-private key pair to produce a first decrypted transaction hash value. The first public-private key pair is associated with a last transaction computing device (e.g., a computing device associated with a last transfer of ownership of the associated longevity-contingent instrument).

Having produced the first decrypted transaction hash value, the processing module **44** hashes a portion of the first blockchain-encoded record utilizing a second public key of a second public-private key pair to produce a candidate transaction hash value. The second public-private key pair is associated with the computing device (e.g., generated by the computing device). Having produced the candidate transaction hash value, the processing module **44** establishes the authenticity indicator **806** to indicate favorable authenticity when the first decrypted transaction hash value compares favorably to the candidate transaction hash value.

When not utilizing the symmetric key signature approach, the processing module **44** applies signature verification to the first signature of the first blockchain-encoded record utilizing the first public key and the second public key to produce the authenticity indicator. The authentication is discussed in greater detail with reference to FIG. **6C**.

FIG. **6C** further illustrates the example of operation of steps of the method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments in accordance with a securely passing process, where, blockchain-encoded records are utilized to securely represent longevity-contingent instruments. In particular, a blockchain of blockchain-encoded records is utilized to record transactions and updates associated with a particular longevity-contingent instrument. For instance, a new blockchain is created when a life insurance policy is initially created by an associated insurance provider and sold to the originally

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insured. As another instance, the blockchain is updated when the life insurance policy is sold by the originally insured in the primary market to a second owner. As yet another instance, the blockchain is updated when life insurance policy is sold by the second owner to a third owner.

Each block of the blockchain includes various fields associated with the blockchain and a transaction field that includes content associated with the corresponding life insurance policy. The content includes one or more of insured name, a longevity status (e.g., living, deceased), policy terms (e.g., initial purchase price, death benefit, premium payment information), insured health records, an estimated life expectancy, a net present value, a current owner, a current holder (e.g., a fiduciary associated with the current owner), and insurance company information. Further information is included as is discussed with reference to FIG. **6D**.

The example blockchain includes blocks **2-4**. Each block includes a header section and a transaction section. The header section includes one or more of a nonce, a hash of a preceding block of the blockchain, where the preceding block was under control of a preceding computing device (e.g., a computing device of a seller) in a chain of control of the blockchain, and a hash of a current block (e.g., a current transaction section). The current block is under control of a current computing device in the chain of control of the blockchain.

The transaction section includes one or more of a public key of the current computing device, a signature of the preceding computing device, request information regarding a record request and change of control from the preceding computing device to the current computing device, and content information from the previous block as received by the previous computing device plus content added by the previous computing device when transferring the current block to the current computing device.

The example further includes computing devices **2-3** (e.g., devices #2 and #3) to facilitate illustration of generation of the blockchain. Each computing device includes a hash function, a signature function, and storage for a public/private key pair generated by the device.

An example of operation of the generating of the blockchain, when the device **2** has control of the blockchain and is passing control of the blockchain to the device **3** (e.g., the device **3** is transacting a transfer of content from device **2**), the device **2** obtains the device **3** public key from device **3**, performs a hash function **2** over the device **3** public key and the transaction **2** to produce a hashing resultant (e.g., preceding transaction to device **2**) and performs a signature function **2** over the hashing resultant utilizing a device **2** private key to produce a device **2** signature.

Having produced the device **2** signature, the device **2** generates the transaction **3** to include the device **3** public key, the device **2** signature, device **3** record request to device **2** information, and the previous content plus content from device **2**. The device **3** record request to device **2** information includes one or more of the actual record request, a query request, background content, and routing instructions from device **3** to device **2** for access to the content. The previous content plus content from device **2** includes one or more of content from an original source, content from any subsequent source after the original source, an identifier of a source of content, a serial number of the content, an expiration date of the content, content utilization rules, and results of previous blockchain validations.

Having produced the transaction **3** section of the block **3** a processing module (e.g., of the device **2**, of the device **3**,

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of a transaction mining computing entity, of a computing device), generates the header section by performing a hashing function over the transaction section 3 to produce a transaction 3 hash, performing the hashing function over the preceding block (e.g., block 2) to produce a block 2 hash. The performing of the hashing function may include generating a nonce such that when performing the hashing function to include the nonce of the header section, a desired characteristic of the resulting hash is achieved (e.g., a desired number of zero's).

Having produced the block 3, the device 2 sends the block 3 to the device 3, where the device 3 initiates control of the blockchain. Having received the block 3, the device 3 validates the received block 3. The validating includes one or more of verifying the device 2 signature over the preceding transaction section (e.g., transaction 2) and the device 3 public key utilizing the device 2 public key (e.g., a re-created signature function result compares favorably to device 2 signature) and verifying that an extracted device 3 public key of the transaction 3 compares favorably to the device 3 public key held by the device 3. The device 3 considers the received block 3 validated when the verifications are favorable (e.g., the authenticity of the associated content is trusted). For instance, the device considers the records intact, valid, and usable to facilitate determination of selection for the set of longevity-contingent instruments.

FIG. 6D further illustrates the example of operation of steps of the method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, having produce the authenticity indicator 806, in a fourth step, when the authenticity indicator for the group of blockchain-encoded records is favorable (e.g., authentic), the processing module 44 selects the first longevity-contingent instrument 722 and the second longevity-contingent instrument 724 based on the rive approach 682 to include in a set of longevity-contingent instruments (e.g., the portfolio). The set of longevity-contingent instruments is associated with a fair market acquisition value (e.g., purchase price based on current status where a common ownership entity owns both the face value benefit and the premium payment stream). The selecting includes a series of sub-steps. The processing module maintains records of the plurality of longevity-contingent instruments as longevity-contingent instrument information 660 within the database 30.

A first sub-step of the series of sub-steps includes extracting first characterization information 808 from the first blockchain-encoded record for the first longevity-contingent instrument to include one or more of a first estimated timeframe for payout of the first face value benefit, a present value of the first face value benefit utilizing the first estimated timeframe, and a present value of the first premium payment stream. A second sub-step includes extracting second characterization information 810 from the second blockchain-encoded record for the second longevity-contingent instrument to include one or more of a second estimated timeframe for payout of the second face value benefit, a present value of the second face value benefit utilizing the second estimated timeframe, and a present value of the second premium payment stream.

A third sub-step includes selecting the first longevity-contingent instrument 722 and the second longevity-contingent instrument 724 to include in the set of longevity-contingent instruments when the first characterization information 808 and the second characterization information 810 compare favorably to the rive approach requirements 714 associated with the rive approach 682. For example, the

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first and second longevity-contingent instruments provide an estimated favorable outcome aligned with the rive approach requirements 714.

Having selected the first and second longevity-contingent instruments, in a fifth step of the method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments, the processing module 44 generates selection information 812 for subsequent updating of the blockchain-encoded records 800 (e.g., to document transfer of ownership and a payment amount). The selection information is generated to include one or more of an identifier of a benefactor computing device associated with the benefit entity, an identifier of a debtor computing device associated with the sponsor entity, an identifier of an associated blockchain-encoded record, an identifier of an associated longevity-contingent instrument, a current purchase transaction value, an ownership entity identifier, a holder identifier, an updated life expectancy value, an updated longevity status indicator, and an identifier of another longevity-contingent instrument of the set of longevity-contingent instruments.

FIG. 6E further illustrates the example of operation of steps of the method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, having generated the selection information 812, in a sixth step, the processing module 44 updates the first blockchain-encoded record for the first longevity-contingent instrument 722 and a second blockchain-encoded record for the second longevity-contingent instrument 724 to include the selection information 812. The group of blockchain-encoded records 800 includes the first and second blockchain-encoded records. The processing module maintains records of the plurality of longevity-contingent instruments as longevity-contingent instrument information 660 within the database 30.

The updating of a blockchain-encoded record includes a series of sub-steps. In a first sub-step the processing module 44 hashes the selection information 812 utilizing a recipient public key of a recipient computing device to produce a next transaction hash value. In a second sub-step the processing module 44 encrypts the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. In a third sub-step the processing module 44 generates a next blockchain-encoded record to include the selection information 812 and the next transaction signature.

Having updated the blockchain-encoded records, in a seventh step of the method for the generating of the portfolio of blockchain-encoded rived longevity-contingent instruments, the processing module 44 rives the first and second longevity-contingent instruments in accordance with the rive approach 682 to produce sub-assets and sub-liabilities. For example, the processing module 44 rives the first longevity-contingent instrument 722 in accordance with the rive approach 682 to reassign the first face value benefit from the first ownership entity to the benefit entity to produce a first sub-asset 728 of a plurality of sub-assets of the set of longevity-contingent instruments. As another example, the processing module 44 further rives the first longevity-contingent instrument 722 in accordance with the rive approach 682 to reassign the first premium payment stream from the first ownership entity to the sponsor entity to produce a first sub-liability 730 of a plurality of sub-liabilities of the set of longevity-contingent instruments.

The plurality of sub-assets is associated with a benefit net present value and the plurality of sub-liabilities is associated with a liability net present value. A beneficial valuation elevation is created such that a sum of the benefit net present

value and the liability net present value is greater than the fair market acquisition value so that the benefit entity and sponsor entity realize the beneficial valuation elevation over direct utilization of selected longevity-contingent instruments of the set of longevity-contingent instruments prior to the riving.

As yet another example of the riving, the processing module 44 rives the second longevity-contingent instrument 724 in accordance with the rive approach 682 to reassign the second face value benefit from the second ownership entity to the benefit entity to produce a second sub-asset 732 of the plurality of sub-assets of the set of longevity-contingent instruments. The processing module 44 further rives the second longevity-contingent instrument 724 in accordance with the rive approach 682 to reassign the second premium payment stream from the second ownership entity to the sponsor entity to produce a second sub-liability 734 of the plurality of sub-liabilities of the set of longevity-contingent instruments. Having produced the plurality of sub-assets and the plurality of sub-liabilities, the processing module 44 stores the sub-assets and the plurality of sub-liabilities as sub-asset information 690 and sub-liability information 726 in the database 30.

The method described above module can alternatively be performed by various modules of the communication system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the communication system 10, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 7A-7C are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for utilizing a portfolio of blockchain-encoded rived longevity-contingent instruments within a computing system. The computing system includes data sources 26-1 through 26-N, a payer computing device 850, the transactional server 18 of FIG. 1, a benefactor computing device 852, and a debtor computing device 854.

In an embodiment, the payer computing device 850 is implemented utilizing the augmentation server 24 FIG. 1. In an embodiment, the benefactor computing device 852 and the debtor computing device 854 are implemented utilizing legacy server 22 of FIG. 1. In an embodiment, the data sources 26-1 through 26-N are implemented utilizing the data source 26 of FIG. 1. The transactional server 18 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1.

FIG. 7A illustrates an example of operation of steps of a method for the utilizing of the portfolio of blockchain-encoded rived longevity-contingent instruments where, in a first step, the processing module 44 obtains a first blockchain-encoded record 864 representing a first longevity-contingent instrument 722. When an insured person passes and a death benefit is provided, availability of a benefit payout is utilized to fund a combination of a cash flow to the benefactor computing device 852, for a benefit entity, and for at least some of a plurality of premium payment streams on behalf of the debtor computing device 854, of a sponsor entity, from the payer computing device 850 in accordance with a rive approach 682. The first blockchain-encoded record 864 includes a notification of the death benefit.

The obtaining includes receiving one or more blockchain-encoded records 860-1 through 860-N from one or more of the data sources 26-1 through 26-N. The obtaining further includes receiving a blockchain-encoded record 862 from the payer computing device 850 when the payer computing device 850 issues the notification of the death benefit (e.g., the life insurance company issues the notice).

Having obtained the first blockchain-encoded record representing the first longevity-contingent instrument 722, a second step of the method for the utilizing of the portfolio of blockchain-encoded rived longevity-contingent instruments includes the processing module 44 verifying authenticity of the first blockchain-encoded record 864 representing the first longevity-contingent instrument 722 of a portfolio of longevity-contingent instruments to produce a verified first blockchain-encoded record. The processing module maintains records of the portfolio of longevity-contingent instruments as longevity-contingent instrument information 660 within the database 30. The portfolio of longevity-contingent instruments is associated with a fair market acquisition value.

The first longevity-contingent instrument 722 is selected and rived in accordance with a rive approach 682 to reassign a first face value benefit from a first ownership entity to the benefit entity to produce a first sub-asset (e.g., death benefit) of a plurality of sub-assets of the portfolio of longevity-contingent instruments. The first longevity-contingent instrument 722 is further selected and rived in accordance with the rive approach 682 to reassign a first premium payment stream from the first ownership entity to the sponsor entity to produce a first sub-liability of a plurality of sub-liabilities of the portfolio of longevity-contingent instruments.

The plurality of sub-assets is associated with a benefit net present value and the plurality of sub-liabilities is associated with a liability net present value. The selecting and riving creates a beneficial valuation elevation such that a sum of the benefit net present value and the liability net present value is greater than the fair market acquisition value.

The verifying of the authenticity includes utilizing a symmetric key signature approach or another approach (e.g., straightforward signature verification). When utilizing the symmetric key signature approach, the processing module 44 decrypts a first signature of the first blockchain-encoded record 864 utilizing a first public key of a first public-private key pair to produce a first decrypted transaction hash value. The first public-private key pair is associated with a last transaction computing device (e.g., a computing device associated with generating the death notification).

Having produced the first decrypted transaction hash value, the processing module 44 hashes a portion of the first blockchain-encoded record utilizing a second public key of a second public-private key pair to produce a candidate transaction hash value. The second public-private key pair is associated with the computing device (e.g., generated by the computing device). Having produced the candidate transaction hash value, the processing module 44 indicates favorable authenticity when the first decrypted transaction hash value compares favorably to the candidate transaction hash value.

When not utilizing the symmetric key signature approach, the processing module 44 applies signature verification to the first signature of the first blockchain-encoded record utilizing the first public key and the second public key to produce the authenticity indicator. The verifying of the authenticity was previously discussed in greater detail with reference to FIG. 6C.

FIG. 7B further illustrates the example of operation of steps of the method for the utilizing of the portfolio of blockchain-encoded rived longevity-contingent instruments where, having verified the authenticity of the first blockchain-encoded record **864**, in a third step, the processing module **44** determines that the first longevity-contingent instrument **722** is associated with an available and unfulfilled benefit status by at least one of several approaches. A first approach includes interpreting the first longevity-contingent instrument **722** to identify a first death-notification of a first insured person identifier. The first insured person identifier is associated with the first longevity-contingent instrument **722**. A second approach includes interpreting the first longevity-contingent instrument **722** to identify the unfulfilled benefit status of the first longevity-contingent instrument **722**. A third approach includes accessing the longevity-contingent instrument information **660** from the database **30** to extract a plurality of insured person identifiers of the plurality of longevity-contingent instruments and identifying the first insured person identifier within the plurality of insured person identifiers.

Having determined that the first longevity-contingent instrument **722** is associated with the available and unfulfilled benefit status, a fourth step of the method for utilizing of the portfolio of blockchain-encoded rived longevity-contingent instruments includes the processing module **44** determining fulfillment information **866** for the first longevity-contingent instrument **722**. The fulfillment information **866** includes a benefit payout **868** of the first sub-asset facilitated by the payer computing device **850** for the benefit entity.

The fulfillment information **866** includes a variety of one or more elements. The elements include an identifier of the computing device, an identifier of the benefactor computing device **852** associated with the benefit entity, an identifier of the debtor computing device **854** associated with the sponsor entity, and an identifier of the payer computing device **850**. The elements of the fulfillment information **866** further includes a request for the payment of the benefit payout **868**, a current purchase transaction value, the benefit payout **868**, and a fulfillment status of the benefit payout **868**.

The elements of the fulfillment information **866** further includes an ownership entity identifier, a holder identifier, an insured person identifier, an identifier of an associated blockchain-encoded record, an identifier of an associated longevity-contingent instrument, a health record, and an updated life expectancy value. The elements of the fulfillment information **866** further includes a death-notification of the insured person identifier, an updated longevity status indicator, and an identifier of another longevity-contingent instrument associated with the first longevity-contingent instrument **722**.

The determining of the fulfillment information **866** includes at least one of a variety of approaches. A first approach includes determining the benefit payout associated with the first sub-asset. A second approach includes generating a request for the payment of the benefit payout. A third approach includes determining a first portion of the benefit payout to associate with a premium cash escrow in accordance with the rive approach **682**. The premium cash escrow is utilized to fund payment of a plurality of premium payment streams associated with the plurality of sub-liabilities of the portfolio of longevity-contingent instruments on behalf of the sponsor entity.

A third approach includes determining a second portion of the benefit payout to associate with a benefit cash account based on the first portion of the payout and in accordance

with the rive approach **682**. The benefit cash account is associated with the benefit entity (e.g., one or more benefactors) associated with the benefactor computing device **852**.

FIG. 7C further illustrates the example of operation of steps of the method for the utilizing of the portfolio of blockchain-encoded rived longevity-contingent instruments where, having produced the fulfillment information **866**, in a fifth step, the processing module **44** updates the first blockchain-encoded record **864** for the first longevity-contingent instrument **722** based on security information (e.g., key pair information) of the payer computing device **850** to include the fulfillment information **866** to produce an updated first blockchain-encoded record **872**.

The updating of the first blockchain-encoded record **864** includes a series of sub-steps. In a first sub-step the processing module **44** hashes the fulfillment information **866** utilizing a recipient public key of a recipient computing device (e.g., of the payer computing device **850**) to produce a next transaction hash value. In a second sub-step the processing module **44** encrypts the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. In a third sub-step the processing module **44** generates a next blockchain-encoded record to include the fulfillment information **866** and the next transaction signature.

Having produced the updated first blockchain-encoded record **872**, in a sixth step of the method of the utilizing of the portfolio of blockchain-encoded rived longevity-contingent instruments, the processing module **44** sends the updated first blockchain-encoded record **872** to the payer computing device **850** to facilitate payment of the benefit payout **868** of the first sub-asset to the benefit entity. The benefit entity and sponsor entity realize the beneficial valuation elevation over direct utilization of selected longevity-contingent instruments of the portfolio of longevity-contingent instruments prior to the riving. The facilitating of the payment includes generating a still further updated representation of the first blockchain-encoded record to include confirmation of payment.

Alternatively, or in addition to, the processing module **44** sends a representation of the updated first blockchain-encoded record **872** to one or more of the benefactor computing device **852** and the debtor computing device **854**. For instance, the processing module **44** further updates the updated first blockchain-encoded record **872** based on security information of at least one of the benefactor computing device **852** and the debtor computing device **854** to include the fulfillment information **866** to produce a further updated first blockchain-encoded record as the representation of the updated first blockchain-encoded record. Having produced the representation, the processing module **44** sends the representation as one or more of an updated first blockchain-encoded record **874** to the benefactor computing device **852** and as an updated first blockchain-encoded record **876** to the debtor computing device **854**.

FIGS. 8A-8D are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for updating a portfolio of blockchain-encoded rived longevity-contingent instruments within a computing system. The computing system includes a benefactor server **700**, a debtor server **702**, user devices **32-1** through **32-N**, longevity-contingent instrument provider servers **704-1** through **704-M**, and the control server **20** of FIG. 1.

In an embodiment, the benefactor server **700** and the debtor server **702** are implemented utilizing the legacy

server **22** of FIG. **1**, where the benefactor server **700** is associated with at least one benefit entity (e.g., pension system) and the debtor server **702** is associated with at least one sponsor entity associated with the at least one benefit entity. In an embodiment, the user devices **32-1** through **32-N** are implemented utilizing the user devices **32** of FIG. **1**.

FIG. **8A** illustrates an example of operation of steps of a method for the updating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, in a first step, the processing module **44** determines to update a set of longevity-contingent instruments (e.g., an existing portfolio of blockchain-encoded rived longevity-contingent instruments).

A first longevity-contingent instrument is further rived in accordance with the rive approach **682** to reassign a first premium payment stream of the first longevity-contingent instrument from the first ownership entity to a sponsor entity to produce a first sub-liability of a plurality of sub-liabilities of the set of longevity-contingent instruments.

The processing module **44** determines to update the set of longevity-contingent instruments utilizing one or more of a variety of approaches including determining that a sum of the benefit net present value and the liability net present value associated with the set of longevity-contingent instruments is less than a low threshold.

FIG. **8B** further illustrates the example of operation of steps of the method for the updating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, in a second step, the processing module **44** verifies authenticity of a blockchain-encoded record **882** representing a second longevity-contingent instrument **724** to produce an authenticity indicator **806**. The second longevity-contingent instrument assigns a second face value benefit of the second longevity-contingent instrument and a second premium payment stream of the second longevity-contingent instrument to a second ownership entity.

The verifying of the authenticity includes obtaining the blockchain-encoded record **882** and analyzing the record for authenticity. The obtaining of the blockchain-encoded record **882** includes accessing one or both of a primary market and a secondary market which includes a series of sub-steps. A first sub-step includes identifying one or more available longevity-contingent instruments by one or more of issuing a solicitation message for longevity-contingent instrument information and receiving the longevity-contingent instrument information. For example, the processing module **44** issues a solicitation message to one or more of the user devices **32-1** through **32-N**, and in response, receives primary market blockchain-encoded records **802**. As another example, the processing module **44** issues the solicitation message to one or more of the longevity-contingent instrument provider servers **704-1** through **704-M**, and in response, receives at least one of secondary market blockchain-encoded records **804-1** through **804-M**. Alternatively, the processing module **44** receives the blockchain-encoded record **882** in an unsolicited fashion.

The analyzing of the blockchain-encoded record **882** for authenticity includes utilizing a symmetric key signature approach or another approach including a straightforward signature verification. The authentication was discussed in greater detail with reference to FIG. **6C**.

FIG. **8C** further illustrates the example of operation of steps of the method for the updating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, having verified the authenticity of the blockchain-encoded record **882** to produce the authenticity indicator

806, in a third step, when the authenticity indicator for the blockchain-encoded record is favorable (e.g., authentic), the processing module **44** determines to include the second longevity-contingent instrument **724** in the set of longevity-contingent instruments to produce an updated set of longevity-contingent instruments in a series of sub-steps.

A first sub-step includes extracting characterization information **884** from the blockchain-encoded record **882** for the second longevity-contingent instrument **724** to include one or more of an estimated timeframe for payout of the second face value benefit, a present value of the second face value benefit utilizing the estimated timeframe, and a present value of the second premium payment stream.

A second sub-step includes indicating to include the second longevity-contingent instrument **724** in the set of longevity-contingent instruments to produce the updated set of longevity-contingent instruments when the characterization information **884** compares favorably to rive approach requirements **714** associated with the rive approach **682**.

Having determined to produce the updated set of longevity-contingent instruments, in a fourth step of the example method, the processing module **44** generates selection information **812** for subsequent updating of the blockchain-encoded records **800** (e.g., to document transfer of ownership and a payment amount).

FIG. **8D** further illustrates the example of operation of steps of the method for the updating of the portfolio of blockchain-encoded rived longevity-contingent instruments where, having produced the selection information **812**, in a fifth step, the processing module **44** updates the blockchain-encoded record **882** for the second longevity-contingent instrument to include the selection information **812**.

Having updated the blockchain-encoded record **882**, in a sixth step of the method for the updating of the portfolio of blockchain-encoded rived longevity-contingent instruments, the processing module **44** rives the second longevity-contingent instrument **724** in accordance with the rive approach **682** to reassign the second face value benefit from the second ownership entity to the benefit entity to produce a second sub-asset **732** of the plurality of sub-assets of the updated set of longevity-contingent instruments, and to reassign the second premium payment stream from the second ownership entity to the sponsor entity to produce a second sub-liability **734** of the plurality of sub-liabilities of the updated set of longevity-contingent instruments.

Having produced the plurality of sub-assets and the plurality of sub-liabilities, the processing module **44** stores the sub-assets and the plurality of sub-liabilities as sub-asset information **690** and sub-liability information **726** in the database **30**. A beneficial valuation elevation is created such that a sum of the benefit net present value and the liability net present value is greater than the fair market acquisition value so that the benefit entity and sponsor entity realize the beneficial valuation elevation over direct utilization of selected longevity-contingent instruments of the updated set of longevity-contingent instruments prior to the riving.

FIGS. **9A-9C** are schematic block diagrams of another embodiment of a communication system illustrating an embodiment of a method for utilizing blockchain-encoded records for rived longevity-contingent instruments within a computing system. The computing system includes a benefactor server **700**, a debtor server **702**, data sources **26-1** through **26-N**, a payer computing device **850**, and the transactional server **18** of FIG. **1**.

In an embodiment, the payer computing device **850** is implemented utilizing the augmentation server **24** FIG. **1**. In an embodiment, the data sources **26-1** through **26-N** are

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implemented utilizing the data source **26** of FIG. **1**. The transactional server **18** includes the processing module **44** of FIG. **1** and the database **30** of FIG. **1**.

In an embodiment, the benefactor server **700** and the debtor server **702** are implemented utilizing the legacy server **22** of FIG. **1**, where the benefactor server **700** is associated with at least one benefit entity (e.g., pension system) and the debtor server **702** is associated with at least one sponsor entity associated with the at least one benefit entity.

FIG. **9A** illustrates an example of operation of steps of a method for the utilizing blockchain-encoded records for rived longevity-contingent instruments where, in a first step, the processing module **44** obtains a first blockchain-encoded record **864** representing a first longevity-contingent instrument **722** of a set of longevity-contingent instruments. When an insured person passes and a death benefit is provided, availability of a benefit payout is utilized to fund a combination of a cash flow to a benefit entity and for at least some of a plurality of premium payment streams on behalf of a sponsor entity, from the payer computing device **850** in accordance with a rive approach **682**. The first blockchain-encoded record **864** includes a notification of the death benefit.

The obtaining includes receiving one or more blockchain-encoded records **860-1** through **860-N** from one or more of the data sources **26-1** through **26-N**. The obtaining further includes receiving a blockchain-encoded record **862** from the payer computing device **850** when the payer computing device **850** issues the notification of the death benefit (e.g., the life insurance company issues the notice).

Having obtained the first blockchain-encoded record **864**, a second step of the method includes the processing module **44** verifying authenticity of the first blockchain-encoded record **864** representing the first longevity-contingent instrument **722** to produce a verified first blockchain-encoded record. The transactional server **18** maintains valuation information **880** within the database **30** to include the fair market acquisition value.

The first longevity-contingent instrument **722** is selected and rived in accordance with the rive approach **682** to reassign a first face value benefit of the first longevity-contingent instrument from a first ownership entity to the benefit entity to produce a first sub-asset of a plurality of sub-assets of the set of longevity-contingent instruments. The transactional server **18** maintains sub-asset information **690** within the database **30** to include information with regards to the plurality of sub-assets.

The first longevity-contingent instrument **722** is further rived in accordance with the rive approach **682** to reassign a first premium payment stream of the first longevity-contingent instrument from the first ownership entity to the sponsor entity to produce a first sub-liability of a plurality of sub-liabilities of the set of longevity-contingent instruments. The transactional server **18** further maintains the valuation information **880** to include the liability net present value. The transactional server **18** maintains sub-liability information **726** within the database **30** to include information with regards to the plurality of sub-liabilities.

The verifying of the authenticity of the first blockchain-encoded record **864** includes utilizing a symmetric key signature approach or another approach (e.g., straightforward signature verification). The verifying of the authenticity was previously discussed in greater detail with reference to FIG. **6C**.

Having verified the authenticity of the first blockchain-encoded record **864**, when the first longevity-contingent

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instrument **722** is associated with an available and unfulfilled benefit status, a third step of the method for the utilizing blockchain-encoded records for rived longevity-contingent instruments includes the processing module **44** determining fulfillment information **863** of the first longevity-contingent instrument. The processing module **44** determines that the first longevity-contingent instrument **722** is associated with an available and unfulfilled benefit status by at least one of several approaches.

A first approach includes interpreting the first longevity-contingent instrument **722** to identify a first death-notification of a first insured person identifier. The first insured person identifier is associated with the first longevity-contingent instrument **722**. A second approach includes interpreting the first longevity-contingent instrument **722** to identify the unfulfilled benefit status of the first longevity-contingent instrument **722**.

Having determined that the first longevity-contingent instrument **722** is associated with the available and unfulfilled benefit status, the processing module **44** determines the fulfillment information **863** for the first longevity-contingent instrument **722**. The fulfillment information **863** includes a benefit payout of the first sub-asset facilitated by the payer computing device **850** for the benefit entity.

The fulfillment information **863** includes a variety of one or more elements. The elements include an identifier of the computing device, an identifier of the benefactor server **700** associated with the benefit entity, an identifier of the debtor server **702** associated with the sponsor entity, and an identifier of the payer computing device **850**.

The elements of the fulfillment information **863** further includes an ownership entity identifier, a holder identifier, an insured person identifier, an identifier of an associated blockchain-encoded record, an identifier of an associated longevity-contingent instrument, a health record, and an updated life expectancy value. The elements of the fulfillment information **863** further includes a death-notification of the insured person identifier, an updated longevity status indicator, and an identifier of another longevity-contingent instrument associated with the first longevity-contingent instrument **722**. The determining of the fulfillment information **863** further includes at least one of a variety of approaches including determining the benefit payout associated with the first sub-asset.

FIG. **9B** further illustrates the example of operation of steps of the method for the utilizing blockchain-encoded records for rived longevity-contingent instruments where, having determined the fulfillment information **863**, in a fourth step the processing module **44** verifies authenticity of an asset blockchain-encoded record **900** representing the plurality of sub-assets to produce a verified asset blockchain-encoded record **904**.

The verifying of the authenticity of the asset blockchain-encoded record **900** includes utilizing a symmetric key signature approach or another approach (e.g., straightforward signature verification). The verifying of the authenticity of blocks of blockchains such as the asset blockchain-encoded record **900** was previously discussed in greater detail with reference to FIG. **6C**.

Having produced the verified asset blockchain-encoded record **904**, a fifth step of the example of operation of the method for the utilizing blockchain-encoded records for rived longevity-contingent instruments includes the processing module **44** verifying authenticity of a liability blockchain-encoded record **902** representing the plurality of sub-liabilities to produce a verified liability blockchain-encoded record **906**.

The verifying of the authenticity of the liability blockchain-encoded record **902** includes utilizing the symmetric key signature approach or the other approach (e.g., straight-forward signature verification). The verifying of the authenticity of blocks of blockchains such as the liability blockchain-encoded record **902** was previously discussed in greater detail with reference to FIG. 6C.

FIG. 9C further illustrates the example of operation of steps of the method for the utilizing blockchain-encoded records for rived longevity-contingent instruments where, having produced the verified asset blockchain-encoded record **904** and the verified liability blockchain-encoded record **906**, in a sixth step the processing module **44** facilitates exclusion of the first longevity-contingent instrument from the set of longevity-contingent instruments in accordance with the fulfillment information **863**. The facilitating of the exclusion includes the processing module **44** excluding the first sub-asset from the plurality of sub-assets to produce an updated plurality of sub-assets.

The facilitating of the exclusion further includes the processing module **44** excluding the first sub-liability from the plurality of sub-liabilities to produce an updated plurality of sub-liabilities.

The facilitating of the exclusion further includes the processing module **44** updating the verified asset blockchain-encoded record **904** to represent the updated plurality of sub-assets and updating the verified liability blockchain-encoded record **906** to represent the updated plurality of sub-liabilities. The updating of the verified asset blockchain-encoded record **904** includes a series of sub-steps. A first sub-step includes generating asset transaction content **912** to include one or more of a variety of elements. The elements include information regarding the fulfillment information **863**, information regarding a second sub-asset, information regarding the first sub-asset, information regarding the updated plurality of sub-assets, an identifier of an owner computing device associated with an ownership entity, and an identifier of a benefactor computing device associated with the benefit entity. The elements further include an identifier of a debtor computing device associated with the sponsor entity, an identifier of an associated blockchain-encoded record, an identifier of an associated longevity-contingent instrument, a current purchase transaction value, and an ownership entity identifier. The elements further include a holder identifier, an updated life expectancy value, an updated longevity status indicator, and an identifier of another longevity-contingent instrument of the set of longevity-contingent instruments.

A second sub-step of the series of sub-steps includes hashing the asset transaction content **912** utilizing a recipient public key of a recipient computing device (e.g., of the benefactor server **700** or of the transactional server **18**) to produce a next transaction hash value. A third sub-step includes encrypting the next transaction hash value utilizing a private key of the transactional server **18** to produce a next transaction signature. A fourth sub-step includes generating a next blockchain-encoded record to include the asset transaction content **912** and the next transaction signature.

The updating of the verified liability blockchain-encoded record **906** includes another series of sub-steps. A first sub-step includes generating liability transaction content **914** to include one or more of a variety of elements. The elements include information regarding the fulfillment information **863**, information regarding a second sub-liability, information regarding the first sub-liability, information regarding the updated plurality of sub-liabilities, the identifier of the owner computing device associated with the

ownership entity, and the identifier of the benefactor computing device associated with the benefit entity.

A second sub-step of the other series of sub-steps includes hashing the liability transaction content **914** utilizing a recipient public key of a recipient computing device (e.g., of the debtor server **702** or of the transactional server **18**) to produce another next transaction hash value. A third sub-step includes encrypting the other next transaction hash value utilizing the private key of the transactional server **18** to produce another next transaction signature. A fourth sub-step includes generating another next blockchain-encoded record to include the liability transaction content **914** and the other next transaction signature.

Having updated the verified asset blockchain-encoded record **904** and the verified liability blockchain-encoded record **906**, the processing module **44** facilitates sharing of the updates. For example, the processing module **44** sends, via the network **28** of FIG. 1, the verified asset blockchain-encoded record **904** to the benefactor server **700**. As another example, the processing module **44** sends, via the network **28** of FIG. 1, the verified liability blockchain-encoded record **906** to the debtor server **702**.

FIGS. 10A-10E are schematic block diagrams of an embodiment of a computing system illustrating an embodiment of a method for utilizing a contingent action token within the computing system. The computing system includes the benefactor server **700** of FIG. 5A, the debtor server **702** of FIG. 5A, the user devices **32-1** through **32-N** of FIG. 5A, the longevity-contingent instrument provider servers **704-1** through **704-M** of FIG. 5A, the control server **20** of FIG. 1, and the transactional server **18** of FIG. 1.

The control server **20** includes the processing module **44** of FIG. 1 and the database **30** of FIG. 1. The benefactor server **700** may be further associated with an outcome recipient identifier **1128** as further discussed below. The debtor server **702** may be further associated with an obligation provider identifier **1146** as further discussed below. The longevity-contingent instrument provider servers **704-1** through **704-M** may further be associated with one or more of an outcome provider identifier **1124** and an obligation recipient identifier **1150** as further discussed below. The transactional server **18** includes the processing module **44** of FIG. 1 and the database **30** of FIG. 1 and functions as a blockchain node of a blockchain associated with an object distributed ledger.

FIG. 10A illustrates an example of the method for utilizing the contingent action token, where a first step of the example method includes the control server **20** identifying a non-fungible token (NFT) of an object distributed ledger that meets minimum NFT requirements. Example content of the contingent action token **1000** is discussed in greater detail with reference to FIG. 10B.

The minimum NFT requirements include a variety of requirements. A first requirement includes a recovered cryptographic token value **1112** of the NFT matches a calculated cryptographic token value of the NFT. A second requirement includes an obligation provider identifier **1146** of the NFT is the same as an original obligation provider identifier of the NFT when the NFT was initially generated. The obligation provider identifier **1146** is associated with providing of an obligation **1148** tied to an obligation recipient identifier **1150** in accordance with obligation requirements **1142**. A positive obligation status **1144** indicates that the obligation **1148** has historically been provided in accordance with the obligation requirements **1142**.

A second requirement includes an outcome recipient identifier **1128** of the NFT is the same as an original outcome

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recipient identifier of the NFT when the NFT was initially generated. The outcome recipient identifier **1128** is associated with a result of a triggered outcome **1126** tied to an outcome provider identifier **1124** in accordance with contingent outcome rules **1118** and the positive obligation status **1144**. A contingency status **1120** indicates whether the triggered outcome **1126** has been triggered for a contingency entity identifier **1122** in accordance with the contingent outcome rules **1118**.

The identifying the NFT of the object distributed ledger that meets the minimum NFT requirements includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** determining whether to indirectly or directly access the object distributed ledger. For example, the processing module **44** determines to indirectly access the object distributed ledger when the blockchain **1004** is not available in the database **30** of the control server **20** and is available in the database **30** of the transactional server **18**. As another example, the processing module **44** of the control server **20** determines to directly access the object distributed ledger when the blockchain **1004** is available within the database **30** of the control server **20**.

When indirectly accessing the object distributed ledger, a second sub-step of the identifying of the NFT includes the processing module **44** of the control server **20** issuing a non-fungible token access request to the transactional server **18** (e.g., an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger). The non-fungible token access request includes a representation of the minimum NFT requirements. The second sub-step further includes the processing module **44** of the control server **20** extracting an NFT identifier of the NFT from a non-fungible token access response that includes the contingency action token **1000** from the transactional server **18**, where the transactional server **18** accesses the object distributed ledger **1002** to identify and NFT block **3** of the blockchain as illustrated in FIG. **10A**.

When directly accessing the object distributed ledger, the second sub-step of the identifying of the NFT includes the processing module **44** of the control server **20** obtaining a copy of the object distributed ledger (e.g., recover the blockchain **1004** from the database **30** of the control server **20**). The second sub-step further includes the processing module **44** of the control server **20** indicating the NFT identifier of the NFT when detecting a block of the copy of the object distributed ledger that matches the representation of the minimum NFT requirements (e.g., original obligation provider identifier and original outcome recipient identifier).

Having obtained the NFT, a third sub-step of the identifying of the NFT that meets the minimum NFT requirements includes the processing module **44** of the control server **20** verifying that the recovered cryptographic token value **1112** of the NFT matches the calculated cryptographic token value of the NFT. For example, the processing module **44** produces the calculated cryptographic token value of the NFT over a corresponding content portion of the obtained NFT and compares that calculated value to the recovered cryptographic token value **1112**. When the cryptographic token values are the same, the processing module **44** indicates that the NFT that meets the minimum NFT requirements has been identified (e.g., by the token identifier **1110**).

FIG. **10B** illustrates an example of elements of the contingency action token **1000** of the example of the method for utilizing the contingent asset token. A token identifier **1110** uniquely identifies the contingency action token **1000**. A cryptographic token value **1112** includes one or more of public keys, a cryptographic signature over a portion of the

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token, a nonce for utilization in a blockchain, and one or more hash values over portions of this token or another of a shared blockchain.

The token further includes blockchain linking information **1114** such as token identifiers and/or links for other tokens of the shared blockchain. A contingency type **1116** indicates a contingency category for the token (e.g., where the contingency, when met, activates the triggered outcome **1126**). Contingent outcome rules **1118** set forth how the activation of the triggered outcome is properly enabled. For example, activation only when the obligation status **1144** is positive and a trigger occurs associated with the contingency type.

A contingency status **1120** indicates whether the contingency has been triggered (e.g., yes or no). A contingency entity identifier (ID) **1122** specifies a unique identifier of an entity associated with the contingency, where the contingent outcome rules dictate what happens when the contingency trigger occurs associated with the contingency entity identifier.

An outcome provider ID **1124** indicates an identifier associated with providing of the triggered outcome when activated. The triggered outcome **1126** occurs when activated and in accordance with the contingent outcome rules. An outcome recipient ID **1128** is associated with receiving of the triggered outcome **1126**.

Obligation requirements **1142** indicate an obligation and timing by an obligation provider identifier to an obligation recipient identifier. The obligation status **1144** indicates historically whether the obligation has been maintained as associated with the obligation provider ID **1146**. The obligation **1148** indicates metrics and/or descriptors associated with an obligation provided in association with the obligation provider ID to an association of the obligation recipient ID **1150**.

The contingent action token **1000** further includes elements associated with evaluating portions of the token in accordance with an evaluation profile **1138** (e.g., risk assumptions, historical probabilities and risks of obligations and outcomes). Major portions of the token include the obligation and the triggered outcome. An obligation evaluation **1156** evaluates the obligation by utilizing an obligation disruption risk assessment **1152** that portrays likelihood of disruption of an obligation stream when the obligation includes a series of sub component obligations. The obligation evaluation is further based on a probability of obligation met **1154** wrapping up a risk assessment of an association of the obligation provider ID faithfully providing the obligation to the association of the obligation recipient ID.

An outcome evaluation **1140** provides an evaluation of the triggered outcome based on an estimated obligation status at estimated outcome triggered date **1130** that portrays risk of a negative obligation status at an expected time of the trigger of the triggered outcome. The outcome evaluation is further based on estimating that outcome triggered date **1134** in accordance with parameters of estimated outcome triggered date **1132** (e.g., historical values of trigger date timing and variables correlated to the contingency entity identifier). The outcome evaluation is further based on an outcome provider probability to perform outcome **1136** that assesses whether an association of the outcome provider ID is able to provide the triggered outcome at the estimated outcome trigger date.

The evaluation of the token further includes a rollup of the obligation evaluation and the outcome evaluation in a combined evaluation of combination of outcome and obligation **1158**. The combined evaluation provides an all-in preponderance of a present evaluation based on an estimated future that concludes with the trigger outcome. Risks associated

with both the obligation and the triggered outcome drive the combined evaluation such that changing of the obligation provider ID can result in an unexpected improvement in the combined evaluation. For example, the outcome evaluation **1140** can rise to an improved evaluation when the obligation provider ID is changed to another obligation provider ID that is associated with a higher probability of obligation met **1154**. With that, the estimated obligation status at estimated outcome triggered date **1130** improves driving up the outcome evaluation **1140**.

A series of related invention embodiments pertain to finding change outs of one or more of the obligation provider ID and the outcome recipient ID to provide an improved combined evaluation. As a result, a beneficial evaluation increase is provided. As a specific example, the token is originally set up with an original obligation provider ID that is the same as the contingency entity ID, an original outcome recipient ID, and an outcome provider ID that matches the obligation recipient ID. Later, the token is updated to replace the obligation provider ID with another obligation provider ID associated with a higher probability of providing the obligation (e.g., that may or may not include the contingency entity ID). As a result, the outcome evaluation **1140** rises to an improved and unexpected level of the invention embodiment.

FIG. **10C** further illustrates the example of the method for utilizing the contingent asset token, where having identify the NFT that meets the minimum NFT requirements, a second step of the example method includes the control server **20** determining whether to select the NFT based on an evaluation of the NFT with regards to an evaluation profile **1138**. A beneficial improvement to the triggered outcome results from a hypothetical change to the obligation provider identifier of the selected NFT. The evaluation of the NFT includes producing the obligation evaluation **1156** based on the evaluation profile **1138**. The obligation evaluation **1156** compares the providing of the obligation in accordance with the obligation requirements (e.g., historical and estimated future). The evaluation of the NFT further includes the outcome evaluation **1140**. The outcome evaluation **1140** estimates the triggered outcome based on the evaluation profile **1138** (e.g., using risks and historical correlations of the evaluation profile).

The determining whether to select the NFT based on the evaluation of the NFT with regards to the evaluation profile includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** determining the evaluation profile to include a required evaluation performance improvement level. For example, the processing module **44** extracts the evaluation profile **1138** from the recovered NFT. As another example, the processing module **44** generates the evaluation profile based on an external input through a user interface (e.g., a request that includes a specific evaluation performance improvement level).

A second sub-step includes the processing module **44** of the control server **20** comparing the providing of the obligation associated with the original obligation provider identifier to providing of the obligation associated with a hypothetical obligation provider identifier in accordance with the evaluation profile to produce the obligation evaluation. For example, the processing module **44** receives identity and risk levels with regards to a new obligation provider identifier and utilizes that to produce the obligation evaluation (e.g., an improved outcome when a risk level of noncompliance to the obligation is lower for the new obligation provider identifier).

A third sub-step includes the processing module **44** of the control server **20** comparing an estimated trigger outcome when the providing of the obligation is associated with the original obligation provider identifier to a hypothetical estimated trigger outcome when the providing of the obligation is associated with the hypothetical obligation provider identifier in accordance with the evaluation profile to produce the outcome evaluation. For example, the processing module **44** produces the outcome evaluation for the new obligation provider (e.g., with the lower risks of not meeting the obligation) and for an estimated trigger date of the triggered outcome.

A fourth sub-step includes the processing module **44** of the control server **20** indicating to select the NFT when the obligation evaluation and the outcome evaluation satisfies the required evaluation performance improvement level. For example, the processing module **44** indicates to select the NFT when the new obligation provider identifier is a catalyst to meet the required evaluation performance improvement level.

FIG. **10D** further illustrates the example of the method for utilizing the contingent asset token, where having selected the NFT to produce a selected NFT, a third step of the example method includes determining reassignment information for the selected NFT. The reassignment information includes a confirmed change of at least one of the obligation provider identifier **1146** and the outcome recipient identifier **1128**.

The determining the reassignment information for the selected NFT includes a variety of approaches. A first approach includes identifying a hypothetical obligation provider identifier based on the evaluation of the selected NFT with regards to the evaluation profile to establish a new obligation provider identifier of the reassignment information. For example, the processing module **44** of the control server **20** replaces or adds a hypothetical obligation provider identifier to the obligation provider identifier **1146** based on a previous analysis of the evaluation of the selected NFT. Acquisition of the hypothetical obligation provider identifier includes extracting the identifier from confirmation information **1170** received from another computing device and extraction from content **1006** of the selected NFT.

A second approach to determine the reassignment information includes, when requested, the processing module **44** of the control server **20** modifying the outcome recipient identifier to establish a new outcome recipient identifier of the reassignment information. For example, the processing module **44** extracts the new outcome recipient identifier from the confirmation information **1170**.

Having determined the reassignment information, a fourth step of the example method of operation includes the processing module **44** of the control server **20** facilitating taking control of the selected NFT of a blockchain of the object distributed ledger as discussed with reference to FIG. **6C**. For example, the processing module **44** exchanges control grant information **1172** with a current controlling entity of the selected NFT to gain the control. For instance, the current controlling entity generates a new block that includes a public key of the processing module **44**, the control request from the processing module **44**, and a cryptographic signature over a portion of the block utilizing a private key of the current controlling entity.

FIG. **10E** further illustrates the example of the method for utilizing the contingent asset token, where having taking control of the selected NFT, a fifth step of the example method includes the processing module **44** of the control server **20** updating the selected NFT utilizing the reassign-

ment information for the NFT to produce an updated NFT **1180**. The updating the selected NFT utilizing the reassignment information for the selected NFT to produce the updated NFT **1180** includes a series of sub-steps. A first sub-step includes the processing module **44** obtaining the selected NFT. For example, the processing module **44** recovers the selected NFT from the blockchain **1004** of the database **30** of the control server **20** when a local copy of the blockchain is up-to-date. As another example, the processing module **44** requests the selected NFT from the transactional server **18**.

A second sub-step includes the processing module **44** replacing corresponding elements of the selected NFT with at least a portion of the reassignment information to produce the updated NFT **1180**. For example, the processing module **44** replaces the obligation provider identifier **1146** with a new obligation provider identifier such that the beneficial valuation level increase is realized subsequently for the triggered outcome upon triggering and when conditions of the contingent outcome rules have been satisfied.

Having produced the updated NFT **1180**, a sixth step of the example method of operation includes the processing module **44** of the control server **20** causing generation of a new block affiliated with the updated NFT via the blockchain of the object distributed ledger. The new block includes the updated NFT.

The causing generation of the new block affiliated with the updated NFT via the blockchain of the object distributed ledger includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** determining whether to indirectly or directly update the object distributed ledger as previously discussed (e.g., using a local copy of the blockchain when available).

When indirectly updating the object distributed ledger, a second sub-step includes the processing module **44** issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the updated NFT **1180**. For example, the processing module **44** of the control server **20** sends the updated NFT **1180** to the transactional server **18** such that the transactional server **18** places an update 1 block **3-1** on the blockchain connected to the NFT as illustrated in FIG. **10E**.

When directly updating the object distributed ledger, the second sub-step includes the processing module **44** obtaining a copy of the object distributed ledger. For example, the processing module **44** recovers the object distributed ledger from the blockchain **1004** of the database **30** of the control server **20**. Having obtained the copy of the object distributed ledger, the second sub-step further includes hashing content of the updated NFT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value. For example, the control module **44** generates a hash value utilizing a cryptographic algorithm over the content of the updated NFT using a public key in possession of receiving entities for subsequent access of the blockchain to produce the next transaction hash value.

Having produced the next transaction hash value, the second sub-step further includes encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. For example, the processing module **44** utilizes a cryptographic encryption algorithm to encrypt the next transaction hash value utilizing the private key of the processing module **44** to produce the next transaction signature.

Having produced the next transaction signature, the second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the content of the updated NFT and the next transaction signature. For example, the processing module **44** populates the cryptographic token value **1112** of the next block with the next transaction signature and populates all the other content fields with the content of the updated NFT to produce the next block.

Having produced the next block, the second sub-step further includes causing inclusion of the next block as the new block in the object distributed ledger. For example, the processing module **44** of the control server **20** adds the block **3-1** as the update 1 associated with the selected NFT on the object distributed ledger **1002** as illustrated in FIG. **10E** by updating the blockchain **1004** and the database **30** of the control server **20** and/or the transactional server **18** and other transactional servers serving as blockchain nodes supporting the object distributed ledger.

Alternatively, or in addition to, the processing module **44** of the control server **20** sends the updated NFT **1182** one or more of the benefactor server **700**, the debtor server **702**, one or more of the user devices **32-1** through **32-N**, and one or more of the longevity-contingent instrument provider servers **704-1** through **704-M**.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. **11A-11B** are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for generating a contingent action token within the computing system. The computing system includes the benefactor server **700** of FIG. **5A**, the debtor server **702** of FIG. **5A**, the user devices **32-1** through **32-N** of FIG. **5A**, accreditation authority servers **1200-1** through **1200-M**, the control server **20** of FIG. **1**, and the transactional server **18** of FIG. **1**. In an embodiment, the accreditation authority server is affiliated with a corresponding longevity-contingent instrument provider server **704** of FIG. **5A**. The control server **20** includes the processing module **44** of FIG. **1** and the database **30** of FIG. **1**. The transactional server **18** includes the processing module **44** FIG. **1** and the database **30** of FIG. **1** and functions as a blockchain node of a blockchain **1004** associated with an object distributed ledger **1002**.

FIG. **11A** illustrates an example method of operation for generating the contingent action token, where a first step of the example method includes the control server **20** interpreting a request to generate a non-fungible token (NFT) **1202** as a contingency action token for the object distributed ledger **1002** to produce baseline content of content **1006** that includes an obligation provider identifier (ID) **1146** and an outcome recipient ID **1128**. The obligation provider ID is associated with providing of an obligation tied to an obligation recipient ID **1150** in accordance with obligation requirements **1142**. The outcome recipient ID **1128** is associated with a result of a triggered outcome **1126** tied to an

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outcome provider ID **1124** in accordance with contingent outcome rules and an obligation status. A positive obligation status indicates that the obligation has historically been provided in accordance with the obligation requirements **1142**. The format of the contingent action token **1000** is discussed in greater detail with reference to FIG. **10B**.

The baseline content for the contingency action token further includes a contingency entity identifier (ID) **1122**, an outcome provider ID **1124**, a value of a triggered outcome **1126**, the outcome recipient ID **1128**, the obligation requirements **1142**, the obligation provider ID **1146**, and the obligation recipient ID **1150**. The interpreting the request to generate the NFT to produce the baseline content includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** extracting, from the request to generate the NFT **1202**, one or more of the contingency entity ID **1122**, the outcome provider ID **1124**, the value of the result of the triggered outcome **1126**, the obligation requirements **1142**, and the obligation recipient ID **1150**.

A second sub-step includes the processing module **44** determining the outcome recipient ID **1128** based on the request to generate the NFT **1202** and benefactor information (e.g., which entity or entities is to receive the result of the triggered outcome **1126**). For example, the processing module **44** interprets the benefactor information to identify an entity that is to receive the result of the triggered outcome and to establish an identifier of the entity as the outcome recipient ID **1128**.

A third sub-step includes the processing module **44** determining the obligation provider ID **1146** based on the request to generate the NFT and debtor information (e.g., which entity or entities are to provide, at least in part, the obligation in accordance with the obligation requirements **1142**). For example, the processing module **44** interprets the debtor information to identify an entity that is to provide the obligation and to establish an identifier of the entity as the obligation provider ID **1146**.

Having produced the baseline content, a second step of the example method of operation includes the processing module **44** verifying with an accreditation authority computing device of the computing system, validity of the baseline content. In an embodiment, one or more of the accreditation authority servers **1200-1** through **1200-M** provide the accreditation authority computing device.

The verifying the validity of the baseline content includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** identifying the accreditation authority computing device based on a first identified corresponding accreditation authority associated with the obligation recipient ID **1150**. For example, the control server **20** accesses the database **30** to retrieve identity of the accreditation authority server **1200-1** that is affiliated with the obligation provider ID.

A second sub-step includes the processing module **44** obtaining baseline validation information **1204** from the accreditation authority computing device for the baseline content. For example, the processing module **44** issues at least some of the baseline content to the accreditation authority server **1200-1** and receives the baseline validation information **1204** in response.

A third sub-step includes the processing module **44** indicating that the baseline content is valid when the baseline validation information is substantially the same as the baseline content. For example, the processing module **44** compares the received baseline validation information **1204** to the baseline content and indicates that the baseline content

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is valid when the comparison indicates a match of at least a minimum set of items for matching (e.g., outcome provider ID **1124**, and obligation recipient ID **1150**).

FIG. **11B** further illustrates the example method of operation of the generating of the contingent action token, where having verified the baseline content as valid, a third step includes the processing module **44** of the control server **20** determining whether a beneficial improvement is provided to the result of the triggered outcome **1126** when a hypothetical change to the obligation provider ID **1146** is made. The hypothetical change includes utilizing a different obligation provider ID with a different risk profile as compared to a baseline risk profile associated with a present obligation provider ID.

The determining whether the beneficial improvement is provided includes producing an obligation evaluation based on an evaluation profile. The obligation evaluation compares the providing of the obligation in accordance with the obligation requirements with the hypothetical change to the obligation provider ID. The determining further includes producing an outcome evaluation. The outcome evaluation estimates the result of the triggered outcome based on the evaluation profile when the hypothetical change to the obligation provider ID is made. For example, an improvement to the result of the triggered outcome may occur when a lower risk profile of the hypothetical change to the obligation provider ID is implemented.

The determining whether the beneficial improvement is provided to the result of the triggered outcome when the hypothetical change to the obligation provider ID is made includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** generating an obligation disruption risk assessment **1152** for the (present) obligation provider ID **1146** based on the evaluation profile to produce a baseline obligation evaluation. For example, the processing module **44** analyzes risks associated with obligation fulfillment associated with the present obligation provider ID.

A second sub-step includes the processing module **44** generating another obligation disruption risk assessment for the hypothetical change to the obligation provider ID based on the evaluation profile to produce a hypothetical baseline obligation evaluation. For example, the processing module **44** analyzes risks associated with the obligation fulfillment associated with the hypothetical change to the obligation provider ID.

A third sub-step includes the processing module **44** generating a baseline outcome evaluation to include an estimate of the result of the triggered outcome utilizing the baseline obligation evaluation. For example, the processing module **44** computes the result of the triggered outcome when obligation risks associated with the present obligation provider ID are considered.

A fourth sub-step includes the processing module **44** generating the outcome evaluation to include another estimate of the result of the triggered outcome utilizing the hypothetical baseline obligation evaluation. For example, the processing module **44** computes another result of the triggered outcome when obligation risks associated with the hypothetical change to the obligation provider ID are considered.

A fifth sub-step includes the processing module **44** indicating that the beneficial improvement is provided to the result of the triggered outcome when the hypothetical change to the obligation provider ID is made when the outcome evaluation is greater than the baseline outcome evaluation. For example, the processing module **44** com-

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compares the results of the triggered outcome and indicates the beneficial improvement is provided when the triggered outcome is greater utilizing the hypothetical change to the obligation provider ID (e.g., due to lowered risks).

Having determined the beneficial improvement when the baseline content as valid, a fourth step of the example method of operation includes the processing module 44 of the control server 20 establishing NFT content based on the outcome evaluation to include the baseline content. The establishing the NFT content based on the outcome evaluation to include the baseline content includes a series of sub-steps. A first sub-step includes the processing module 44 generating the NFT content to include the baseline content. For example, the processing module 44 includes the contingency entity ID 1122, the outcome provider ID 1124, the triggered outcome 1126, the outcome recipient ID 1128, the obligation requirements 1142, the obligation provider ID 1146, and the obligation recipient ID 1150 in the NFT content.

A second sub-step includes the processing module 44 updating the NFT content to replace the obligation provider ID 1146 with an updated obligation provider ID associated with the hypothetical change to the obligation provider ID when the beneficial improvement is provided to the result of the triggered outcome from the hypothetical change to the obligation provider ID.

Having established the NFT content, a fifth step of the example method of operation includes the processing module 44 of the control server 20 causing generation of a new block affiliated with the NFT via the blockchain 1004 of the object distributed ledger 1002 as discussed with reference to FIG. 6C. The new block includes the NFT content. The causing generation of the new block includes a series of sub-steps. A first sub-step includes the processing module 44 determining whether to indirectly or directly update the object distributed ledger as previously discussed (e.g., using a local copy of the blockchain when available).

When indirectly updating the object distributed ledger, a second sub-step includes the processing module 44 of the control server 20 issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the NFT 1210. For example, the processing module 44 sends the updated NFT 1210 to the transactional server 18 such that the transactional server 18 places a NFT3 block 3 on the blockchain connected to the NFT 2 as illustrated in FIG. 11B.

When directly updating the object distributed ledger, the second sub-step includes the processing module 44 obtaining a copy of the object distributed ledger. For example, the processing module 44 recovers the object distributed ledger from the blockchain 1004 of the database 30 of the control server 20. Having obtained the copy of the object distributed ledger, the second sub-step further includes hashing NFT content of the NFT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value. For example, the control module 44 generates a hash value utilizing a cryptographic algorithm over the NFT content of the NFT using a public key in possession of receiving entities for subsequent access of the blockchain to produce the next transaction hash value.

Having produced the next transaction hash value, the second sub-step further includes encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. For example, the processing module 44 utilizes a cryptographic encryption algorithm to encrypt the next transaction hash value

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utilizing the private key of the processing module 44 to produce the next transaction signature.

Having produced the next transaction signature, the second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the NFT content of the NFT and the next transaction signature. For example, the processing module 44 populates the cryptographic token value of the next block with the next transaction signature and populates all the other content fields with the content of the NFT to produce the next block.

Having produced the next block, the second sub-step further includes causing inclusion of the next block as the new block in the object distributed ledger. For example, the processing module 44 of the control server 20 adds the block 3 on the object distributed ledger 1002 as illustrated in FIG. 11B by updating the blockchain 1004 and the database 30 of the control server 20 and/or the transactional server 18 and other transactional servers serving as blockchain nodes supporting the object distributed ledger.

Alternatively, or in addition to, the processing module 44 of the control server 20 sends the NFT 1210 to one or more of the benefactor server 700, the debtor server 702, one or more of the user devices 32-1 through 32-N, and one or more of the accreditation authority servers 1200-1 through 1200-M.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 12A-12B are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for utilizing a contingent action token within the computing system. The computing system includes the benefactor server 700 of FIG. 5A, the debtor server 702 of FIG. 5A, the user devices 32-1 through 32-N of FIG. 5A, accreditation authority servers 1200-1 through 1200-M, the control server 20 of FIG. 1, and the transactional server 18 of FIG. 1. In an embodiment, the accreditation authority server is affiliated with a corresponding longevity-contingent instrument provider server 704 of FIG. 5A. The control server 20 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1. The transactional server 18 includes the processing module 44 FIG. 1 and the database 30 of FIG. 1 and functions as a blockchain node of a blockchain 1004 associated with an object distributed ledger 1002.

FIG. 12A illustrates an example of the method for utilizing the contingent action token, where a first step of the example method includes the control server 20 interpreting a contingency status message from another computing device that indicates the change of contingency status for a contingency entity identifier (ID) 1122 common to the contingency status message and a first contingency action token. For example, the processing module 44 of the control server 20 receives contingency status 1120 from of the benefactor server 700, the debtor server 702, the user devices 32-1 through 32-N and the accreditation authority servers 1200-1 through 1200-M.

Having interpreted the contingency status message that indicates the change of contingency status, a second step of the example method of operation includes obtaining a first block **1209** of the blockchain from the object distributed ledger based on the contingency entity ID. For example, the processing module **44** of the control server **20** recovers the first block **1209** (e.g., NFT block **3** as illustrated in FIG. **12A**) that includes a matching contingency entity ID from the blockchain **1004** of the object distributed ledger **1002**.

Having produced the change of contingency status and obtained the first block, the second step of the further example method of operation includes obtaining, in accordance with a securely passing process, control over the first block of the blockchain of the object distributed ledger in response to the change of contingency status of the first block. The first block is associated with a first contingency action token of a multitude of contingency action tokens.

The first contingency action token includes an obligation provider identifier (ID) and an outcome recipient ID **1128**. The obligation provider ID is associated with providing of an obligation tied to an obligation recipient ID in accordance with obligation requirements. The outcome recipient ID is associated with a result of a triggered outcome tied to an outcome provider ID in accordance with contingent outcome rules and an obligation status **1144**. A positive obligation status indicates that the obligation has historically been provided in accordance with the obligation requirements. Only a device possessing control over the first block of the blockchain may modify the first block of the blockchain.

The securely passing process includes obtaining control as either an originator of the first block and being authorized as to possess control from a current entity that possesses the control. The current owner includes a public key of the next owner in a transaction section of the block and generates the signature over the transaction section utilizing both the private key of the current owner device and the public key of the next owner device as discussed in greater detail with reference to FIG. **6C**.

The obtaining, in accordance with the securely passing process, the control over the first block includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** receiving an indication of the control over the first block from a requesting computing entity. For example, the processing module **44** receives contingency status **1120**, where the request includes the first block that includes a private key associated with the benefactor server **700**.

A second sub-step includes establishing the identity of the computing device (e.g., the control server **20**) to have the control over the first block of the blockchain. For example, the processing module **44** modifies the first block **2** include the identity of the control server **20** as having the control over the first block.

Having obtained control over the first block, a third step of the example method of operation includes determining whether the triggered outcome **1126** is valid for the first contingency action token based on the change of contingency status **1120** of the first block. The determining whether the triggered outcome **1126** is valid for the first contingency action token based on the change of contingency status **1120** of the first block includes a series of sub-steps. A first sub-step includes verifying that the outcome recipient ID **1128** is associated with the result of the triggered outcome **1126** tied to the outcome provider ID **1124** in accordance with contingent outcome rules **1118**. For example, the processing module **44** of the control server **20** matches the outcome recipient ID of the contingency status

1120 to the outcome recipient ID of the first block and matches the triggered outcome to the outcome provider ID.

A second sub-step includes verifying that the positive obligation status indicates that the obligation has historically been provided in accordance with the obligation requirements. For example, the processing module **44** accesses the obligation status of the first block for the contingency action token to verify that the obligation has been historically made in a positive fashion in accordance with the obligation requirements (e.g., completed timely transfer of monthly payment obligations).

FIG. **12B** further illustrates the example method of operation of the utilization of the contingency action token, where, having determined that the triggered outcome is valid, a fourth step includes the control server **20**, when the triggered outcome is valid for the first contingency action token, updating the first contingency action token to indicate that the triggered outcome is valid for the first contingency action token to produce an updated first contingency action token. For example, the processing module **44** of the control server **20** sets a flag in the first block **2** indicate that the triggered outcome is valid.

Having produced the updated first contingency action token, a fifth step of the example method of operation includes the control server **20** generating, by the computing device, a new block for the blockchain of the object distributed ledger in accordance with the securely passing process to represent the updated first contingency action token. The generating the new block includes a series of sub-steps.

A first sub-step includes the processing module **44** determining whether to indirectly or directly update the object distributed ledger as previously discussed (e.g., using a local copy of the blockchain when available).

When indirectly updating the object distributed ledger, a second sub-step includes the processing module **44** of the control server **20** issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the new block **1211**. For example, the processing module **44** sends the updated new block **1211** to the transactional server **18** such that the transactional server **18** places a block **3-1** on the blockchain connected to the NFT **3** as illustrated in FIG. **12B**.

When directly updating the object distributed ledger, the second sub-step includes the processing module **44** obtaining a copy of the object distributed ledger. For example, the processing module **44** recovers the object distributed ledger from the blockchain **1004** of the database **30** of the control server **20**. Having obtained the copy of the object distributed ledger, the second sub-step further includes hashing content of the updated first contingency action token utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value. For example, the control module **44** generates a hash value utilizing a cryptographic algorithm over the content of the updated first contingency action token using a public key in possession of receiving entities for subsequent access of the blockchain to produce the next transaction hash value.

Having produced the next transaction hash value, the second sub-step further includes encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. For example, the processing module **44** utilizes a cryptographic encryption algorithm to encrypt the next transaction hash value utilizing the private key of the processing module **44** to produce the next transaction signature.

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Having produced the next transaction signature, the second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the content of the updated first contingency action token and the next transaction signature. For example, the processing module 44 populates the cryptographic token value of the next block with the next transaction signature and populates all the other content fields with the content of the updated first contingency action token to produce the next block.

Having produced the next block, the second sub-step further includes causing inclusion of the next block as the new block in the object distributed ledger. For example, the processing module 44 of the control server 20 adds the block 3-1 on the object distributed ledger 1002 as illustrated in FIG. 12B by updating the blockchain 1004 and the database 30 of the control server 20 and/or the transactional server 18 and other transactional servers serving as blockchain nodes supporting the object distributed ledger.

Alternatively, or in addition to, the processing module 44 of the control server 20 sends the new block 1211 to one or more of the benefactor server 700, the debtor server 702, one or more of the user devices 32-1 through 32-N, and one or more of the accreditation authority servers 1200-1 through 1200-M. Alternatively or in addition to, having generated the new block, in a sixth step of the example method of operation the control server 20 securely passes control over the new block via the object distributed ledger to another computing device when the new block is further processed to complete the triggered outcome. For example, the processing module 44 of the control server 20 utilizes the process discussed with reference to FIG. 6C to pass control of the new block to the other computing device when the other computing device is responsible to complete payment to the outcome recipient ID 1128 for the triggered outcome 1126.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 13A-13D are schematic block diagrams of an embodiment of a computing system illustrating an embodiment of a method for selecting a contingent action token within the computing system. The computing system includes the benefactor server 700 of FIG. 5A, the debtor server 702 of FIG. 5A, the user devices 32-1 through 32-N of FIG. 5A, the longevity-contingent instrument provider servers 704-1 through 704-M of FIG. 5A, the control server 20 of FIG. 1, and the transactional server 18 of FIG. 1.

The control server 20 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1. The benefactor server 700 may be further associated with an outcome recipient identifier 1128 as further discussed below. The debtor server 702 may be further associated with an obligation provider identifier 1146 as further discussed below. The longevity-contingent instrument provider servers 704-1 through 704-M may further be associated with one or more of an outcome provider identifier 1124 and an obligation recipient identifier 1150 as further discussed below. The

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transactional server 18 includes the processing module 44 FIG. 1 and the database 30 of FIG. 1 and functions as a blockchain node of a blockchain 1004 associated with an object distributed ledger 1002.

FIG. 13A illustrates an example of the method for selecting the contingent action token, 1000 where a first step of the example method includes the control server 20 identifying the contingency-action token (CAT) of the object distributed ledger 1002 that meets minimum CAT requirements. The CAT includes one of a non-fungible token (NFT) and another token linked to the NFT by the object distributed ledger when the CAT is not the NFT as illustrated in FIG. 10A. Example content of the contingent action token 1000 is discussed in greater detail with reference to FIG. 10B.

The minimum CAT requirements include a variety of requirements. A first requirement includes a recovered cryptographic token value 1112 of the CAT matches a calculated cryptographic token value of the CAT. A second requirement includes an obligation provider identifier record of the CAT that includes an original obligation provider identifier 1146 of the CAT when the NFT was initially generated. The obligation provider identifier 1146 is associated with providing of an obligation 1148 tied to an obligation recipient identifier 1150 in accordance with obligation requirements 1142. A positive obligation status 1144 indicates that the obligation 1148 has historically been provided in accordance with the obligation requirements 1142.

A third requirement includes an outcome recipient identifier record of the CAT that includes an original outcome recipient identifier of the NFT when the NFT was initially generated. The outcome recipient identifier 1128 is associated with a result of a triggered outcome 1126 tied to an outcome provider identifier 1124 in accordance with contingent outcome rules 1118 and the positive obligation status 1144. A contingency status 1120 indicates whether the triggered outcome 1126 has been triggered for a contingency entity identifier 1122 in accordance with the contingent outcome rules 1118.

The identifying the CAT of the object distributed ledger that meets the minimum CAT requirements includes a series of sub-steps. A first sub-step includes the processing module 44 of the control server 20 determining whether to indirectly or directly access the object distributed ledger. For example, the processing module 44 determines to indirectly access the object distributed ledger when the blockchain 1004 is not available in the database 30 of the control server 20 and is available in the database 30 of the transactional server 18. As another example, the processing module 44 of the control server 20 determines to directly access the object distributed ledger when the blockchain 1004 is available within the database 30 of the control server 20.

When indirectly accessing the object distributed ledger, a second sub-step of the identifying of the CAT includes the processing module 44 of the control server 20 issuing a CAT access request to the transactional server 18 (e.g., an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger). The CAT access request includes a representation of the minimum CAT requirements. The second sub-step further includes the processing module 44 of the control server 20 extracting a CAT identifier of the CAT from a CAT access response from the transactional server 18, where the transactional server 18 accesses the object distributed ledger 1002 to identify an NFT block 3 and/or a linked block 3-1 of the blockchain as the CAT as illustrated in FIG. 13A.

When directly accessing the object distributed ledger, the second sub-step of the identifying of the CAT includes the

processing module 44 of the control server 20 obtaining a copy of the object distributed ledger (e.g., recover the blockchain 1004 from the database 30 of the control server 20). The second sub-step further includes the processing module 44 of the control server 20 indicating the CAT identifier of the CAT when detecting a block of the copy of the object distributed ledger that matches the representation of the minimum CAT requirements (e.g., original obligation provider identifier and original outcome recipient identifier or a linked identifiers subsequent to the originals).

Having obtained the CAT, a third sub-step of the identifying of the CAT that meets the minimum CAT requirements includes the processing module 44 of the control server 20 verifying that the recovered cryptographic token value 1112 of the CAT matches the calculated cryptographic token value of the CAT. For example, the processing module 44 produces the calculated cryptographic token value of the CAT over a corresponding content portion of the obtained CAT and compares that calculated value to the recovered cryptographic token value 1112. When the cryptographic token values are the same, the processing module 44 indicates that the CAT that meets the minimum CAT requirements has been identified (e.g., by the token identifier 1110).

FIG. 13B further illustrates the example of the method for selecting the contingent asset token, where having identified the CAT that meets the minimum CAT requirements, a second step of the example method includes the control server 20 determining whether to select the CAT based on an evaluation of the CAT with regards to an evaluation profile 1138. A beneficial improvement to a present estimate of the triggered outcome results from a hypothetical change to the obligation provider identifier of the selected CAT. The evaluation of the CAT includes producing the obligation evaluation 1156 based on the evaluation profile 1138 (e.g., using risks and historical correlations of the evaluation profile). The obligation evaluation 1156 compares the providing of the obligation in accordance with the obligation requirements (e.g., historical and estimated future). The evaluation of the CAT further includes the outcome evaluation 1140. The outcome evaluation 1140 produces the present estimate of the triggered outcome based on the evaluation profile.

The determining whether to select the CAT based on the evaluation of the CAT with regards to the evaluation profile includes a series of sub-steps. A first sub-step includes the processing module 44 of the control server 20 determining the evaluation profile to include a required evaluation performance improvement level. For example, the processing module 44 extracts the evaluation profile 1138 from the recovered CAT. As another example, the processing module 44 generates the evaluation profile based on an external input through a user interface (e.g., a request that includes a specific evaluation performance improvement level).

A second sub-step includes the processing module 44 of the control server 20 comparing the providing of the obligation associated with the obligation provider identifier (e.g., original or subsequent) to providing of the obligation associated with a hypothetical obligation provider identifier in accordance with the evaluation profile to produce the obligation evaluation. For example, the processing module 44 receives identity and risk levels with regards to a new obligation provider identifier and utilizes that to produce the obligation evaluation (e.g., an improved outcome when a risk level of noncompliance to the obligation is lower for the new obligation provider identifier).

A third sub-step includes the processing module 44 of the control server 20 comparing an estimated trigger outcome

when the providing of the obligation is associated with the obligation provider identifier to a hypothetical estimated trigger outcome when the providing of the obligation is associated with the hypothetical obligation provider identifier in accordance with the evaluation profile to produce the outcome evaluation. For example, the processing module 44 produces the outcome evaluation for the new obligation provider (e.g., with the lower risks of not meeting the obligation) and for an estimated trigger date of the triggered outcome.

A fourth sub-step includes the processing module 44 of the control server 20 indicating to select the CAT when the obligation evaluation and the outcome evaluation satisfies the required evaluation performance improvement level. For example, the processing module 44 indicates to select the CAT when the new obligation provider identifier is a catalyst to meet the required evaluation performance improvement level.

FIG. 13C further illustrates the example of the method for selecting the contingent asset token, where having selected the CAT to produce a selected CAT, a third step of the example method includes determining reassignment information for the selected CAT. The reassignment information includes a confirmed change of at least one of the obligation provider identifier 1146 and the outcome recipient identifier 1128.

The determining the reassignment information for the selected CAT includes a variety of approaches. A first approach includes identifying a hypothetical obligation provider identifier based on the evaluation of the selected CAT with regards to the evaluation profile to establish a new obligation provider identifier of the reassignment information. For example, the processing module 44 of the control server 20 replaces or adds a hypothetical obligation provider identifier to the obligation provider identifier 1146 based on a previous analysis of the evaluation of the selected CAT. Acquisition of the hypothetical obligation provider identifier includes extracting the identifier from confirmation information 1170 received from another computing device and extraction from content 1006 of the selected CAT.

A second approach to determine the reassignment information includes, when requested, the processing module 44 of the control server 20 modifying the outcome recipient identifier to establish a new outcome recipient identifier of the reassignment information. For example, the processing module 44 extracts the new outcome recipient identifier from the confirmation information 1170.

Having determined the reassignment information, a fourth step of the example method of operation includes the processing module 44 of the control server 20 facilitating taking control of the selected CAT of a blockchain of the object distributed ledger as discussed with reference to FIG. 6C. For example, the processing module 44 exchanges control grant information 1172 with a current controlling entity of the selected CAT to gain the control. For instance, the current controlling entity generates a new block that includes a public key of the processing module 44, the control request from the processing module 44, and a cryptographic signature over a portion of the block utilizing a private key of the current controlling entity.

FIG. 13D further illustrates the example of the method for selecting the contingent asset token, where having taking control of the selected CAT, a fifth step of the example method includes the processing module 44 of the control server 20 updating the selected CAT utilizing the reassignment information for the CAT to produce an updated CAT 1181. The updating the selected CAT utilizing the reassign-

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ment information for the selected CAT to produce the updated CAT **1181** includes a series of sub-steps. A first sub-step includes the processing module **44** obtaining the selected CAT. For example, the processing module **44** recovers the selected CAT from the blockchain **1004** of the database **30** of the control server **20** when a local copy of the blockchain is up-to-date. As another example, the processing module **44** requests the selected CAT from the transactional server **18**.

A second sub-step includes the processing module **44** replacing corresponding elements of the selected CAT with at least a portion of the reassignment information to produce the updated CAT **1181**. For example, the processing module **44** replaces the obligation provider identifier **1146** with a new obligation provider identifier such that a beneficial valuation level increase is realized subsequently for the triggered outcome upon triggering and when conditions of the contingent outcome rules have been satisfied.

Having produced the updated CAT **1181**, a sixth step of the example method of operation includes the processing module **44** of the control server **20** causing generation of a new block affiliated with the updated CAT via the blockchain of the object distributed ledger. The new block includes the updated CAT.

The causing generation of the new block affiliated with the updated CAT via the blockchain of the object distributed ledger includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** determining whether to indirectly or directly update the object distributed ledger as previously discussed (e.g., using a local copy of the blockchain when available).

When indirectly updating the object distributed ledger, a second sub-step includes the processing module **44** issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the updated CAT **1181**. For example, the processing module **44** of the control server **20** sends the updated CAT **1181** to the transactional server **18** such that the transactional server **18** places an update 2 block 3-2 on the blockchain connected to the CAT as illustrated in FIG. 13D.

When directly updating the object distributed ledger, the second sub-step includes the processing module **44** obtaining a copy of the object distributed ledger. For example, the processing module **44** recovers the object distributed ledger from the blockchain **1004** of the database **30** of the control server **20**. Having obtained the copy of the object distributed ledger, the second sub-step further includes hashing content of the updated CAT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value. For example, the control module **44** generates a hash value utilizing a cryptographic algorithm over the content of the updated CAT using a public key in possession of receiving entities for subsequent access of the blockchain to produce the next transaction hash value.

Having produced the next transaction hash value, the second sub-step further includes encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. For example, the processing module **44** utilizes a cryptographic encryption algorithm to encrypt the next transaction hash value utilizing the private key of the processing module **44** to produce the next transaction signature.

Having produced the next transaction signature, the second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the

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content of the updated CAT and the next transaction signature. For example, the processing module **44** populates the cryptographic token value **1112** of the next block with the next transaction signature and populates all the other content fields with the content of the updated CAT to produce the next block.

Having produced the next block, the second sub-step further includes causing inclusion of the next block as the new block in the object distributed ledger. For example, the processing module **44** of the control server **20** adds the block 3-2 as an update 1 associated with the selected CAT on the object distributed ledger **1002** as illustrated in FIG. 13D by updating the blockchain **1004** and the database **30** of the control server **20** and/or the transactional server **18** and other transactional servers serving as blockchain nodes supporting the object distributed ledger.

Alternatively, or in addition to, the processing module **44** of the control server **20** sends the updated CAT **1181** to one or more of the benefactor server **700**, the debtor server **702**, one or more of the user devices **32-1** through **32-N**, and one or more of the longevity-contingent instrument provider servers **704-1** through **704-M**.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 14A-14B are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for establishing a contingent action token within the computing system. The computing system includes the benefactor server **700** of FIG. 5A, the debtor server **702** of FIG. 5A, the user devices **32-1** through **32-N** of FIG. 5A, accreditation authority servers **1200-1** through **1200-M**, the control server **20** of FIG. 1, and the transactional server **18** of FIG. 1. In an embodiment, the accreditation authority server is affiliated with a corresponding longevity-contingent instrument provider server **704** of FIG. 5A. The control server **20** includes the processing module **44** of FIG. 1 and the database **30** of FIG. 1. The transactional server **18** includes the processing module **44** FIG. 1 and the database **30** of FIG. 1 and functions as a blockchain node of a blockchain **1004** associated with an object distributed ledger **1002**.

FIG. 14A illustrates an example method of operation for generating the contingent action token, where a first step of the example method includes the control server **20** interpreting a request to generate a non-fungible token (NFT) **1202** as a contingency action token for the object distributed ledger **1002** to produce baseline content of content **1006** that includes an obligation provider identifier (ID) **1146** and an outcome recipient ID **1128**. The obligation provider ID is associated with providing of an obligation tied to an obligation recipient ID **1150** in accordance with obligation requirements **1142**. The outcome recipient ID **1128** is associated with a result of a triggered outcome **1126** tied to an outcome provider ID **1124** in accordance with contingent outcome rules and an obligation status. A positive obligation status indicates that the obligation has historically been

provided in accordance with the obligation requirements **1142**. The format of the contingent action token **1000** is discussed in greater detail with reference to FIG. **10B**.

The baseline content for the contingency action token further includes a contingency entity identifier (ID) **1122**, an outcome provider ID **1124**, a value of a triggered outcome **1126**, the outcome recipient ID **1128**, the obligation requirements **1142**, the obligation provider ID **1146**, and the obligation recipient ID **1150**. The interpreting the request to generate the NFT to produce the baseline content includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** extracting, from the request to generate the NFT **1202**, one or more of the contingency entity ID **1122**, the outcome provider ID **1124**, the value of the result of the triggered outcome **1126**, the obligation requirements **1142**, and the obligation recipient ID **1150**.

A second sub-step includes the processing module **44** determining the outcome recipient ID **1128** based on the request to generate the NFT **1202** and benefactor information (e.g., which entity or entities is to receive the result of the triggered outcome **1126**). For example, the processing module **44** interprets the benefactor information to identify an entity that is to receive the result of the triggered outcome and to establish an identifier of the entity as the outcome recipient ID **1128**.

A third sub-step includes the processing module **44** determining the obligation provider ID **1146** based on the request to generate the NFT and debtor information (e.g., which entity or entities are to provide, at least in part, the obligation in accordance with the obligation requirements **1142**). For example, the processing module **44** interprets the debtor information to identify an entity that is to provide the obligation and to establish an identifier of the entity as the obligation provider ID **1146**.

Having produced the baseline content, a second step of the example method of operation includes the processing module **44** verifying with an accreditation authority computing device of the computing system, validity of the baseline content. In an embodiment, one or more of the accreditation authority servers **1200-1** through **1200-M** provide the accreditation authority computing device.

The verifying the validity of the baseline content includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** identifying the accreditation authority computing device based on a first identified corresponding accreditation authority associated with the obligation recipient ID **1150**. For example, the control server **20** accesses the database **30** to retrieve identity of the accreditation authority server **1200-1** that is affiliated with the obligation provider ID.

A second sub-step includes the processing module **44** obtaining baseline validation information **1204** from the accreditation authority computing device for the baseline content. For example, the processing module **44** issues at least some of the baseline content to the accreditation authority server **1200-1** and receives the baseline validation information **1204** in response.

A third sub-step includes the processing module **44** indicating that the baseline content is valid when the baseline validation information is substantially the same as the baseline content. For example, the processing module **44** compares the received baseline validation information **1204** to the baseline content and indicates that the baseline content is valid when the comparison indicates a match of at least a minimum set of items for matching (e.g., outcome provider ID **1124**, and obligation recipient ID **1150**).

FIG. **14B** further illustrates the example method of operation of the establishing of the contingent action token, where having verified the baseline content as valid, a third step includes the processing module **44** of the control server **20** determining whether a beneficial improvement to a present estimate of the triggered outcome **1126** results from a hypothetical change to the obligation provider ID **1146**. The hypothetical change includes utilizing a different obligation provider ID with a different risk profile as compared to a baseline risk profile associated with a present obligation provider ID.

The determining whether the beneficial improvement to the present estimate of the triggered outcome from the hypothetical change to the obligation provider ID includes producing an obligation evaluation based on an evaluation profile. The obligation evaluation compares the providing of the obligation in accordance with the obligation requirements with the hypothetical change to the obligation provider ID. The determining further includes producing an outcome evaluation. The outcome evaluation is a present estimate of the triggered outcome based on the evaluation profile when the hypothetical change to the obligation provider ID is made. For example, an improvement to the result of the triggered outcome may occur when a lower risk profile of the hypothetical change to the obligation provider ID is implemented.

The determining whether the beneficial improvement is provided to the result of the triggered outcome when the hypothetical change to the obligation provider ID is made includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** generating an obligation disruption risk assessment **1152** for the (present) obligation provider ID **1146** based on the evaluation profile to produce a baseline obligation evaluation. For example, the processing module **44** analyzes risks associated with obligation fulfillment associated with the present obligation provider ID.

A second sub-step includes the processing module **44** generating another obligation disruption risk assessment for the hypothetical change to the obligation provider ID based on the evaluation profile to produce a hypothetical baseline obligation evaluation. For example, the processing module **44** analyzes risks associated with the obligation fulfillment associated with the hypothetical change to the obligation provider ID.

A third sub-step includes the processing module **44** generating a baseline outcome evaluation to include a present estimate of the result of the triggered outcome utilizing the baseline obligation evaluation. For example, the processing module **44** computes the result of the triggered outcome when obligation risks associated with the present obligation provider ID are considered in a present sense (e.g., value presently considering the risk).

A fourth sub-step includes the processing module **44** generating the outcome evaluation to include another present estimate of the result of the triggered outcome utilizing the hypothetical baseline obligation evaluation. For example, the processing module **44** computes another present result of the triggered outcome when obligation risks associated with the hypothetical change to the obligation provider ID are considered.

A fifth sub-step includes the processing module **44** indicating that the beneficial improvement is provided to the present estimate of the triggered outcome when the hypothetical change to the obligation provider ID is made when the outcome evaluation is greater than the baseline outcome evaluation. For example, the processing module **44** com-

compares the results of the triggered outcome and indicates the beneficial improvement to the present estimate is provided when the triggered outcome is greater utilizing the hypothetical change to the obligation provider ID (e.g., due to lowered risks).

Having determined the beneficial improvement when the baseline content as valid, a fourth step of the example method of operation includes the processing module 44 of the control server 20 establishing NFT content based on the outcome evaluation to include the baseline content. The establishing the NFT content based on the outcome evaluation to include the baseline content includes a series of sub-steps. A first sub-step includes the processing module 44 generating the NFT content to include the baseline content. For example, the processing module 44 includes the contingency entity ID 1122, the outcome provider ID 1124, the triggered outcome 1126, the outcome recipient ID 1128, the obligation requirements 1142, the obligation provider ID 1146, and the obligation recipient ID 1150 in the NFT content.

A second sub-step includes the processing module 44 updating the NFT content to replace the obligation provider ID 1146 with an updated obligation provider ID associated with the hypothetical change to the obligation provider ID when the beneficial improvement to the present estimate is provided to the result of the triggered outcome from the hypothetical change to the obligation provider ID.

Having established the NFT content, a fifth step of the example method of operation includes the processing module 44 of the control server 20 causing generation of a new block affiliated with the NFT via the blockchain 1004 of the object distributed ledger 1002 as discussed with reference to FIG. 6C. The new block includes the NFT content. The causing generation of the new block includes a series of sub-steps. A first sub-step includes the processing module 44 determining whether to indirectly or directly update the object distributed ledger as previously discussed (e.g., using a local copy of the blockchain when available).

When indirectly updating the object distributed ledger, a second sub-step includes the processing module 44 of the control server 20 issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the NFT 1210. For example, the processing module 44 sends the updated NFT 1210 to the transactional server 18 such that the transactional server 18 places a NFT 3 block 3 on the blockchain connected to the NFT 2 as illustrated in FIG. 14B.

When directly updating the object distributed ledger, the second sub-step includes the processing module 44 obtaining a copy of the object distributed ledger. For example, the processing module 44 recovers the object distributed ledger from the blockchain 1004 of the database 30 of the control server 20. Having obtained the copy of the object distributed ledger, the second sub-step further includes hashing NFT content of the NFT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value. For example, the control module 44 generates a hash value utilizing a cryptographic algorithm over the NFT content of the NFT using a public key in possession of receiving entities for subsequent access of the blockchain to produce the next transaction hash value.

Having produced the next transaction hash value, the second sub-step further includes encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. For example, the processing module 44 utilizes a cryptographic encryption

algorithm to encrypt the next transaction hash value utilizing the private key of the processing module 44 to produce the next transaction signature.

Having produced the next transaction signature, the second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the NFT content of the NFT and the next transaction signature. For example, the processing module 44 populates the cryptographic token value of the next block with the next transaction signature and populates all the other content fields with the content of the NFT to produce the next block.

Having produced the next block, the second sub-step further includes causing inclusion of the next block as the new block in the object distributed ledger. For example, the processing module 44 of the control server 20 adds the block 3 on the object distributed ledger 1002 as illustrated in FIG. 14B by updating the blockchain 1004 and the database 30 of the control server 20 and/or the transactional server 18 and other transactional servers serving as blockchain nodes supporting the object distributed ledger. Alternatively, or in addition to, the processing module 44 of the control server 20 sends the NFT 1210 to one or more of the benefactor server 700, the debtor server 702, one or more of the user devices 32-1 through 32-N, and one or more of the accreditation authority servers 1200-1 through 1200-M.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 15A-15B are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for generating a contingent action token within the computing system. In particular, a non-fungible token (NFT) form of the contingent action token is established. The computing system includes the benefactor server 700 of FIG. 5A, the debtor server 702 of FIG. 5A, the user devices 32-1 through 32-N of FIG. 5A, accreditation authority servers 1200-1 through 1200-M, the control server 20 of FIG. 1, and the transactional server 18 of FIG. 1. In an embodiment, the accreditation authority server is affiliated with a corresponding longevity-contingent instrument provider server 704 of FIG. 5A. The control server 20 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1. The transactional server 18 includes the processing module 44 of FIG. 1 and the database 30 of FIG. 1 and functions as a blockchain node of a blockchain 1004 associated with an object distributed ledger 1002.

FIG. 15A illustrates an example method of operation for generating the contingent action token, where a first step of the example method includes the control server 20 determining whether baseline content of content 1006 associated with a longevity-contingent instrument is qualified for a proposed non-fungible token (NFT) for an object distributed ledger 1002 that includes an obligation provider identifier (ID) 1146 and an outcome recipient ID 1128. The obligation provider ID is associated with providing of an obligation tied to an obligation recipient ID 1150 in accordance with obligation requirements 1142. The outcome recipient ID

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1128 is associated with a result of a triggered outcome 1126 tied to an outcome provider ID 1124 in accordance with contingent outcome rules and an obligation status. A positive obligation status indicates that the obligation has historically been provided in accordance with the obligation requirements 1142. The format of much of the baseline content is represented by the contingent action token 1000 is discussed in greater detail with reference to FIG. 10B.

The baseline content for the contingency action token further includes a contingency entity identifier (ID) 1122, an outcome provider ID 1124, a value of a triggered outcome 1126, the outcome recipient ID 1128, the obligation requirements 1142, the obligation provider ID 1146, and the obligation recipient ID 1150. The determining whether the baseline content associated with the longevity-contingent instrument is qualified for the proposed nonfungible token includes a series of sub-steps.

A first sub-step includes the processing module 44 of the control server 20 extracting, from an evaluation request 1250 from the user device 32-1 with regards to the longevity-contingent instrument, one or more of a contingency entity ID, the outcome provider ID, a value of the result of the triggered outcome, obligation requirements, and the obligation recipient ID. A second sub-step includes the processing module 44 verifying with an accreditation authority computing device of the computing system, validity of the baseline content. In an embodiment, one or more of the accreditation authority servers 1200-1 through 1200-M provide the accreditation authority computing device.

The verifying the validity of the baseline content includes a series of further-steps. A first further-step includes the processing module 44 of the control server 20 identifying the accreditation authority computing device based on a first identified corresponding accreditation authority associated with the obligation recipient ID 1150. For example, the control server 20 accesses the database 30 to retrieve identity of the accreditation authority server 1200-1 that is affiliated with the obligation provider ID.

A second further-step includes the processing module 44 obtaining baseline validation information 1204 from the accreditation authority computing device for the baseline content. For example, the processing module 44 issues at least some of the baseline content to the accreditation authority server 1200-1 and receives the baseline validation information 1204 in response.

A third further-step includes the processing module 44 indicating that the baseline content is valid when the baseline validation information is substantially the same as the baseline content. For example, the processing module 44 compares the received baseline validation information 1204 to the baseline content and indicates that the baseline content is valid when the comparison indicates a match of at least a minimum set of items for matching (e.g., outcome provider ID 1124, and obligation recipient ID 1150).

Having verified with the accreditation authority computing device, the validity of the baseline content, a third sub-step of the determining whether the baseline content is qualified for the proposed NFT includes identifying an approval for the proposed NFT. For example, the processing module 44 of the control server 20 further interprets the evaluation request 1250 to extract an indicator of approval to establish an NFT corresponding to the longevity-contingent instrument associated with a user of the user device 32-1. Alternatively, when interpreting another indicator of disapproval to establish the NFT, the processing module 44 indicates that approval for the proposed NFT has not been

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obtained and hence a subsequent process to produce the proposed NFT is not carried out. In such a scenario, an alternative process is triggered including one or more of ending the overall process, providing further details an explanation of the benefits of generating the NFT to the user, and seeking approval from the user.

When qualified, the example method of operation further continues in a second step that includes the control server 20 determining an exposure level for the NFT based on an availability status of the longevity-contingent instrument. The determining the exposure level for the NFT based on the availability status of the longevity-contingent instrument includes a series of sub-steps.

A first sub-step includes identifying the availability status based on one or more of the baseline content and a user input with regards to disposition of the longevity-contingent instrument. The disposition of the longevity-contingent instrument includes idling without a utilization, a life insurance investment utilization, and a life settlement utilization. The availability status includes at least two of unavailable, available within a first timeframe for a set of utilizations, and available within a second timeframe for a second set of utilizations. For example, the processing module 44 the control server 20 evaluates the baseline information to determine optional dispositions (e.g., whether the instrument is viable as a life insurance investment or a life settlement, or neither).

For instance, the processing module 44 produces the combined evaluation of the combination of outcome and obligation 1158 is discussed with FIG. 10B to determine the optional dispositions. As a further instance, the processing module 44 indicates positive viability for the life insurance investment when the estimated outcome triggered date 1134 of FIG. 10B is greater than a life insurance threshold value (e.g., 15 plus years) and an input from the user indicates a willingness to offer the longevity-contingent instrument as the life insurance investment. As a still further instance, the processing module 44 indicates positive viability for the life settlement when the estimated outcome triggered date 1134 is less than a life settlement threshold value (e.g., 10 years) and a further input from the user indicates a willingness to offer the longevity-contingent instrument as the life settlement.

A second sub-step includes interpreting the availability status to produce the exposure level. The exposure level includes one of hidden without exposure, a private portion of the object distributed ledger (e.g., accessible via a pass-code and/or encryption key), and a public portion of the object distributed ledger (e.g., accessible by computers able to obtain the blockchain of the object distributed ledger). For example, the processing module 44 of the control server 20 establishes the exposure level as hidden when the user disagrees to offer the longevity-contingent instrument. As another example, the processing module 44 establishes the exposure level as the private portion of the object distributed ledger when the user indicates a desire for restrictive offering of the longevity-contingent instrument for the purposes of the life insurance investment when a particular purchaser for the life insurance investment purposes has been identified. As yet another example, the processing module 44 establishes the exposure level as the public portion of the object distributed ledger when the user indicates a desire for a wider offering of the longevity-contingent instrument for the purposes of the life settlement (e.g., to offer widely to multiple life settlement brokers).

FIG. 15B further illustrates the example method of operation of the establishing of the contingent action token, where

having determined that the baseline content is qualified and having produce the exposure level for the NFT, a third step includes the processing module **44** of the control server **20** establishing NFT content to include the baseline content and the exposure level for the NFT. The establishing the NFT content to include the baseline content and the exposure level for the NFT includes a series of sub-steps.

The first sub-step includes identifying a portion of the object distributed ledger based on the exposure level. For example, the processing module **44** of the control server **20** interprets the exposure level (e.g., hidden, private, public) to produce and indicator for the portion (e.g., none, encrypted portion, open portion).

The second sub-step includes generating the NFT content for the portion of the object distributed ledger utilizing the baseline content. For example, when hidden, no action is taken. As another example, when private, the processing module **44** of the control server **20** encrypts the baseline content for storage in the encrypted portion of the object distributed ledger as the NFT content. As yet another example, one public, the processing module **44** indicates establishment of the open portion of the object distributed ledger for storage of the baseline content as the NFT content.

Having established the NFT content, a fourth step of the example method of operation includes the processing module **44** of the control server **20** causing generation of a new block affiliated with the NFT via the blockchain **1004** of the object distributed ledger **1002** as discussed with reference to FIG. 6C. The new block includes the NFT content. The causing generation of the new block includes a series of sub-steps. A first sub-step includes the processing module **44** determining whether to indirectly or directly update the object distributed ledger as previously discussed (e.g., using a local copy of the blockchain when available).

When indirectly updating the object distributed ledger, a second sub-step includes the processing module **44** of the control server **20** issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the NFT **1210**. For example, the processing module **44** sends the updated NFT **1210** to the transactional server **18** such that the transactional server **18** places a NFT 3 block **3** on the blockchain connected to the NFT **2** as illustrated in FIG. 15B.

When directly updating the object distributed ledger, the second sub-step includes the processing module **44** obtaining a copy of the object distributed ledger. For example, the processing module **44** recovers the object distributed ledger from the blockchain **1004** of the database **30** of the control server **20**. Having obtained the copy of the object distributed ledger, the second sub-step further includes hashing NFT content of the NFT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value. For example, the control module **44** generates a hash value utilizing a cryptographic algorithm over the NFT content of the NFT using a public key in possession of receiving entities for subsequent access of the blockchain to produce the next transaction hash value.

Having produced the next transaction hash value, the second sub-step further includes encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. For example, the processing module **44** utilizes a cryptographic encryption algorithm to encrypt the next transaction hash value utilizing the private key of the processing module **44** of the control server **20** to produce the next transaction signature.

Having produced the next transaction signature, the second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the NFT content of the NFT and the next transaction signature. For example, the processing module **44** populates the cryptographic token value of the next block with the next transaction signature and populates all the other content fields with the content of the NFT to produce the next block.

Having produced the next block, the second sub-step further includes causing inclusion of the next block as the new block in the object distributed ledger. For example, the processing module **44** of the control server **20** adds the block **3** on the object distributed ledger **1002** as illustrated in FIG. 15B by updating the blockchain **1004** and the database **30** of the control server **20** and/or the transactional server **18** and other transactional servers serving as blockchain nodes supporting the object distributed ledger. Alternatively, or in addition to, the processing module **44** of the control server **20** sends the NFT **1210** to one or more of the benefactor server **700**, the debtor server **702**, one or more of the user devices **32-1** through **32-N**, and one or more of the accreditation authority servers **1200-1** through **1200-M** to establish a record of this new NFT.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. 16A-16D are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for securely transitioning purpose of a contingent action token within the computing system. The computing system includes the benefactor server **700** of FIG. 5A, the debtor server **702** of FIG. 5A, the user devices **32-1** through **32-N** of FIG. 5A, the longevity-contingent instrument provider servers **704-1** through **704-M** of FIG. 5A, the control server **20** of FIG. 1, and the transactional server **18** of FIG. 1.

The control server **20** includes the processing module **44** of FIG. 1 and the database **30** of FIG. 1. The benefactor server **700** may be further associated with an outcome recipient identifier **1128** as further discussed below. The debtor server **702** may be further associated with an obligation provider identifier **1146** as further discussed below. The longevity-contingent instrument provider servers **704-1** through **704-M** may further be associated with one or more of an outcome provider identifier **1124** and an obligation recipient identifier **1150** as further discussed below. The transactional server **18** includes the processing module **44** FIG. 1 and the database **30** of FIG. 1 and functions as a blockchain node of a blockchain **1004** associated with an object distributed ledger **1002**.

FIG. 16A illustrates an example of the method for securely transitioning purpose of the contingent action token **1000**, where a first step of the example method includes the control server **20** identifying the contingency-action token (CAT) of the object distributed ledger **1002** that meets minimum CAT requirements. The CAT includes one of a non-fungible token (NFT) and another token linked to the

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NFT by the object distributed ledger when the CAT is not the NFT as illustrated in FIG. 16A. Example content of the contingent action token 1000 is discussed in greater detail with reference to FIG. 10B.

The minimum CAT requirements include a variety of requirements. A first requirement includes a recovered cryptographic token value 1112 of the CAT matches a calculated cryptographic token value of the CAT. A second requirement includes an obligation provider identifier record of the CAT that includes an original obligation provider identifier 1146 of the CAT when the NFT was initially generated. The obligation provider identifier 1146 is associated with providing of an obligation 1148 tied to an obligation recipient identifier 1150 in accordance with obligation requirements 1142. A positive obligation status 1144 indicates that the obligation 1148 has historically been provided in accordance with the obligation requirements 1142.

A third requirement includes an outcome recipient identifier record of the CAT that includes an original outcome recipient identifier of the NFT when the NFT was initially generated. The outcome recipient identifier 1128 is associated with a result of a triggered outcome 1126 tied to an outcome provider identifier 1124 in accordance with contingent outcome rules 1118 and the positive obligation status 1144. A contingency status 1120 indicates whether the triggered outcome 1126 has been triggered for a contingency entity identifier 1122 in accordance with the contingent outcome rules 1118.

A fourth requirement includes a utilization status of the CAT that indicates one of a life insurance investment utilization and a non-utilization yet available as previously discussed. For example, the processing module 44 of the control server 20 interprets the contingency action token 1000 to determine the utilization status of the CAT. For instance, the processing module 44 interprets the life insurance investment utilization as the utilization status for the CAT when a user associated with a longevity-contingent instrument of the CAT approved utilization of the longevity-contingent instrument for the life insurance investment. Alternatively, the processing module 44 interprets the non-utilization yet available status of the CAT when the user associated with the longevity-contingent instrument of the CAT has not yet approved utilization for either the life insurance investment utilization or for a life settlement utilization.

The identifying the CAT of the object distributed ledger that meets the minimum CAT requirements includes a series of sub-steps. A first sub-step includes the processing module 44 of the control server 20 determining whether to indirectly or directly access the object distributed ledger. For example, the processing module 44 determines to indirectly access the object distributed ledger when the blockchain 1004 is not available in the database 30 of the control server 20 and is available in the database 30 of the transactional server 18. As another example, the processing module 44 of the control server 20 determines to directly access the object distributed ledger when the blockchain 1004 is available within the database 30 of the control server 20.

When indirectly accessing the object distributed ledger, a second sub-step of the identifying of the CAT includes the processing module 44 of the control server 20 issuing a CAT access request to the transactional server 18 (e.g., an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger). The CAT access request includes a representation of the minimum CAT requirements. The second sub-step further includes the processing module 44 of the control server 20 extracting a

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CAT identifier of the CAT from a CAT access response from the transactional server 18, where the transactional server 18 accesses the object distributed ledger 1002 to identify an NFT block 3 and/or a linked block 3-1 of the blockchain as the CAT as illustrated in FIG. 16A.

When directly accessing the object distributed ledger, the second sub-step of the identifying of the CAT includes the processing module 44 of the control server 20 obtaining a copy of the object distributed ledger (e.g., recover the blockchain 1004 from the database 30 of the control server 20). The second sub-step further includes the processing module 44 of the control server 20 indicating the CAT identifier of the CAT when detecting a block of the copy of the object distributed ledger that matches the representation of the minimum CAT requirements (e.g., original obligation provider identifier and original outcome recipient identifier or a linked identifiers subsequent to the originals).

Having obtained the CAT, a third sub-step of the identifying of the CAT that meets the minimum CAT requirements includes the processing module 44 of the control server 20 verifying that the recovered cryptographic token value 1112 of the CAT matches the calculated cryptographic token value of the CAT. For example, the processing module 44 produces the calculated cryptographic token value of the CAT over a corresponding content portion of the obtained CAT and compares that calculated value to the recovered cryptographic token value 1112. When the cryptographic token values are the same, the processing module 44 indicates that the CAT that meets the minimum CAT requirements has been identified (e.g., by the token identifier 1110).

Having verified that the recovered cryptographic token value of the CAT matches the calculated cryptographic token value of the CAT, a fourth sub-step of the identifying of the CAT that meets the minimum CAT requirements includes the processing module 44 of the control server 20 extracting the utilization status of the CAT. The fourth sub-step further includes verifying that the utilization status of the CAT indicates the one of the life insurance investment utilization and the non-utilization yet available.

FIG. 16B further illustrates the example of the method for securely transitioning the purpose of the contingent action token 1000, where having identified the CAT that meets the minimum CAT requirements, a second step of the example method includes the control server 20 determining whether to select the CAT based on an evaluation of the CAT with regards to an evaluation profile 1138. A beneficial improvement to a present estimate of the triggered outcome results from a hypothetical change to the obligation provider identifier of the selected CAT. The evaluation of the CAT includes producing the obligation evaluation 1156 based on the evaluation profile 1138 (e.g., using risks and historical correlations of the evaluation profile). The obligation evaluation 1156 compares the providing of the obligation in accordance with the obligation requirements (e.g., historical and estimated future). The evaluation of the CAT further includes the outcome evaluation 1140. The outcome evaluation 1140 produces the present estimate of the triggered outcome based on the evaluation profile.

The determining whether to select the CAT based on the evaluation of the CAT with regards to the evaluation profile includes a series of sub-steps. A first sub-step includes the processing module 44 of the control server 20 determining the evaluation profile to include a required evaluation performance improvement level. For example, the processing module 44 extracts the evaluation profile 1138 from the recovered CAT. As another example, the processing module 44 generates the evaluation profile based on an external

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input through a user interface (e.g., a request that includes a specific evaluation performance improvement level).

A second sub-step includes the processing module 44 of the control server 20 comparing the providing of the obligation associated with the obligation provider identifier (e.g., original or subsequent) to providing of the obligation associated with a hypothetical obligation provider identifier in accordance with the evaluation profile to produce the obligation evaluation. For example, the processing module 44 receives identity and risk levels with regards to a new obligation provider identifier and utilizes that to produce the obligation evaluation (e.g., an improved outcome when a risk level of noncompliance to the obligation is lower for the new obligation provider identifier).

A third sub-step includes the processing module 44 of the control server 20 comparing an estimated trigger outcome when the providing of the obligation is associated with the obligation provider identifier to a hypothetical estimated trigger outcome when the providing of the obligation is associated with the hypothetical obligation provider identifier in accordance with the evaluation profile to produce the outcome evaluation. For example, the processing module 44 produces the outcome evaluation for the new obligation provider (e.g., with the lower risks of not meeting the obligation) and for an estimated trigger date of the triggered outcome.

A fourth sub-step includes the processing module 44 of the control server 20 indicating to select the CAT when the obligation evaluation and the outcome evaluation satisfies the required evaluation performance improvement level. For example, the processing module 44 indicates to select the CAT when the new obligation provider identifier is a catalyst to meet the required evaluation performance improvement level.

FIG. 16C further illustrates the example of the method for securely transitioning the purpose of the contingent action token 1000, where having selected the CAT to produce a selected CAT, a third step of the example method includes determining reassignment information for the selected CAT. The reassignment information includes a confirmed change of at least one of the obligation provider identifier 1146 and the outcome recipient identifier 1128.

The determining the reassignment information for the selected CAT includes a variety of approaches. A first approach includes identifying a hypothetical obligation provider identifier based on the evaluation of the selected CAT with regards to the evaluation profile to establish a new obligation provider identifier of the reassignment information. For example, the processing module 44 of the control server 20 replaces or adds a hypothetical obligation provider identifier to the obligation provider identifier 1146 based on a previous analysis of the evaluation of the selected CAT. Acquisition of the hypothetical obligation provider identifier includes extracting the identifier from confirmation information 1170 received from another computing device and extraction from content 1006 of the selected CAT.

A second approach to determine the reassignment information includes, when approved (e.g., from a user associated with the longevity-contingent instrument), the processing module 44 of the control server 20 modifying the outcome recipient identifier to establish a new outcome recipient identifier of the reassignment information. For example, the processing module 44 extracts the new outcome recipient identifier from the confirmation information 1170.

Having determined the reassignment information, a fourth step of the example method of operation includes the

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processing module 44 of the control server 20 facilitating taking control of the selected CAT of a blockchain of the object distributed ledger in accordance with a securely passing process as discussed with reference to FIG. 6C. For example, the processing module 44 exchanges control grant information 1172 with a current controlling entity of the selected CAT to gain the control. For instance, the current controlling entity generates a new block that includes a public key of the processing module 44, the control request from the processing module 44, and a cryptographic signature over a portion of the block utilizing a private key of the current controlling entity.

FIG. 16D further illustrates the example of the method for securely transitioning the purpose of the contingent action token 1000, where having taking control of the selected CAT, a fifth step of the example method includes the processing module 44 of the control server 20 updating the selected CAT utilizing the reassignment information for the CAT to produce an updated CAT 1181. The updating the selected CAT utilizing the reassignment information for the selected CAT to produce the updated CAT 1181 includes a series of sub-steps. A first sub-step includes the processing module 44 obtaining the selected CAT. For example, the processing module 44 recovers the selected CAT from the blockchain 1004 of the database 30 of the control server 20 when a local copy of the blockchain is up-to-date. As another example, the processing module 44 requests the selected CAT from the transactional server 18.

A second sub-step includes the processing module 44 replacing corresponding elements of the selected CAT with at least a portion of the reassignment information to produce the updated CAT 1181. For example, the processing module 44 replaces the obligation provider identifier 1146 with a new obligation provider identifier such that a beneficial valuation level increase is realized subsequently for the triggered outcome upon triggering and when conditions of the contingent outcome rules have been satisfied.

Having produced the updated CAT 1181, a sixth step of the example method of operation includes the processing module 44 of the control server 20 causing generation of a new block affiliated with the updated CAT via the blockchain of the object distributed ledger in accordance with the securely passing process. The new block includes the updated CAT.

The causing generation of the new block affiliated with the updated CAT via the blockchain of the object distributed ledger includes a series of sub-steps. A first sub-step includes the processing module 44 of the control server 20 determining whether to indirectly or directly update the object distributed ledger as previously discussed (e.g., using a local copy of the blockchain when available).

When indirectly updating the object distributed ledger, a second sub-step includes the processing module 44 issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the updated CAT 1181. For example, the processing module 44 of the control server 20 sends the updated CAT 1181 to the transactional server 18 such that the transactional server 18 places an update 2 block 3-2 on the blockchain connected to the CAT as illustrated in FIG. 16D.

When directly updating the object distributed ledger, the second sub-step includes the processing module 44 obtaining a copy of the object distributed ledger. For example, the processing module 44 recovers the object distributed ledger from the blockchain 1004 of the database 30 of the control

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server **20**. Having obtained the copy of the object distributed ledger, the second sub-step further includes hashing content of the updated CAT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value. For example, the control module **44** generates a hash value utilizing a cryptographic algorithm over the content of the updated CAT using a public key in possession of receiving entities for subsequent access of the blockchain to produce the next transaction hash value.

Having produced the next transaction hash value, the second sub-step further includes encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature. For example, the processing module **44** utilizes a cryptographic encryption algorithm to encrypt the next transaction hash value utilizing the private key of the processing module **44** to produce the next transaction signature.

Having produced the next transaction signature, the second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the content of the updated CAT and the next transaction signature. For example, the processing module **44** populates the cryptographic token value **1112** of the next block with the next transaction signature and populates all the other content fields with the content of the updated CAT to produce the next block.

Having produced the next block, the second sub-step further includes causing inclusion of the next block as the new block in the object distributed ledger. For example, the processing module **44** of the control server **20** adds the block **3-2** as an update 1 associated with the selected CAT on the object distributed ledger **1002** as illustrated in FIG. **16D** by updating the blockchain **1004** and the database **30** of the control server **20** and/or the transactional server **18** and other transactional servers serving as blockchain nodes supporting the object distributed ledger.

Alternatively, or in addition to, the processing module **44** of the control server **20** sends the updated CAT **1181** to one or more of the benefactor server **700**, the debtor server **702**, one or more of the user devices **32-1** through **32-N**, and one or more of the longevity-contingent instrument provider servers **704-1** through **704-M**.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

FIGS. **17A-17D** are schematic block diagrams of another embodiment of a computing system illustrating another embodiment of a method for securely processing a contingent action token within the computing system. In particular, a nonfungible token (NFT) form of the contingent action token is utilized. The computing system includes the benefactor server **700** of FIG. **5A**, the debtor server **702** of FIG. **5A**, the user devices **32-1** through **32-N** of FIG. **5A**, the longevity-contingent instrument provider servers **704-1** through **704-M** of FIG. **5A**, the control server **20** of FIG. **1**, and the transactional server **18** of FIG. **1**.

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The control server **20** includes the processing module **44** of FIG. **1** and the database **30** of FIG. **1**. The benefactor server **700** may be further associated with an outcome recipient identifier **1128** as further discussed below. The debtor server **702** may be further associated with an obligation provider identifier **1146** as further discussed below. The longevity-contingent instrument provider servers **704-1** through **704-M** may further be associated with one or more of an outcome provider identifier **1124** and an obligation recipient identifier **1150** as further discussed below. The transactional server **18** includes the processing module **44** of FIG. **1** and the database **30** of FIG. **1** and functions as a blockchain node of one or more blockchains **1004** associated with one or more object distributed ledgers **1002**. In an instance the transactional server **18** is utilized as part of a secure exchange to support the exchange of tokens representing various longevity-contingent instruments.

FIG. **17A** illustrates an example of the method for securely processing the contingent action token **1000**, where a first step of the example method includes the control server **20** identifying a set of contingency-action tokens (CATs) **1000** of the object distributed ledger **1002** that meets minimum CAT requirements. Each CAT includes one of a non-fungible token (NFT) and another token linked to the NFT by the object distributed ledger when the CAT is not the NFT as illustrated in FIG. **17A**. Example content of the contingent action token **1000** is discussed in greater detail with reference to FIG. **10B**.

The minimum CAT requirements include a variety of requirements. A first requirement includes a recovered cryptographic token value **1112** of the CAT matches a calculated cryptographic token value of the CAT. A second requirement includes an obligation provider identifier record of the CAT that includes an original obligation provider identifier **1146** of the CAT when the NFT was initially generated. The obligation provider identifier **1146** is associated with providing of an obligation **1148** tied to an obligation recipient identifier **1150** in accordance with obligation requirements **1142**. A positive obligation status **1144** indicates that the obligation **1148** has historically been provided in accordance with the obligation requirements **1142**.

A third requirement includes an outcome recipient identifier record of the CAT that includes an original outcome recipient identifier of the NFT when the NFT was initially generated. The outcome recipient identifier **1128** is associated with a result of a triggered outcome **1126** tied to an outcome provider identifier **1124** in accordance with contingent outcome rules **1118** and the positive obligation status **1144**. A contingency status **1120** indicates whether the triggered outcome **1126** has been triggered for a contingency entity identifier **1122** in accordance with the contingent outcome rules **1118**.

The identifying the set of CATs of the object distributed ledger that meets the minimum CAT requirements includes a series of sub-steps for each CAT. A first sub-step includes the processing module **44** of the control server **20** determining whether to indirectly or directly access the object distributed ledger. For example, the processing module **44** determines to indirectly access the object distributed ledger when the blockchain **1004** is not available in the database **30** of the control server **20** and is available in the database **30** of the transactional server **18**. As another example, the processing module **44** of the control server **20** determines to directly access the object distributed ledger when the blockchain **1004** is available within the database **30** of the control server **20**.

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When indirectly accessing the object distributed ledger, a second sub-step of the identifying of each CAT includes the processing module **44** of the control server **20** issuing a CAT access request to the transactional server **18** (e.g., an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger). The CAT access request includes a representation of the minimum CAT requirements. The second sub-step further includes the processing module **44** extracting a CAT identifier of a particular CAT from a CAT access response from the transactional server **18**, where the transactional server **18** accesses the object distributed ledger **1002** to identify an NFT block **3** and/or a linked block **3-1** of the blockchain as the CAT as illustrated in FIG. **17A**.

When directly accessing the object distributed ledger, the second sub-step of the identifying of the CAT includes the processing module **44** of the control server **20** obtaining a copy of the object distributed ledger (e.g., recover the blockchain **1004** from the database **30** of the control server **20**). The second sub-step further includes the processing module **44** indicating the CAT identifier of the particular CAT when detecting a block of the copy of the object distributed ledger that matches the representation of the minimum CAT requirements (e.g., original obligation provider identifier and original outcome recipient identifier or a linked identifiers subsequent to the originals).

Having obtained the CAT, a third sub-step of the identifying of the CAT that meets the minimum CAT requirements includes the processing module **44** verifying that the recovered cryptographic token value **1112** of the particular CAT matches the calculated cryptographic token value of the particular CAT. For example, the processing module **44** produces the calculated cryptographic token value of the CAT over a corresponding content portion of the obtained CAT and compares that calculated value to the recovered cryptographic token value **1112**. When the cryptographic token values are the same, the processing module **44** indicates that the CAT that meets the minimum CAT requirements has been identified (e.g., by the token identifier **1110**).

FIG. **17B** further illustrates the example of the method for securely processing the contingent action token **1000**, where having identified the set of CATs that meets the minimum CAT requirements, a second step of the example method includes the control server **20** determining whether to update the set of CATs based on an evaluation of each CAT with regards to an evaluation profile **1138**. A beneficial improvement to a present estimate of the triggered outcome results from a hypothetical change to the obligation provider identifier of the CAT. The evaluation of the CAT includes producing the obligation evaluation **1156** based on the evaluation profile **1138** (e.g., using risks and historical correlations of the evaluation profile). The obligation evaluation **1156** compares the providing of the obligation in accordance with the obligation requirements (e.g., historical and estimated future). The evaluation of the CAT further includes the outcome evaluation **1140**. The outcome evaluation **1140** produces the present estimate of the triggered outcome based on the evaluation profile.

The determining whether to update the set of CATs based on the evaluation of each CAT with regards to the evaluation profile includes a series of sub-steps. A first sub-step includes the processing module **44** of the control server **20** determining the evaluation profile to include a required evaluation performance improvement level. For example, the processing module **44** extracts the evaluation profile **1138** from the recovered CAT. As another example, the processing module **44** generates the evaluation profile based

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on an external input through a user interface (e.g., a request that includes a specific evaluation performance improvement level).

A second sub-step includes the processing module **44** comparing the providing of the obligation associated with the obligation provider identifier (e.g., original or subsequent) to providing of the obligation associated with a hypothetical obligation provider identifier in accordance with the evaluation profile to produce the obligation evaluation. For example, the processing module **44** receives identity and risk levels with regards to a new obligation provider identifier and utilizes that to produce the obligation evaluation (e.g., an improved outcome when a risk level of noncompliance to the obligation is lower for the new obligation provider identifier).

A third sub-step includes the processing module **44** of the control server **20** comparing an estimated trigger outcome when the providing of the obligation is associated with the obligation provider identifier to a hypothetical estimated trigger outcome when the providing of the obligation is associated with the hypothetical obligation provider identifier in accordance with the evaluation profile to produce the outcome evaluation. For example, the processing module **44** produces the outcome evaluation for the new obligation provider (e.g., with the lower risks of not meeting the obligation) and for an estimated trigger date of the triggered outcome.

A fourth sub-step includes the processing module **44** indicating to update the CAT when the obligation evaluation and the outcome evaluation satisfies the required evaluation performance improvement level. For example, the processing module **44** indicates to update the CAT when the new obligation provider identifier is a catalyst to meet the required evaluation performance improvement level when reestablishing an existing pool (e.g., a set of CATs) of CATs to an equal weight reassignment approach as further discussed below.

FIG. **17C** further illustrates the example of the method for the processing of the contingent action token **1000**, where having determined to update the set of CATs, a third step of the example method includes the control server **20** determining reassignment information for each CAT of the set of CATs. The reassignment information includes a confirmed change of at least one of the obligation provider identifier **1146** and the outcome recipient identifier **1128**. The reassignment information further specifies a unique percentage-of-face-value level for each triggered outcome in accordance with an assignment approach **1147**.

The determining the reassignment information for each CAT of the set of CATs includes one or more of a variety of approaches. A first approach includes the processing module **44** identifying a hypothetical obligation provider identifier based on the evaluation of the CAT with regards to the evaluation profile to establish a new obligation provider identifier of the reassignment information (e.g., de-risk with new provider too).

A second approach includes, when approved, the processing module **44** modifying the outcome recipient identifier **1128** to establish a new outcome recipient identifier of the reassignment information. For example, a common recipient for the set of CATs.

A third approach includes the processing module **44** determining the assignment approach **1147**. The assignment approach includes one of an equal weight approach, a pass-through approach, and an open approach. The equal weight approach includes establishing the unique percentage-of-face-value level for a corresponding triggered out-

come such that one of an identical percentage-of-face-value level and a constant value of each triggered outcome associated with the set of CATs is assigned to a corresponding outcome recipient ID.

The pass-through approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome such that one of a variable percentage-of-face-value level and a variable value of each triggered outcome associated with the set of CATs is assigned to the corresponding outcome recipient ID. The open approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome upon detection of the corresponding triggered outcome.

A third approach includes the processing module **44** determining the unique percentage-of-face-value level for a particular corresponding triggered outcome for each CAT of the set of CATs in accordance with the assignment approach. For example, the processing module selects different percentages such that the constant value of each triggered outcome associated with the set of CATs is assigned to the corresponding outcome recipient ID when the equal weight approach is selected via confirmation information **1170** from the benefactor server **700**.

Having produced the reassignment information, a fourth step of the example method of operation includes the processing module **44** facilitating, in accordance with a securely passing process, taking control of each CAT of the set of CATs of a blockchain **1004** of the object distributed ledger **1002** utilizing the reassignment information for each CAT to produce a corresponding updated CAT.

Only a device possessing control over the CAT may modify the CAT in accordance with the securely passing process previously discussed with reference to FIG. **6C**. A technological improvement is provided over prior art communication and computing systems associated with records management since only the device possessing control over the CAT may modify the CAT as part of such a tightly integrated overall digital records process described in this section for the present invention. Only a present trusted device may pass the control to a next trusted device that is part of the overall records management process.

The facilitating, in accordance with the securely passing process, the taking control over each CAT of the set of CATs includes a series of sub-steps. A first sub-step includes the processing module **44** determining whether to indirectly or directly access the object distributed ledger as previously discussed.

When indirectly accessing the object distributed ledger, a second sub-step includes the processing module **44** issuing a blockchain access request to the transactional server **18** functioning as an object ledger computing device serving as a blockchain node of the object distributed ledger. As another example, when directly accessing the object distributed ledger, the processing module **44** obtains a copy of the object distributed ledger (e.g., from the database **30**). The processing module **44** hashes content of the CAT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value and encrypts the next transaction hash value utilizing a private key of the control server **20** to produce a next transaction signature. The processing module **44** generates a next block (e.g., block **3-2**) of the blockchain of the object distributed ledger to include the content of the CAT and the next transaction signature. The processing module **44** causes inclusion of the next block in the object distributed ledger as illustrated in FIG. **17C**.

FIG. **17D** further illustrates the example of the method for the processing of the contingent action token **1000**, where having taking control of each CAT of the set of CATs, a fifth step of the example method includes the processing module **44** of the control server **20** updating each CAT utilizing the reassignment information for the CAT to produce a corresponding updated CAT **1181-1** through **1181-N**.

Having produced each updated CAT, a sixth step of the example method of operation includes the processing module **44** causing, in accordance with the securely passing process, generation of a new block affiliated with the corresponding updated CAT via the blockchain of the object distributed ledger. The new block includes the corresponding updated CAT.

The causing, in accordance with the securely passing process, the generation of the new block affiliated with the corresponding updated CAT via the blockchain of the object distributed ledger includes a series of sub-steps. A first sub-step includes the processing module **44** determining whether to indirectly or directly access the object distributed ledger as previously discussed. When indirectly accessing the object distributed ledger, a second sub-step includes the processing module **44** issuing a blockchain update request to an object ledger computing device (e.g., the transactional server **18**) of the computing system serving as a blockchain node of the object distributed ledger. The blockchain update request includes the corresponding updated CAT.

When directly accessing the object distributed ledger, the second sub-step includes the processing module **44** obtaining a copy of the object distributed ledger (e.g., from the database **30**). The second sub-step further includes hashing content of the corresponding updated CAT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value and encrypting the next transaction hash value utilizing a private key of the control server **20** to produce a next transaction signature. The second sub-step further includes generating a next block of the blockchain of the object distributed ledger to include the content of the corresponding updated CAT and the next transaction signature and causing inclusion of the next block in the object distributed ledger. For example, the block **3-2** is inserted into the object distributed ledger **1002** as the updated CAT **1181-1** as illustrated in FIG. **17D**.

Alternatively, or in addition to, the processing module **44** of the control server **20** sends the updated CAT **1181-1** to one or more of the benefactor server **700**, the debtor server **702**, one or more of the user devices **32-1** through **32-N**, and one or more of the longevity-contingent instrument provider servers **704-1** through **704-M**.

The method described above module can alternatively be performed by various modules of the computing system or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers) of the computing system, cause the one or more computing devices to perform any or all of the steps described above.

It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a number

of desired types (e.g., data, video, speech, text, graphics, audio, etc. any of which may generally be referred to as 'data').

As may be used herein, the terms "substantially" and "approximately" provides an industry-accepted tolerance for its corresponding term and/or relativity between items. For some industries, an industry-accepted tolerance is less than one percent and, for other industries, the industry-accepted tolerance is 10 percent or more. Other examples of industry-accepted tolerance range from less than one percent to fifty percent. Industry-accepted tolerances correspond to, but are not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, thermal noise, dimensions, signaling errors, dropped packets, temperatures, pressures, material compositions, and/or performance metrics. Within an industry, tolerance variances of accepted tolerances may be more or less than a percentage level (e.g., dimension tolerance of less than $\pm 1\%$). Some relativity between items may range from a difference of less than a percentage level to a few percent. Other relativity between items may range from a difference of a few percent to magnitude of differences.

As may also be used herein, the term(s) "configured to", "operably coupled to", "coupled to", and/or "coupling" includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as "coupled to".

As may even further be used herein, the term "configured to", "operable to", "coupled to", or "operably coupled to" indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term "associated with", includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

As may be used herein, the term "compares favorably", indicates that a comparison between two or more items, signals, etc., provides a desired relationship. For example, when the desired relationship is that signal 1 has a greater magnitude than signal 2, a favorable comparison may be achieved when the magnitude of signal 1 is greater than that of signal 2 or when the magnitude of signal 2 is less than that of signal 1. As may be used herein, the term "compares unfavorably", indicates that a comparison between two or more items, signals, etc., fails to provide the desired relationship.

As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase "at least one of a, b, and c" or of this generic form "at least one of a, b, or c", with more or less elements than "a", "b", and "c". In either phrasing, the phrases are to be interpreted identically. In particular, "at least one of a, b, and c" is equivalent to "at least one of a, b, or c" and shall mean a, b, and/or c. As an example, it means: "a" only, "b" only, "c" only, "a" and "b", "a" and "c", "b" and "c", and/or "a", "b", and "c".

As may also be used herein, the terms "processing module", "processing circuit", "processor", "processing circuitry", and/or "processing unit" may be a single processing

device or a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, processing circuitry, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, processing circuitry, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, processing circuitry, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processing module, module, processing circuit, processing circuitry and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, processing circuitry and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.

To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate definitions of both functional building blocks and flow diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules and components herein, can be implemented as illustrated or by discrete components, appli-

cation specific integrated circuits, processors executing appropriate software and the like or any combination thereof.

In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with one or more other routines. In addition, a flow diagram may include an “end” and/or “continue” indication. The “end” and/or “continue” indications reflect that the steps presented can end as described and shown or optionally be incorporated in or otherwise used in conjunction with one or more other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

Unless specifically stated to the contra, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.

The term “module” is used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules.

As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, a quantum register or other quantum memory and/or any other device that stores data in a non-transitory manner. Furthermore, the memory device may be in a form of a solid-state memory, a hard drive memory or other disk storage, cloud memory, thumb drive,

server memory, computing device memory, and/or other non-transitory medium for storing data. The storage of data includes temporary storage (i.e., data is lost when power is removed from the memory element) and/or persistent storage (i.e., data is retained when power is removed from the memory element). As used herein, a transitory medium shall mean one or more of: (a) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for temporary storage or persistent storage; (b) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for temporary storage or persistent storage; (c) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for processing the data by the other computing device; and (d) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for processing the data by the other element of the computing device. As may be used herein, a non-transitory computer readable memory is substantially equivalent to a computer readable memory. A non-transitory computer readable memory can also be referred to as a non-transitory computer readable storage medium.

While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

What is claimed is:

1. A computer-implemented method of using a computing system, the method comprises:

identifying, by a computing device of the computing system, a set of contingency-action tokens (CATs) of an object distributed ledger, wherein each contingency-action token (CAT) includes one of a non-fungible token (NFT) and another token linked to the NFT by the object distributed ledger when the CAT is not the NFT, wherein each CAT meets minimum CAT requirements that includes:

a recovered cryptographic token value of the CAT matches a calculated cryptographic token value of the CAT,

an obligation provider identifier record of the CAT that includes an original obligation provider identifier of the NFT when the NFT was initially generated, wherein the obligation provider identifier is associated with providing of an obligation tied to an obligation recipient identifier in accordance with obligation requirements, wherein a positive obligation status indicates that the obligation has historically been provided in accordance with the obligation requirements, and

an outcome recipient identifier record of the CAT that includes an original outcome recipient identifier of the NFT when the NFT was initially generated, wherein the outcome recipient identifier is associated with a result of a triggered outcome tied to an outcome provider identifier in accordance with contingent outcome rules and the positive obligation status, wherein a contingency status indicates whether the triggered outcome has been triggered for a contingency entity identifier in accordance with the contingent outcome rules;

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determining, by the computing device, whether to update the set of CATs based on an evaluation of each CAT with regards to an evaluation profile, wherein a beneficial improvement to a present estimate of the triggered outcome result from a hypothetical change to the obligation provider identifier of the CAT, wherein the evaluation of the CAT includes producing:

- an obligation evaluation based on the evaluation profile, wherein the obligation evaluation compares the providing of the obligation in accordance with the obligation requirements, and
- an outcome evaluation, wherein the outcome evaluation produces the present estimate of the triggered outcome based on the evaluation profile; and

when updating the set of CATs to produce an updated set of CATs:

- determining, by the computing device, reassignment information for each CAT of the set of CATs, wherein the reassignment information includes a confirmed change of at least one of the obligation provider identifier and the outcome recipient identifier, wherein the reassignment information further specifies a unique percentage-of-face-value level for each triggered outcome in accordance with an assignment approach;
- facilitating, by the computing, taking control of each CAT of the set of CATs of a blockchain of the object distributed ledger, wherein the facilitating taking control of each CAT includes:
 - obtaining a copy of the object distributed ledger,
 - hashing content of the CAT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value,
 - encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature,
 - generating a next block of the blockchain of the object distributed ledger to include the content of the CAT and the next transaction signature, and
 - causing inclusion of the next block in the object distributed ledger;
- updating, by the computing device, each CAT utilizing the reassignment information for the CAT to produce a corresponding updated CAT; and
- causing, by the computing device, generation of a new block affiliated with the corresponding updated CAT via the blockchain of the object distributed ledger, wherein the new block includes the corresponding updated CAT, wherein the causing generation of the new block includes:
 - issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger, wherein the blockchain update request includes the corresponding updated CAT.

2. The method of claim 1, wherein the identifying the set of CATs of the object distributed ledger comprises:

- determining whether to indirectly or directly access the object distributed ledger;
- when indirectly accessing the object distributed ledger:
 - issuing a CAT access request to the object ledger computing device of the computing system serving as the blockchain node of the object distributed ledger, wherein the CAT access request includes a representation of the minimum CAT requirements, and

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- extracting a CAT identifier of a particular CAT from a CAT access response from the object ledger computing device;
- when directly accessing the object distributed ledger:
 - obtaining the copy of the object distributed ledger, and indicating the CAT identifier of the particular CAT when detecting a block of the copy of the object distributed ledger that matches the representation of the minimum CAT requirements; and
- verifying that the recovered cryptographic token value of the particular CAT matches the calculated cryptographic token value of the particular CAT.

3. The method of claim 1, wherein the determining whether to update the set of CATs based on the evaluation of each CAT with regards to the evaluation profile comprises:

- determining the evaluation profile to include a required evaluation performance improvement level;
- comparing the providing of the obligation associated with the obligation provider identifier to providing of the obligation associated with a hypothetical obligation provider identifier in accordance with the evaluation profile to produce the obligation evaluation;
- comparing an estimated trigger outcome when the providing of the obligation is associated with the obligation provider identifier to a hypothetical estimated trigger outcome when the providing of the obligation is associated with the hypothetical obligation provider identifier in accordance with the evaluation profile to produce the outcome evaluation; and
- indicating to update the CAT when the obligation evaluation and the outcome evaluation satisfies the required evaluation performance improvement level.

4. The method of claim 1, wherein the determining the reassignment information for each CAT of the set of CATs comprises at least one of:

- identifying a hypothetical obligation provider identifier based on the evaluation of the CAT with regards to the evaluation profile to establish a new obligation provider identifier of the reassignment information;
- when approved, modifying the outcome recipient identifier to establish a new outcome recipient identifier of the reassignment information;
- determining the assignment approach, wherein the assignment approach includes one of an equal weight approach, a pass-through approach, and an open approach, wherein the equal weight approach includes establishing the unique percentage-of-face-value level for a corresponding triggered outcome such that one of an identical percentage-of-face-value level and a constant value of each triggered outcome associated with the set of CATs is assigned to a corresponding outcome recipient ID, wherein the pass-through approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome such that one of a variable percentage-of-face-value level and a variable value of each triggered outcome associated with the set of CATs is assigned to the corresponding outcome recipient ID, wherein the open approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome upon detection of the corresponding triggered outcome; and
- determining the unique percentage-of-face-value level for a particular corresponding triggered outcome for each CAT of the set of CATs in accordance with the assignment approach.

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5. A computing device of a computing system, the computing device comprises:

- an interface;
- a local memory; and
- a processor operably coupled to the interface and the local memory, wherein the processor executes operational instructions stored in the local memory functions to:
 - identify a set of contingency-action tokens (CATs) of an object distributed ledger, wherein each contingency-action token (CAT) includes one of a non-fungible token (NFT) and another token linked to the NFT by the object distributed ledger when the CAT is not the NFT, wherein each CAT meets minimum CAT requirements that includes:
 - a recovered cryptographic token value of the CAT matches a calculated cryptographic token value of the CAT;
 - an obligation provider identifier record of the CAT that includes an original obligation provider identifier of the NFT when the NFT was initially generated, wherein the obligation provider identifier is associated with providing of an obligation tied to an obligation recipient identifier in accordance with obligation requirements, wherein a positive obligation status indicates that the obligation has historically been provided in accordance with the obligation requirements, and
 - an outcome recipient identifier record of the CAT that includes an original outcome recipient identifier of the NFT when the NFT was initially generated, wherein the outcome recipient identifier is associated with a result of a triggered outcome tied to an outcome provider identifier in accordance with contingent outcome rules and the positive obligation status, wherein a contingency status indicates whether the triggered outcome has been triggered for a contingency entity identifier in accordance with the contingent outcome rules;
 - determine whether to update each CAT of the set of CATs based on an evaluation of each CAT with regards to an evaluation profile, wherein a beneficial improvement to a present estimate of the triggered outcome result from a hypothetical change to the obligation provider identifier of the CAT, wherein the evaluation of the CAT includes producing:
 - an obligation evaluation based on the evaluation profile, wherein the obligation evaluation compares the providing of the obligation in accordance with the obligation requirements, and
 - an outcome evaluation, wherein the outcome evaluation produces the present estimate of the triggered outcome based on the evaluation profile; and
 - when updating the set of CATs to produce an updated set of CATs:
 - determine reassignment information for each CAT of the set of CATs, wherein the reassignment information includes a confirmed change of at least one of the obligation provider identifier and the outcome recipient identifier, wherein the reassignment information further specifies a unique percentage-of-face-value level for each triggered outcome in accordance with an assignment approach;
 - facilitate taking control of each CAT of the set of CATs of a blockchain of the object distributed ledger, wherein the processor facilitates taking control of each CAT by:
 - obtaining a copy of the object distributed ledger,

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- hashing content of the CAT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value,
 - encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature,
 - generating a next block of the blockchain of the object distributed ledger to include the content of the CAT and the next transaction signature, and
 - causing inclusion of the next block in the object distributed ledger;
- update each CAT utilizing the reassignment information for the CAT to produce a corresponding updated CAT; and
- cause generation of a new block affiliated with the corresponding updated CAT via the blockchain of the object distributed ledger, wherein the new block includes the corresponding updated CAT, wherein the processor causes generation of the new block by:
 - issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger, wherein the blockchain update request includes the corresponding updated CAT.

6. The computing device of claim 5, wherein the processor executes the operational instructions stored in the local memory to identify the set of CATs of the object distributed ledger by:

- determining whether to indirectly or directly access the object distributed ledger;
- when indirectly accessing the object distributed ledger:
 - issuing, via the interface, a CAT access request to the object ledger computing device of the computing system serving as the blockchain node of the object distributed ledger, wherein the CAT access request includes a representation of the minimum CAT requirements, and
 - extracting a CAT identifier of a particular CAT from a CAT access response from the object ledger computing device;
- when directly accessing the object distributed ledger:
 - obtaining, via the interface, the copy of the object distributed ledger, and
 - indicating the CAT identifier of the particular CAT when detecting a block of the copy of the object distributed ledger that matches the representation of the minimum CAT requirements; and
- verifying that the recovered cryptographic token value of the particular CAT matches the calculated cryptographic token value of the particular CAT.

7. The computing device of claim 5, wherein the processor executes the operational instructions stored in the local memory to determine whether to update the set of CATs based on the evaluation of each CAT with regards to the evaluation profile by:

- determining the evaluation profile to include a required evaluation performance improvement level;
- comparing the providing of the obligation associated with the obligation provider identifier to providing of the obligation associated with a hypothetical obligation provider identifier in accordance with the evaluation profile to produce the obligation evaluation;
- comparing an estimated trigger outcome when the providing of the obligation is associated with the obligation provider identifier to a hypothetical estimated trigger outcome when the providing of the obligation is associated with the hypothetical obligation provider

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identifier in accordance with the evaluation profile to produce the outcome evaluation; and
 indicating to update the CAT when the obligation evaluation and the outcome evaluation satisfies the required evaluation performance improvement level.

8. The computing device of claim 7, wherein the processor executes the operational instructions stored in the local memory to determine the reassignment information for each CAT of the set of CATs by at least one of:

identifying a hypothetical obligation provider identifier based on the evaluation of the CAT with regards to the evaluation profile to establish a new obligation provider identifier of the reassignment information;

when approved, modifying the outcome recipient identifier to establish a new outcome recipient identifier of the reassignment information;

determining the assignment approach, wherein the assignment approach includes one of an equal weight approach, a pass-through approach, and an open approach, wherein the equal weight approach includes establishing the unique percentage-of-face-value level for a corresponding triggered outcome such that one of an identical percentage-of-face-value level and a constant value of each triggered outcome associated with the set of CATs is assigned to a corresponding outcome recipient ID, wherein the pass-through approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome such that one of a variable percentage-of-face-value level and a variable value of each triggered outcome associated with the set of CATs is assigned to the corresponding outcome recipient ID, wherein the open approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome upon detection of the corresponding triggered outcome; and

determining the unique percentage-of-face-value level for a particular corresponding triggered outcome for each CAT of the set of CATs in accordance with the assignment approach.

9. A non-transitory computer readable memory comprises:

a first memory element that stores operational instructions that, when executed by a processing module of a computing device of a computing system, causes the processing module to:

identify a set of contingency-action tokens (CATs) of an object distributed ledger, wherein each contingency-action token (CAT) includes one of a non-fungible token (NFT) and another token linked to the NFT by the object distributed ledger when the CAT is not the NFT, wherein each CAT meets minimum CAT requirements that includes:

a recovered cryptographic token value of the CAT matches a calculated cryptographic token value of the CAT,

an obligation provider identifier record of the CAT that includes an original obligation provider identifier of the NFT when the NFT was initially generated, wherein the obligation provider identifier is associated with providing of an obligation tied to an obligation recipient identifier in accordance with obligation requirements, wherein a positive obligation status indicates that the obligation has historically been provided in accordance with the obligation requirements, and

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an outcome recipient identifier record of the CAT that includes an original outcome recipient identifier of the NFT when the NFT was initially generated, wherein the outcome recipient identifier is associated with a result of a triggered outcome tied to an outcome provider identifier in accordance with contingent outcome rules and the positive obligation status, wherein a contingency status indicates whether the triggered outcome has been triggered for a contingency entity identifier in accordance with the contingent outcome rules;

a second memory element that stores operational instructions that, when executed by the processing module, causes the processing module to:

determine whether to update each CAT of the set of CATs based on an evaluation of each CAT with regards to an evaluation profile, wherein a beneficial improvement to a present estimate of the triggered outcome result from a hypothetical change to the obligation provider identifier of the CAT, wherein the evaluation of the CAT includes producing:

an obligation evaluation based on the evaluation profile, wherein the obligation evaluation compares the providing of the obligation in accordance with the obligation requirements, and

an outcome evaluation, wherein the outcome evaluation produces the present estimate of the triggered outcome based on the evaluation profile; and

a third memory element that stores operational instructions that, when executed by the processing module, causes the processing module to:

when updating the set of CATs to produce an updated set of CATs:

determine reassignment information for each CAT of the set of CATs, wherein the reassignment information includes a confirmed change of at least one of the obligation provider identifier and the outcome recipient identifier, wherein the reassignment information further specifies a unique percentage-of-face-value level for each triggered outcome in accordance with an assignment approach;

facilitate taking control of each CAT of the set of CATs of a blockchain of the object distributed ledger, wherein the processing module facilitates taking control of each CAT by:

obtaining a copy of the object distributed ledger, hashing content of the CAT utilizing a receiving public key of the object distributed ledger to produce a next transaction hash value, encrypting the next transaction hash value utilizing a private key of the computing device to produce a next transaction signature, generating a next block of the blockchain of the object distributed ledger to include the content of the CAT and the next transaction signature, and

causing inclusion of the next block in the object distributed ledger;

update each CAT utilizing the reassignment information for the CAT to produce a corresponding updated CAT; and

cause generation of a new block affiliated with the corresponding updated CAT via the blockchain of the object distributed ledger, wherein the new

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block includes the corresponding updated CAT, wherein the processing module causes generation of the new block by:

issuing a blockchain update request to an object ledger computing device of the computing system serving as a blockchain node of the object distributed ledger, wherein the blockchain update request includes the corresponding updated CAT.

10. The non-transitory computer readable memory of claim 9, wherein the processing module functions to execute the operational instructions stored by the first memory element to cause the processing module to identify the set of CATs of the object distributed ledger by:

determining whether to indirectly or directly access the object distributed ledger;

when indirectly accessing the object distributed ledger:

issuing a CAT access request to the object ledger computing device of the computing system serving as the blockchain node of the object distributed ledger, wherein the CAT access request includes a representation of the minimum CAT requirements, and

extracting a CAT identifier of a particular CAT from a CAT access response from the object ledger computing device;

when directly accessing the object distributed ledger:

obtaining the copy of the object distributed ledger, and indicating the CAT identifier of the particular CAT when detecting a block of the copy of the object distributed ledger that matches the representation of the minimum CAT requirements; and

verifying that the recovered cryptographic token value of the particular CAT matches the calculated cryptographic token value of the particular CAT.

11. The non-transitory computer readable memory of claim 9, wherein the processing module functions to execute the operational instructions stored by the second memory element to cause the processing module to determine whether to update the set of CATs based on the evaluation of each CAT with regards to the evaluation profile by:

determining the evaluation profile to include a required evaluation performance improvement level;

comparing the providing of the obligation associated with the obligation provider identifier to providing of the obligation associated with a hypothetical obligation provider identifier in accordance with the evaluation profile to produce the obligation evaluation;

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comparing an estimated trigger outcome when the providing of the obligation is associated with the obligation provider identifier to a hypothetical estimated trigger outcome when the providing of the obligation is associated with the hypothetical obligation provider identifier in accordance with the evaluation profile to produce the outcome evaluation; and

indicating to update the CAT when the obligation evaluation and the outcome evaluation satisfies the required evaluation performance improvement level.

12. The non-transitory computer readable memory of claim 9, wherein the processing module functions to execute the operational instructions stored by the third memory element to cause the processing module to determine the reassignment information for each CAT of the set of CATs by at least one of:

identifying a hypothetical obligation provider identifier based on the evaluation of the CAT with regards to the evaluation profile to establish a new obligation provider identifier of the reassignment information;

when approved, modifying the outcome recipient identifier to establish a new outcome recipient identifier of the reassignment information;

determining the assignment approach, wherein the assignment approach includes one of an equal weight approach, a pass-through approach, and an open approach, wherein the equal weight approach includes establishing the unique percentage-of-face-value level for a corresponding triggered outcome such that one of an identical percentage-of-face-value level and a constant value of each triggered outcome associated with the set of CATs is assigned to a corresponding outcome recipient ID, wherein the pass-through approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome such that one of a variable percentage-of-face-value level and a variable value of each triggered outcome associated with the set of CATs is assigned to the corresponding outcome recipient ID, wherein the open approach includes establishing the unique percentage-of-face-value level for each corresponding triggered outcome upon detection of the corresponding triggered outcome; and

determining the unique percentage-of-face-value level for a particular corresponding triggered outcome for each CAT of the set of CATs in accordance with the assignment approach.

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